



Lakshya JEE 2027

JEE Mains (PYQ) : KPP 01

Electrostatics

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PROPERTIES OF ELECTRIC CHARGE AND COULOMB'S LAW AND PRINCIPLE OF SUPERPOSITION

1. A small uncharged conducting sphere is placed in contact with an identical sphere but having 4×10^{-8} C charge and then removed to a distance such that the force of repulsion between them is 9×10^{-3} N. The distance between them is:

(Take $\frac{1}{4\pi\epsilon_0}$ as 9×10^9 in SI units).

[JEE MAINS Jan. 24, 2025 (II)]

- (1) 2 cm (2) 3 cm
(3) 4 cm (4) 1 cm

2. A positive ion A and a negative ion B has charges 6.67×10^{-19} C and 9.6×10^{-10} C, and masses 19.2×10^{-27} kg and 9×10^{-27} kg respectively. At an instant, the ions are separated by a certain distance r . At that instant the ratio of the magnitudes of electrostatic force to gravitational force is $P \times 10^{44}$, where the value of P is _____.

(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-1}$ and universal gravitational constant as $6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$).

[JEE MAINS Jan. 23, 2025 (I)]

3. Two identical conducting spheres P and S with charge Q on each, repel each other with a force 16 N. A third identical uncharged conducting sphere R is successively brought in contact with the two spheres. The new force of repulsion between P and S is:

[JEE MAINS April 6, 2024 (II)]

- (1) 4N (2) 6N
(3) 1N (4) 12N

4. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle θ with each other. When suspended in water the angle remains the same. If density of the material of the sphere is 1.5 g/cc, the dielectric constant of water will be _____ (Take density of water = 1 g/cc).

[JEE MAINS Feb. 1, 2024 (II)]

5. Force between two point charges q_1 , and q_2 placed in vacuum at ' r ' cm apart is F. Force between them when placed in a medium having dielectric $K = 5$ at ' $r/5$ ' cm apart will be: [JEE MAINS Jan. 31, 2024 (II)]

- (1) $F/25$ (2) $5F$
(3) $F/5$ (4) $25F$

6. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 37° with each other. When suspended in a liquid of density 0.7 g/cm^3 , the angle remains same. If density of material of the sphere is 1.4 g/cm^3 , the dielectric constant of the liquid is _____. $\left(\tan 37^\circ = \frac{3}{4} \right)$

[JEE MAINS Jan. 30, 2024 (II)]

7. A thin metallic wire having cross sectional area of 10^{-4} m^2 is used to make a ring of radius 30 cm. A positive charge of $2\pi\text{C}$ is uniformly distributed over the ring, while another positive charge of 30 pC is kept at the centre of the ring. The tension in the ring is _____ N; provided that the ring does not get deformed (neglect the influence of gravity). (given, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI units}$).

[JEE MAINS Jan. 27, 2024 (I)]

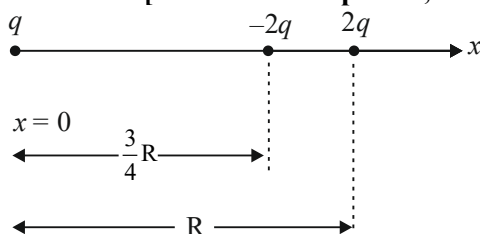
8. A $10 \mu\text{C}$ charge is divided into two parts and placed at 1 cm distance so that the repulsive force between them is maximum. The charges of the two parts are:

[JEE MAINS April 13, 2023 (II)]

- (1) $9 \mu\text{C}$, $1 \mu\text{C}$ (2) $5 \mu\text{C}$, $5 \mu\text{C}$
(3) $7 \mu\text{C}$, $3 \mu\text{C}$ (4) $8 \mu\text{C}$, $2 \mu\text{C}$

9. Three point charges q , $-2q$ and $2q$ are placed on x -axis at a distance $x = 0$, $x = \frac{3}{4}R$ and $x = R$ respectively from origin as shown. If $q = 2 \times 10^{-6} \text{ C}$ and $R = 2 \text{ cm}$, the magnitude of net force experienced by the charge $-2q$ is _____ N.

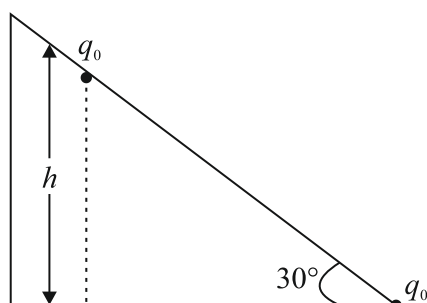
[JEE MAINS April 13, 2023 (II)]



10. As shown in the figure, a configuration of two equal point charges ($q_0 = +2\mu \text{ C}$) is placed on an inclined plane. Mass of each point charge is 20 g . Assume that there is no friction between charge and plane. For the system of two point charges to be in equilibrium (at rest) the height $h = x \times 10^{-3} \text{ m}$. The value of x is _____.

[JEE MAINS April 11, 2023 (I)]

(Take $\frac{1}{4\pi\epsilon_0}$ as $9 \times 10^9 \text{ N m}^2\text{C}^{-2}$, $g = 10 \text{ ms}^{-1}$)



11. Two equal positive point charges are separated by a distance $2a$. The distance of a point from the centre of the line joining two charges on the equatorial line (perpendicular bisector) at which force experienced by a test charge q_0 becomes maximum is $\frac{a}{\sqrt{x}}$. The value of x is _____.

[JEE MAINS Feb. 1, 2023 (I)]

12. A point charge $q_1 = 4q_0$, is placed at origin. Another point charge $q_1 = -q_0$ is placed at $x = 12 \text{ cm}$. Charge of proton is q_0 . The proton is placed on x -axis so that the electrostatic force on the proton is zero. In this situation, the position of the proton from the origin is _____ cm.

[JEE MAINS Jan. 29, 2023 (I)]

13. If two charges q_1 and q_2 are separated with distance ' d ' and placed in a medium of dielectric constant K . What will be the equivalent distance between charges in air for the same electrostatic force?

[JEE MAINS Jan. 24, 2023 (I)]

- (1) $d\sqrt{K}$ (2) $K\sqrt{d}$
(3) $1.5d\sqrt{K}$ (4) $2d\sqrt{K}$

14. Two identical metallic spheres A and B when placed at certain distance in air repel each other with a force of F . Another identical uncharged sphere C is first placed in contact with A and then in contact with B and finally placed at midpoint between spheres A and B. The force experienced by sphere C will be:

[JEE MAINS July 29, 2022 (II)]

- (1) $3F/2$ (2) $3F/4$
(3) F (4) $2F$

15. Two identical positive charges Q each are fixed at a distance of ' $2a$ ' apart from each other. Another point charge q_0 with mass ' m ' is placed at midpoint between two fixed charges. For a small displacement along the line joining the fixed charges, the charge q_0 executes SHM. The time period of oscillation of charge q_0 , will be:

[JEE MAINS July 27, 2022 (I)]

- (1) $\sqrt{\frac{4\pi^3\epsilon_0 ma^3}{q_0 Q}}$ (2) $\sqrt{\frac{q_0 Q}{4\pi^3\epsilon_0 ma^3}}$
(3) $\sqrt{\frac{2\pi^2\epsilon_0 ma^3}{q_0 Q}}$ (4) $\sqrt{\frac{8\pi^3\epsilon_0 ma^3}{q_0 Q}}$

16. A charge of $4 \mu\text{C}$ is to be divided into two parts. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be:

[JEE MAINS July 27, 2022 (II)]

- (1) $1 \mu\text{C}$ and $3 \mu\text{C}$ (2) $2 \mu\text{C}$ and $2 \mu\text{C}$
(3) $0 \mu\text{C}$ and $4 \mu\text{C}$ (4) $1.5 \mu\text{C}$ and $2.5 \mu\text{C}$

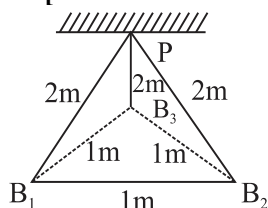
17. Two point charges Q each are placed at a distance d apart. A third point charge q is placed at a distance x from mid-point on the perpendicular bisector. The value of x at which charge q will experience the maximum Coulomb's force is:

[JEE MAINS June 29, 2022 (II)]

- (1) $x = d$ (2) $x = \frac{d}{2}$
(3) $x = \frac{d}{\sqrt{2}}$ (4) $x = \frac{d}{2\sqrt{2}}$

18. Three identical charged balls each of charge 2 C are suspended from a common point P by silk threads of 2 m each (as shown in figure). They form an equilateral triangle of side 1 m . The ratio of net force on a charged ball to the force between any two charged balls will be:

[JEE MAINS June 27, 2022 (II)]



- (1) $1 : 1$ (2) $1 : 4$
(3) $\sqrt{3} : 2$ (4) $\sqrt{3} : 1$

19. Sixty four conducting drops each of radius 0.02 m and each carrying a charge of $5\text{ }\mu\text{C}$ are combined to form a bigger drop. The ratio of surface density of bigger drop to the smaller drop will be:

[JEE MAINS June 26, 2022 (II)]

- (1) $1 : 4$ (2) $4 : 1$
(3) $1 : 8$ (4) $8 : 1$

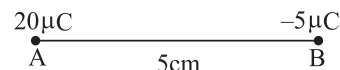
20. Two identical charged particles each having a mass 10 g and charge $2.0 \times 10^{-7}\text{ C}$ placed on a horizontal table with a separation of L between them such that they stay in limited equilibrium. If the coefficient of friction between each particle and the table is 0.25 , find the value of L . [Use $g = 10\text{ ms}^{-2}$]

[JEE MAINS June 24, 2022 (II)]

- (1) 12 cm (2) 10 cm
(3) 8 cm (4) 5 cm

21. Two particles A and B having charges $20\text{ }\mu\text{C}$ and $-5\text{ }\mu\text{C}$ respectively are held fixed with a separation of 5 cm . At what position a third charged particle should be placed so that it does not experience a net electric force?

[JEE MAINS Aug. 31, 2021 (I)]



- (1) At 5 cm from $20\text{ }\mu\text{C}$ on the left side of system
(2) At 5 cm from $-5\text{ }\mu\text{C}$ on the right side
(3) At 1.25 cm from $-5\text{ }\mu\text{C}$ between two charges
(4) At midpoint between two charges

22. Two identical tennis balls each having mass ' m ' and charge ' q ' are suspended from a fixed point by threads of length ' l '. What is the equilibrium separation when each thread makes a small angle ' θ ' with the vertical?

[JEE MAINS July 27, 2021 (I)]

- (1) $d = \left(\frac{q^2 l}{2\pi\epsilon_0 m g} \right)^{\frac{1}{2}}$ (2) $d = \left(\frac{q^2 l}{2\pi\epsilon_0 m g} \right)^{\frac{1}{3}}$
(3) $d = \left(\frac{q^2 l^2}{2\pi\epsilon_0 m^2 g} \right)^{\frac{1}{3}}$ (4) $d = \left(\frac{q^2 l^2}{2\pi\epsilon_0 m^2 g^2} \right)^{\frac{1}{3}}$

23. A certain charge Q is divided into two parts q and $(Q - q)$. How should the charges Q and q be divided so that q and $(Q - q)$ placed at a certain distance apart experience maximum electrostatic repulsion?

[JEE MAINS July 20, 2021 (I)]

- (1) $Q = \frac{q}{2}$ (2) $Q = 2q$
(3) $Q = 4q$ (4) $Q = 3q$

24. An infinite number of point charges, each carrying $1\text{ }\mu\text{C}$ charge, are placed along the y -axis at $y = 1\text{ m}$, 2 m , 4 m , 8 m _____.

The total force on a 1 C point charge, placed at the origin, is $x \times 10^3\text{ N}$.

The value of x , to the nearest integer, is _____.

[Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\text{ Nm}^2/\text{C}^2$]

[JEE MAINS March 18, 2021 (I)]

25. Two identical conducting spheres with negligible volume have 2.1 nC and -0.1 nC charges, respectively. They are brought into contact and then separated by a distance of 0.5 m. The electrostatic force acting between the spheres is $\times 10^{-9}$ N.

[Given: $4\pi\epsilon_0 = \frac{1}{9 \times 10^9}$ SI unit]

[JEE MAINS Feb. 25, 2021 (I)]

26. Three charges $+Q$, q , $+Q$ are placed respectively, at origin, $d/2$ and d from the origin, on the x -axis. If the net force experienced by $+Q$, placed at $x = 0$, is zero, then value of q is:

[JEE MAINS Jan. 9, 2019 (I)]

- (1) $-Q/4$ (2) $+Q/2$
(3) $+Q/4$ (4) $-Q/2$

27. Two identical conducting spheres A and B, carry equal charge. They are separated by a distance much larger than their diameter, and the force between them is F . A third identical conducting sphere, C, is uncharged. Sphere C is first touched to A, then to B, and then removed. As a result, the force between A and B would be equal to.

[JEE MAINS 2018]

- (1) $\frac{3F}{4}$ (2) $\frac{F}{2}$
(3) F (4) $\frac{3F}{8}$

28. Two charges, each equal to q , are kept at $x = -a$ and $x = a$ on the x -axis. A particle of mass m and charge $q_0 = \frac{q}{2}$ is placed at the origin. If charge q_0 , is given a small displacement ($y \ll a$) along the y -axis, the net force acting on the particle is proportional to

[JEE MAINS 2013]

- (1) y (2) $-y$
(3) $\frac{1}{y}$ (4) $-\frac{1}{y}$

29. Two identical charged spheres suspended from a common point by two massless strings of length l are initially a distance d ($d \ll l$) apart because of their mutual repulsion. The charge begins to leak from both the spheres at a constant rate. As a result charges approach each other with a velocity v . Then as a function of distance x between them,

[JEE MAINS 2011]

- (1) $v \propto x^{-1}$ (2) $v \propto x^{1/2}$
(3) $v \propto x$ (4) $v \propto x^{-1/2}$

30. A charge Q is placed at each of the opposite corners of a square. A charge q is placed at each of the other two corners. If the net electrical force on Q is zero, then Q/q equals:

[JEE MAINS 2009]

- (1) -1 (2) 1
(3) $-\frac{1}{\sqrt{2}}$ (4) $-2\sqrt{2}$

31. If g_E and g_M are the accelerations due to gravity on the surfaces of the earth and the moon respectively and if Millikan's oil drop experiment could be performed on the two surfaces, one will find the ratio $\frac{\text{electronic charge on the moon}}{\text{electronic charge on the earth}}$ to be:

[JEE MAINS 2007]

- (1) g_M/g_E (2) 1
(3) 0 (4) g_E/g_M

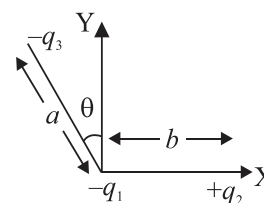
32. Two spherical conductors B and C having equal radii and carrying equal charges on them repel each other with a force F when kept apart at some distance. A third spherical conductor having same radius as that of B but uncharged is brought in contact with B, then brought in contact with C and finally removed away from both. The new force of repulsion between B and C is:

[JEE MAINS 2004]

- (1) $F/8$ (2) $3F/4$
(3) $F/4$ (4) $3F/8$

33. Three charges $-q_1$, $+q_2$, and $-q_3$, are placed as shown in the figure. The x -component of the force on $-q_1$ is proportional to.

[JEE MAINS 2003]



- (1) $\frac{q_2}{b^2} - \frac{q_3}{a^2} \cos \theta$ (2) $\frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta$
(3) $\frac{q_2}{b^2} + \frac{q_3}{a^2} \cos \theta$ (4) $\frac{q_2}{b^2} - \frac{q_3}{a^2} \sin \theta$

34. If a charge q is placed at the centre of the line joining two equal charges Q such that the system is in equilibrium then the value of q is:

[JEE MAINS 2002]

- (1) $Q/2$ (2) $-Q/2$
(3) $Q/4$ (4) $-Q/4$

35. An oil drop of radius 2 mm with a density 3 g cm^{-3} is held stationary under a constant electric field $3.55 \times 10^5 \text{ V m}^{-1}$ in the Millikan's oil drop experiment. What is the number of excess electrons that the oil drop will process? Consider $g = 9.81 \text{ m/s}^2$

[JEE MAINS March 18, 2021 (I)]

- (1) 1.73×10^{12} (2) 1.73×10^{10}
(3) 48.8×10^{11} (4) 17.3×10^{10}

36. Four charges equal to $-Q$ are placed at the four corners of a square and a charge q is at its centre. If the system is in equilibrium the value of q is.

[JEE MAINS 2004]

- (1) $-\frac{Q}{2}(1+2\sqrt{2})$
(2) $\frac{Q}{4}(1+2\sqrt{2})$
(3) $-\frac{Q}{4}(1+2\sqrt{2})$
(4) $\frac{Q}{2}(1+2\sqrt{2})$

37. A charged oil drop is suspended in a uniform field of $3 \times 10^4 \text{ v/m}$ so that it neither falls nor rises. The charge on the drop will be (Take the mass of the charge $= 9.9 \times 10^{-15} \text{ kg}$ and $g = 10 \text{ m/s}^2$)

[JEE MAINS 2004]

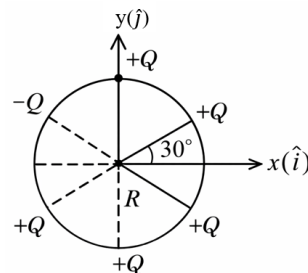
- (1) $1.6 \times 10^{-18} \text{ C}$
(2) $3.2 \times 10^{-18} \text{ C}$
(3) $3.3 \times 10^{-18} \text{ C}$
(4) $4.8 \times 10^{-18} \text{ C}$

ELECTRIC FIELD DUE TO A POINT CHARGE AND SUPERPOSITION PRINCIPLE

38. Six point charges are kept 60° apart from each other on the circumference of a circle of radius R as shown in figure. The net electric field at the center of the circle is _____.

(ϵ_0 is permittivity of free space)

[JEE MAINS Jan. 22, 2026 (I)]



- (1) $-\left(\frac{5Q}{8\pi\epsilon_0 R^2}\right)(\hat{i} - 3\hat{j})$
(2) $\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i} - \hat{j})$
(3) $-\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i} - \hat{j})$
(4) $-\frac{5Q}{8\pi\epsilon_0 R^2}(\hat{i} + \sqrt{3}\hat{j})$

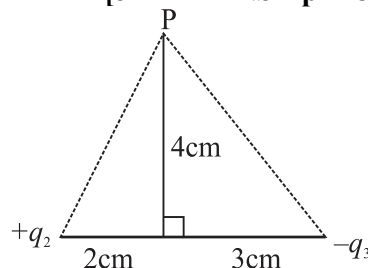
39. A point charge $+q$ is placed at the origin. A second point charge $+9q$ is placed at $(d, 0, 0)$ in Cartesian coordinate system. The point in between them where the electric field vanishes is:

[JEE MAINS April 2, 2025 (I)]

- (1) $(4d/3, 0, 0)$ (2) $(d/4, 0, 0)$
(3) $(3d/4, 0, 0)$ (4) $(d/3, 0, 0)$

40. If the net electric field at point P (as shown in figure) along Y axis is zero, then the ratio of $\left|\frac{q_2}{q_3}\right|$ is $\frac{8}{5\sqrt{x}}$, where $x =$ _____.

[JEE MAINS April 8, 2024 (II)]



41. Two charges q and $3q$ are separated by a distance ' r ' in air. At a distance x from charge q , the resultant electric field is zero. The value of x is:

[JEE MAINS Jan. 31, 2024 (I)]

- (1) $\frac{r}{(1+\sqrt{3})}$ (2) $\frac{(1+\sqrt{3})}{r}$
(3) $r(1+\sqrt{3})$ (4) $\frac{r}{3(1+\sqrt{3})}$

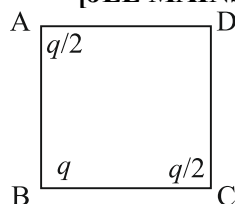
42. Point charge of $10 \mu\text{C}$ is placed at the origin. At what location on the X-axis should a point charge of $40 \mu\text{C}$ be placed so that the net electric field is zero at $x = 2 \text{ cm}$ on the X-axis?

[JEE MAINS Jan. 25, 2023 (II)]

- (1) $x = 6 \text{ cm}$ (2) $x = 4 \text{ cm}$
(3) $x = 8 \text{ cm}$ (4) $x = -4 \text{ cm}$

43. The three charges $q/2$, q and $q/2$ are placed at the corners A, B and C of a square of side ' a ' as shown in figure. The magnitude of electric field (E) at the corner D of the square, is

[JEE MAINS June 28, 2022 (I)]



- (1) $\frac{q}{4\pi\epsilon_0 a^2} \left(\frac{1}{\sqrt{2}} + \frac{1}{2} \right)$
(2) $\frac{q}{4\pi\epsilon_0 a^2} \left(1 + \frac{1}{\sqrt{2}} \right)$
(3) $\frac{q}{4\pi\epsilon_0 a^2} \left(1 - \frac{1}{\sqrt{2}} \right)$
(4) $\frac{q}{4\pi\epsilon_0 a^2} \left(\frac{1}{\sqrt{2}} - \frac{1}{2} \right)$

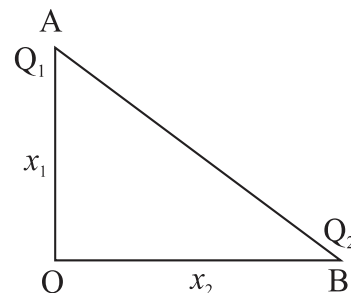
44. Two point charges A and B of magnitude $+8 \times 10^{-6} \text{ C}$ and $-8 \times 10^{-6} \text{ C}$ respectively are placed at a distance apart. The electric field at the middle point O between the charges is $6.4 \times 10^4 \text{ NC}^{-1}$. The distance ' d ' between the point charges A and B is:

[JEE MAINS June 28, 2022 (II)]

- (1) 2.0 m (2) 3.0 m
(3) 1.0 m (4) 4.0 m

45. Charges Q_1 and Q_2 are at points A and B of a right angle triangle OAB (see figure). The resultant electric field at point O is perpendicular to the hypotenuse, then Q_1/Q_2 is proportional to:

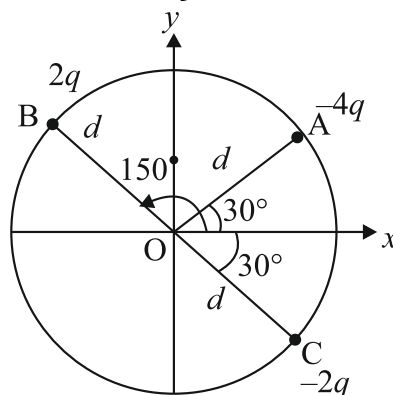
[JEE MAINS Sep. 06, 2020 (I)]



- (1) $\frac{x_1^3}{x_2^3}$ (2) $\frac{x_2}{x_1}$
(3) $\frac{x_1}{x_2}$ (4) $\frac{x_2^2}{x_1^2}$

46. Three charged particles A, B and C with charges $-4q$, $2q$ and $-2q$ are present on the circumference of a circle of radius d . The charged particles A, C and centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x-direction is:

[JEE MAINS Jan. 8, 2020 (I)]



- (1) $\frac{\sqrt{3}q}{\pi\epsilon_0 d^2}$ (2) $\frac{2\sqrt{3}q}{\pi\epsilon_0 d^2}$
(3) $\frac{\sqrt{3}q}{4\pi\epsilon_0 d^2}$ (4) $\frac{3\sqrt{3}q}{4\pi\epsilon_0 d^2}$

47. Four point charges $-q, +q, +q$ and $-q$ are placed on y -axis at $y = -2d, y = +d, y = +d$ and $y = +2d$, respectively. The magnitude of the electric field E at a point on the x -axis at $x = D$, with $D \gg d$, will behave as: [JEE MAINS April 09, 2019 (II)]

- (1) $E \propto \frac{1}{D^3}$ (2) $E \propto \frac{1}{D}$
(3) $E \propto \frac{1}{D^4}$ (4) $E \propto \frac{1}{D^2}$

48. Two point charges $q_1, (\sqrt{10} \mu\text{C})$ and $q_2 (-25 \mu\text{C})$ are placed on the x -axis at $x = 1 \text{ m}$ and $x = 4 \text{ m}$ respectively. The electric field (in V/m) at a point $y = 3 \text{ m}$ on y -axis is,

$$\left[\text{take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right]$$

[JEE MAINS Jan. 09, 2019 (II)]

- (1) $(63\hat{i} - 27\hat{j}) \times 10^2$
(2) $(-63\hat{i} + 27\hat{j}) \times 10^2$
(3) $(81\hat{i} - 81\hat{j}) \times 10^2$
(4) $(-81\hat{i} + 81\hat{j}) \times 10^2$

49. Two point charges $+8q$ and $-2q$ are located at $x = 0$ and $x = L$ respectively. The location of a point on the x -axis at which the net electric field due to these two point charges is zero is: [JEE MAINS 2005]

- (1) $\frac{L}{4}$ (2) $2L$
(3) $4L$ (4) $8L$

50. Three point charges of magnitude $5\mu\text{C}, 0.16\mu\text{C}$ and $0.3\mu\text{C}$ are located at the vertices A, B, C of a right angled triangle whose sides are $AB = 3 \text{ cm}$, $BC = 3\sqrt{2} \text{ cm}$ and $CA = 3 \text{ cm}$ and point A is the right angle corner. Charge at point A experiences _____ N of electrostatic force due to the other two charges.

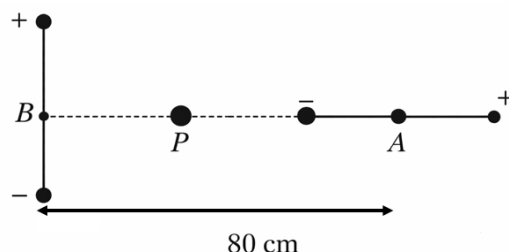
[JEE MAINS July 26, 2022 (II)]

ELECTRIC FIELD DUE TO AN ELECTRIC DIPOLE, DIPOLE IN A UNIFORM FIELD, POTENTIAL ENERGY ASSOCIATED WITH DIPOLE

51. Three charges $+2q, +3q$ and $-4q$ are situated at $(0, -3a), (2a, 0)$ and $(-2a, 0)$ respectively in the xy plane. The resultant dipole moment about origin is _____. [JEE MAINS Jan. 24, 2026 (I)]

- (1) $2qa(7\hat{i} - 3\hat{j})$ (2) $2qa(3\hat{j} - 7\hat{i})$
(3) $2qa(3\hat{i} - 7\hat{j})$ (4) $2qa(3\hat{j} - \hat{i})$

52. Two short dipoles (A, B), A having charges $\pm 2\mu\text{C}$ and length 1 cm and B having charges $\pm 4\mu\text{C}$ and length 1 cm are placed with their centres 80 cm apart as shown in the figure. The electric field at a point P, equi-distant from the centres of both dipoles is ____ N/C . [JEE MAINS Jan. 23, 2026 (II)]



- (1) $\frac{9}{16}\sqrt{2} \times 10^5$ (2) $\frac{9}{16}\sqrt{2} \times 10^4$
(3) $9\sqrt{2} \times 10^4$ (4) $4.5\sqrt{2} \times 10^4$

53. A simple pendulum has a bob with mass m and charge q . The pendulum string has negligible mass. When a uniform and horizontal electric field $\vec{E}$ is applied, the tension in the string changes. The final tension in the string, when pendulum attains an equilibrium position is _____. [JEE MAINS Jan. 22, 2026 (I)]

- (1) $mg - qE$
(2) $\sqrt{m^2g^2 + q^2E^2}$
(3) $mg + qE$
(4) $\sqrt{m^2g^2 - q^2E^2}$

54. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Net dipole moment of a polar linear isotropic dielectric substance is not zero even in the absence of an external electric field.

Reason (R): In absence of an external electric field, the different permanent dipoles of a polar dielectric substance are oriented in random directions.

In the light of the above statements, choose the most appropriate answer from the options given below:

[JEE MAINS April 2, 2025 (II)]

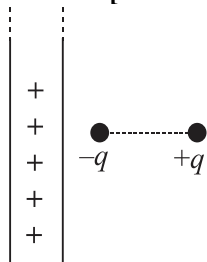
- (1) (A) is correct but (R) is not correct
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (4) (A) is not correct but (R) is correct

55. An electric dipole of moment $\vec{p} = (\hat{i} - 3\hat{j} + 2\hat{k}) \times 10^{-29}$ Cm is at the origin (0, 0, 0). The electric field due to this dipole at $\vec{r} = +\hat{i} + 3\hat{j} + 5\hat{k}$ (note that $\vec{r} \cdot \vec{p} = 0$) is parallel to: [JEE MAINS Jan. 09, 2021 (I)]

- (1) $(+\hat{i} - 3\hat{j} - 2\hat{k})$
- (2) $(-\hat{i} + 3\hat{j} - 2\hat{k})$
- (3) $(+\hat{i} + 3\hat{j} - 2\hat{k})$
- (4) $(-\hat{i} - 3\hat{j} + 2\hat{k})$

56. An electric dipole is placed at a distance of 2 cm from an infinite plane sheet having positive charge density σ_0 . Choose the correct option from the following.

[JEE MAINS Jan. 29, 2025]



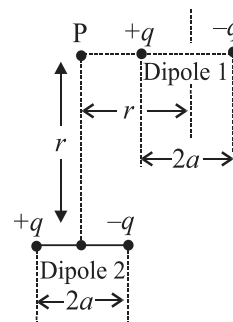
- (1) Torque on dipole is zero and net force is directed away from the sheet.
- (2) Torque on dipole is zero and net force acts towards the sheet.
- (3) Potential energy of dipole is minimum and torque is zero.
- (4) Potential energy and torque both are maximum

57. An electric dipole of mass m , charge q , and length l is placed in a uniform electric field $\vec{E} = E_0 \hat{i}$. When the dipole is rotated slightly from its equilibrium position and released, the time period of its oscillations will be [JEE MAINS Jan. 29, 2025 (I)]

- (1) $\frac{1}{2\pi} \sqrt{\frac{2ml}{qE_0}}$
- (2) $2\pi \sqrt{\frac{ml}{qE_0}}$
- (3) $\frac{1}{2\pi} \sqrt{\frac{2ml}{2qE_0}}$
- (4) $2\pi \sqrt{\frac{ml}{2qE_0}}$

58. A point particle of charge Q is located at P along the axis of an electric dipole 1 at a distance r as shown in the figure. The point P is also on the equatorial plane of a second electric dipole 2 at a distance r . The dipoles are made of opposite charge q separated by a distance $2a$. For the charge particle at P not to experience any net force, which of the following correctly describes the situation?

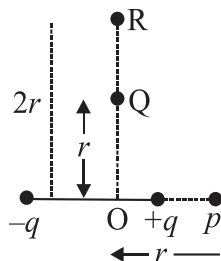
[JEE MAINS Jan. 23, 2025 (I)]



- (1) $\frac{a}{r} \approx 20$
- (2) $\frac{a}{r} \approx 10$
- (3) $\frac{a}{r} \approx 0.5$
- (4) $\frac{a}{r} \approx 3$

59. The electric field at point p due to an electric dipole is E . The electric field at point R on equatorial line will be $\frac{E}{x}$. The value of x :

[JEE MAINS April 05, 2024 (II)]



60. Two charges of $-4 \mu\text{C}$ and $+4 \mu\text{C}$ are placed at the points $A(1, 0, 4)\text{m}$ and $B(2, -1, 5)\text{m}$ located in an electric field $\vec{E} = 0.20\hat{i} \text{ V/cm}$. The magnitude of the torque acting on the dipole is $8\sqrt{\alpha} \times 10^{-5} \text{ Nm}$, Where $\alpha = \underline{\hspace{1cm}}$. [JEE MAINS Jan. 27, 2024 (II)]

61. The electric field due to a short electric dipole at a large distance (r) from center of dipole on the equatorial plane varies with distance as:

[JEE MAINS April 15, 2023 (I)]

- (1) r (2) $\frac{1}{r}$
(3) $\frac{1}{r^3}$ (4) $\frac{1}{r^2}$

62. Two charges each of magnitude 0.01 C and separated by a distance of 0.4 mm constitute an electric dipole. If the dipole is placed in a uniform electric field ' $\vec{E}$ ' of 10 dyne/C making 30° angle with $\vec{E}$, the magnitude of torque acting on dipole is:

[JEE MAINS April 13, 2023 (I)]

- (1) $4.0 \times 10^{-10} \text{ Nm}$ (2) $2.0 \times 10^{-10} \text{ Nm}$
(3) $1.0 \times 10^{-8} \text{ Nm}$ (4) $1.5 \times 10^{-9} \text{ Nm}$

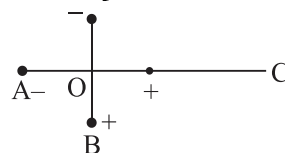
63. Two electric dipoles of dipole moments $1.2 \times 10^{-30} \text{ cm}$ and $2.4 \times 10^{-30} \text{ cm}$ are placed in two difference uniform electric fields of strengths $5 \times 10^4 \text{ NC}^{-1}$ and $15 \times 10^4 \text{ NC}^{-1}$ respectively. The ratio of maximum torque experienced by the electric dipoles will be $\frac{1}{x}$. The value of x is $\underline{\hspace{1cm}}$.

[JEE MAINS July 28, 2022 (I)]

64. Two ideal electric dipoles A and B, having their dipole moment p_1 and p_2 respectively are placed on a plane with their centres at O as shown in the figure. At point C on the axis of dipole A, the resultant electric field is making an angle of 37° with the axis. The ratio of the dipole moment of A and B, $\frac{p_1}{p_2}$ is:

(take $\sin 37^\circ = \frac{3}{5}$).

[JEE MAINS July 25, 2021 (II)]



- (1) $3/8$ (2) $3/2$
(3) $2/3$ (4) $4/3$

65. Two identical electric point dipoles have dipole moments $\vec{P}_1 = P\hat{i}$ and $\vec{P}_2 = -P\hat{i}$ and are held on the x axis at distance ' a ' from each other. When released, they move along x -axis with the direction of their dipole moments remaining unchanged. If the mass of each dipole is ' m ', their speed when they are infinitely far apart is: [JEE MAINS Sep. 06, 2020 (II)]

- (1) $\frac{P}{a} \sqrt{\frac{1}{\pi\epsilon_0 ma}}$
(2) $\frac{P}{a} \sqrt{\frac{1}{2\pi\epsilon_0 ma}}$
(3) $\frac{P}{a} \sqrt{\frac{2}{\pi\epsilon_0 ma}}$
(4) $\frac{P}{a} \sqrt{\frac{2}{2\pi\epsilon_0 ma}}$

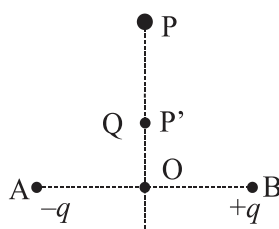
66. An electric field of 1000 V/m is applied to an electric dipole at angle of 45° . The value of electric dipole moment is 10^{-29} C.m . What is the potential energy of the electric dipole?

[JEE MAINS Jan 11, 2019 (II)]

- (1) $-20 \times 10^{-18} \text{ J}$ (2) $-7 \times 10^{-27} \text{ J}$
(3) $-10 \times 10^{-29} \text{ J}$ (4) $-9 \times 10^{-20} \text{ J}$

67. Charges $-q$ and $+q$ located at A and B, respectively, constitute an electric dipole. Distance $AB = 2a$, O is the mid point of the dipole and OP is perpendicular to AB. A charge Q is placed at P where $OP = y$ and $y \gg 2a$. The charge Q experiences an electrostatic force F. If Q is now moved along the equatorial line to P' such that $OP' = \left(\frac{y}{3}\right)$, the force on Q will be close to: $\left(\frac{y}{3} \gg 2a\right)$.

[JEE MAINS Jan 10, 2019 (II)]



68. An electric dipole has a fixed dipole moment $\vec{p}$, which makes angle θ with respect to x -axis. When subjected to an electric field $\vec{E}_1 = E\hat{i}$, it experiences a torque $\vec{T}_1 = \tau\hat{k}$. When subjected to another electric field $\vec{E}_2 = \sqrt{3}E_1\hat{j}$ it experiences torque $\vec{T}_2 = -\vec{T}_1$. The angle θ is: [JEE MAINS 2017]

- (1) $3F$ (2) $\frac{F}{3}$
(3) $9F$ (4) $27F$
69. An electric dipole is placed at an angle of 30° to a non-uniform electric field. The dipole will experience [JEE: MAINS 2006]
- (1) a translational force only in the direction of the field
(2) a translational force only in a direction normal to the direction of the field
(3) a torque as well as a translational force
(4) a torque only

SHORT DIPOLE IN NON-UNIFORM ELECTRIC FIELD, DIPOLE OSCILLATION IN UNIFORM ELECTRIC FIELD, MOTION OF CHARGE PARTICLE IN UNIFORM FIELD

70. Consider two identical metallic spheres of radius R each having charge Q and mass m . Their centers have an initial separation of $4R$. Both the spheres are given an initial speed of u towards each other. The minimum value of u , so that they can just touch each other is:

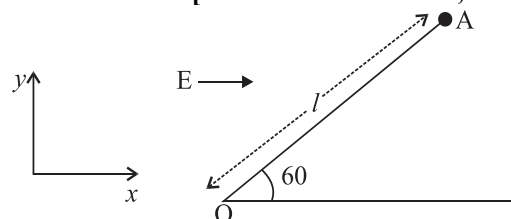
(Take $k = \frac{1}{4\pi\epsilon_0}$ and assume $kQ^2 > Gm^2$ where G is the Gravitational constant).

[JEE MAINS Jan. 21, 2026 (II)]

- (1) $\sqrt{\frac{kQ^2}{4mR} \left(1 + \frac{Gm^2}{kQ^2}\right)}$
(2) $\sqrt{\frac{kQ^2}{2mR} \left(1 - \frac{Gm^2}{2kQ^2}\right)}$
(3) $\sqrt{\frac{kQ^2}{2mR} \left(1 - \frac{Gm^2}{kQ^2}\right)}$
(4) $\sqrt{\frac{kQ^2}{4mR} \left(1 - \frac{Gm^2}{kQ^2}\right)}$

71. A particle of mass ' m ' and charge ' q ' is fastened to one end 'A' of a massless string having equilibrium length ' l ' whose other end is fixed at point 'O'. The whole system is placed on a frictionless horizontal plane and is initially at rest. If uniform electric field is switched on along the direction as shown in figure, then the speed of the particle when it crosses the x -axis is

[JEE MAINS Jan. 28, 2025 (I)]



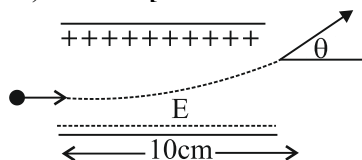
- (1) $\sqrt{\frac{2qEl}{m}}$ (2) $\sqrt{\frac{qEl}{4m}}$
(3) $\sqrt{\frac{qEl}{m}}$ (4) $\sqrt{\frac{qEl}{2m}}$

72. An electron is made to enter symmetrically between two parallel and equally but oppositely charged metal plates, each of 10 cm length. The electron emerges out of the field region with a horizontal component of velocity 10^6 m/s. If the magnitude of the electric field between the plates is 9.1 V/cm, then the vertical component of velocity of electron is: (mass of electron = 9.1×10^{-31} kg and charge of electron = 1.6×10^{-19} C). [JEE MAINS Jan. 22, 2025 (I)]
- (1) 1×10^6 m/s (2) 0
(3) 16×10^6 m/s (4) 16×10^4 m/s

73. A point charge 2×10^{-2} C is moved from P to S in a uniform electric field of 30 NC^{-1} directed along positive x-axis. If coordinates of P and S are (1, 2, 0) m and (0, 0, 0) m respectively, the work done by electric field will be:

[JEE MAINS Jan. 29, 2023 (II)]

- (1) 1200 mJ (2) 600 mJ
(3) -600 mJ (4) -1200 mJ
74. A uniform electric field of 10 N/C is created between two parallel charged plates (as shown in figure). An electron enters the field symmetrically between the plates with a kinetic energy 0.5 eV. The length of each plate is 10 cm. The angle (θ) of deviation of the path of electron as it comes out of the field is _____ (in degree). [JEE MAINS Jan. 25, 2023 (I)]

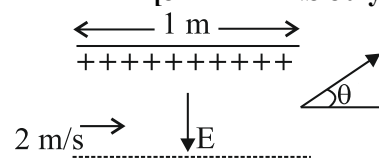


75. A stream of a positively charged particles having $\frac{q}{m} = 2 \times 10^{11} \frac{\text{C}}{\text{kg}}$ and velocity $\vec{v}_0 = 3 \times 10^7 \hat{i} \text{ m/s}$ is deflected by an electric field $1.8 \hat{j} \text{ KV/m}$. The electric field exists in a region of 10 cm along x direction. Due to the electric field, the deflection of the charge particles in y direction is _____ mm.

[JEE MAINS Jan. 24, 2023 (I)]

76. A uniform electric field $E = (8m/e) \text{ V/m}$ is created between two parallel plates of length 1 m as shown in figure, (where m = mass of electron and e = charge of electron). An electron enters the field symmetrically between the plates with a speed of 2 m/s. The angle of the deviation (θ) of the path of the electron as it comes out of the field will be _____.

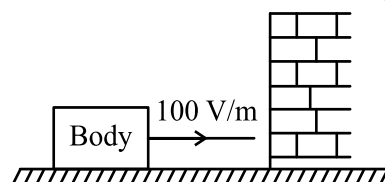
[JEE MAINS July 28, 2022 (II)]



- (1) $\tan^{-1}(4)$ (2) $\tan^{-1}(2)$
(3) $\tan^{-1}\left(\frac{1}{3}\right)$ (4) $\tan^{-1}(3)$
77. A positive charge particle of $100 \mu\text{g}$ is thrown in opposite direction to a uniform electric field of strength $1 \times 10^5 \text{ NC}^{-1}$. If the charge on the particle is $40 \mu\text{C}$ and the initial velocity is 200 ms^{-1} , how much distance it will travel before coming to the rest momentarily: [JEE MAINS June 29, 2022 (I)]
- (1) 1 m (2) 5 m
(3) 10 m (4) 0.5 m
78. A vertical electric field of magnitude $4.9 \times 10^5 \text{ N/C}$ just prevents a water droplet of a mass 0.1 g from falling. The value of charge on the droplet will be: (Given $g = 9.8 \text{ m/s}^2$). [JEE MAINS June 24, 2022 (I)]
- (1) $1.6 \times 10^{-9} \text{ C}$ (2) $2.0 \times 10^{-9} \text{ C}$
(3) $3.2 \times 10^{-9} \text{ C}$ (4) $0.5 \times 10^{-9} \text{ C}$

79. A body having specific charge $8 \mu\text{C/g}$ is resting on a frictionless plane at a distance 10 cm from the wall (as shown in the figure). It starts moving towards the wall when a uniform electric field of 100 V/m is applied horizontally towards the wall. If the collision of the body with the wall is perfectly elastic, then the time period of the motion will be _____ s.

[JEE MAINS July 20, 2021 (I)]



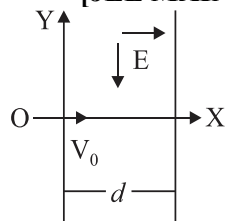
80. A particle of charge q and mass m is subjected to an electric field $E = E_0(1 - ax^2)$ in the x -direction, where a and E_0 are constants. Initially the particle was at rest at $x = 0$. Other than the initial position the kinetic energy of the particle becomes zero when the distance of the particle from the origin is:

[JEE MAINS Sep. 04, 2020 (II)]

- (1) a (2) $\sqrt{\frac{2}{a}}$
(3) $\sqrt{\frac{3}{a}}$ (4) $\sqrt{\frac{1}{a}}$

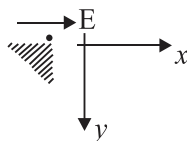
81. A charged particle (mass m and charge q) moves along X axis with velocity V_0 . When it passes through the origin it enters a region having uniform electric field $\vec{E} = -E\hat{j}$ which extends upto $x = d$. Equation of path of electron in the region $x > d$ is:

[JEE MAINS Sep. 02, 2020 (I)]



- (1) $y = \frac{qEd}{mV_0^2}(x-d)$ (2) $y = \frac{qEd}{mV_0^2}\left(\frac{d}{2} - x\right)$
(3) $y = \frac{qEd}{mV_0^2}x$ (4) $y = \frac{qEd}{mV_0^2}x$

82. A small point mass carrying some positive charge on it, is released from the edge of a table. There is a uniform electric field in this region in the horizontal direction.



Which of the following options then correctly describe the trajectory of the mass? (Curves are drawn schematically and are not to scale).

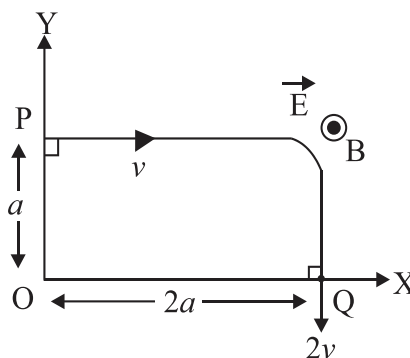
[JEE MAINS Sep. 02, 2020 (II)]

- (1) (2)
(3) (4)

83. A charged particle of mass ' m ' and charge ' q ' moving under the influence of uniform electric field $E\hat{i}$ and a uniform magnetic field $B\hat{k}$ follows a trajectory from point P to Q as shown in figure.

The velocities at P and Q are respectively, $v\hat{i}$ and $-2v\hat{j}$. Then which of the following statements (A, B, C, D) are the correct? (Trajectory shown is schematic and not to scale).

[JEE MAINS Jan. 9, 2020 (I)]



(A) $E = \frac{3}{4}\left(\frac{mv^2}{qa}\right)$

- (B) Rate of work done by the electric field at P is

$\frac{3}{4}\left(\frac{mv^3}{a}\right)$

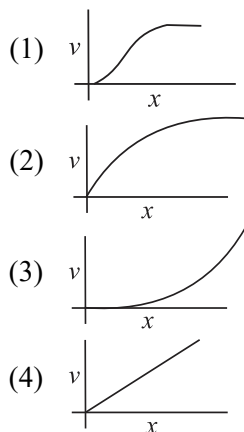
- (C) Rate of work done by both the fields at Q is zero

- (D) The difference between the magnitude of angular momentum of the particle at P and Q is $2mav$.

- (1) (A), (C), (D)
(2) (B), (C), (D)
(3) (A), (B), (C)
(4) (A), (B), (C), (D)

84. A particle of mass m and charge q is released from rest in a uniform electric field. If there is no other force on the particle, the dependence of its speed v on the distance x travelled by it is correctly given by (graphs are schematic and not drawn to scale).

[JEE MAINS Jan. 8, 2020 (II)]



85. A particle of mass m and charge q has an initial velocity $\vec{v} = v_0 \hat{j}$. If an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = B_0 \hat{i}$ act on the particle, its speed will double after a time:

[JEE MAINS Jan. 07, 2020 (II)]

- (1) $\frac{2mv_0}{qE_0}$ (2) $\frac{3mv_0}{qE_0}$
 (3) $\frac{\sqrt{3}mv_0}{qE_0}$ (4) $\frac{\sqrt{2}mv_0}{qE_0}$

86. A body of mass M and charge q is connected to a spring of spring constant k . It is oscillating along x -direction about its equilibrium position, taken to be at $x = 0$, with an amplitude A . An electric field E is applied along the x -direction. Which of the following statements is correct? [JEE MAINS 2018]

- (1) The total energy of the system is $\frac{1}{2}m\omega^2 A^2 + \frac{1}{2} \frac{q^2 E^2}{k}$
 (2) The new equilibrium position is at a distance: $\frac{2qE}{k}$ from $x = 0$
 (3) The new equilibrium position is at a distance: $\frac{qE}{2k}$ From $x = 0$
 (4) The total energy of the system is $\frac{1}{2}m\omega^2 A^2 - \frac{1}{2} \frac{q^2 E^2}{k}$

87. A liquid drop having 6 excess electrons is kept stationary under a uniform electric field of 25.5 k Vm^{-1} . The density of liquid is $1.26 \times 10^3 \text{ kg m}^{-3}$. The radius of the drop is (neglect buoyancy). [JEE MAINS 2013]

- (1) $4.3 \times 10^{-7} \text{ m}$ (2) $7.8 \times 10^{-7} \text{ m}$
 (3) $0.078 \times 10^{-7} \text{ m}$ (4) $3.4 \times 10^{-7} \text{ m}$

88. This question contains Statement-1 and Statement-2. of the four choices given after the statements, choose the one that best describes the two statements.

Statement-1: For a charged particle moving from point P to point Q, the net work done by an electrostatic field on the particle is independent of the path connecting point P to point Q.

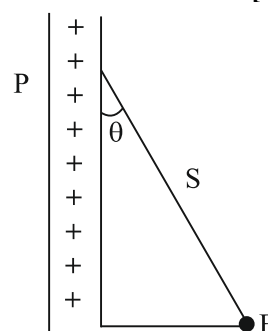
Statement-2: The net work done by a conservative force on an object moving along a closed loop is zero.

[JEE MAINS 2009]

- (1) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation of Statement-1.
 (2) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation of Statement-1.
 (3) Statement-1 is false, Statement-2 is true.
 (4) Statement-1 is true, Statement-2 is false.

89. A charged ball B hangs from a silk thread S, which makes an angle θ with a large charged conducting sheet P, as shown in the figure. The surface charge density σ of the sheet is proportional to:

[JEE MAINS 2005]



- (1) $\cot \theta$ (2) $\cos \theta$
 (3) $\tan \theta$ (4) $\sin \theta$

90. An electron is moving under the influence of the electric field of a uniformly charged infinite plane sheet having surface charge density $+\sigma$. The electron at $t = 0$ is at a distance of 1 m from S and has a speed of 1 m/s. The maximum value of σ if the electron strikes S at $t = 1$ s is $\alpha \left[\frac{m\epsilon_0}{e} \right] \frac{C}{m^2}$, the value of α is _____. [JEE MAINS Jan. 29, 2024 (I)]

91. A dipole comprises of two charged particles of identical magnitude q and opposite in nature. The mass ' m ' of the positive charged particle is half of the mass of the negative charged particle. The two charges are separated by a distance ' l '. If the dipole is placed in a uniform electric field ' $\vec{E}$ '; in such a way that dipole axis makes a very small angle with the electric field, ' $\vec{E}$ '. The angular frequency of the oscillations of the dipole when released is given by:

[JEE MAINS April 06, 2023 (II)]

- (1) $\sqrt{\frac{8qE}{3ml}}$ (2) $\sqrt{\frac{4qE}{ml}}$
(3) $\sqrt{\frac{3qE}{2ml}}$ (4) $\sqrt{\frac{8qE}{ml}}$

92. Given below are two statements:

Statement-I: A point charge is brought in an electric field. The value of electric field at a point near to the charge may increase if the charge is positive.

Statement-II: An electric dipole is placed in a non-uniform electric field. The net electric force on the dipole will not be zero.

Choose the correct answer from the options given below: [JEE MAINS June 28, 2022 (I)]

- (1) Both statement-I and statement-II are true.
(2) Both statement-I and statement-II are false.
(3) Statement-I is true but statement-II is false.
(4) Statement-I is false but statement-II is true.

93. An electric dipole is formed by two equal and opposite charges q with separation d . The charges have same mass m . It is kept in a uniform electric field E . If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is:

[JEE MAINS April 08, 2019 (II)]

- (1) $\sqrt{\frac{qE}{md}}$ (2) $\sqrt{\frac{2qE}{md}}$
(3) $\sqrt{\frac{qE}{md}}$ (4) $\sqrt{\frac{qE}{2md}}$

ELECTRIC FLUX, GAUSS'S LAW AND ITS APPLICATION

94. Identify the correct statements:

- A. Electrostatic field lines form closed loops.
B. The electric field lines point radially outward when charge is greater than zero.
C. The Gauss - Law is valid only for inverse-square force.
D. The work done in moving a charged particle in a static electric field around a closed path is zero.
E. The motion of a particle under Coulomb's force must take place in a plane.

Choose the correct answer from the options given below: [JEE MAINS Jan 28, 2026 (II)]

- (1) A, C, E Only
(2) A, B, D, E Only
(3) B, C, D, E Only
(4) A, B, C, D Only

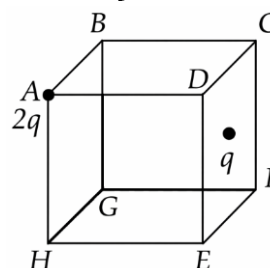
95. The electrostatic potential in a charged spherical region of radius r varies as $V = ar^3 + b$, where a and b are constants. The total charge in the sphere of unit radius is $\alpha \times \pi\epsilon_0$. The value of α is _____. (permittivity of vacuum is ϵ_0).

[JEE MAINS Jan 24, 2026 (I)]

- (1) -12 (2) -8
(3) -9 (4) -6

96. Two point charges $2q$ and q are placed at vertex A and centre of face CDEF of the cube as shown in figure. The electric flux passing through the cube is:

[JEE MAINS Jan 23, 2026 (I)]



- (1) $\frac{3q}{2\epsilon_0}$ (2) $\frac{3q}{4\epsilon_0}$
(3) $\frac{q}{\epsilon_0}$ (4) $\frac{3q}{\epsilon_0}$

97. An infinitely long wire has uniform linear charge density $\lambda = 2 \text{ nC/m}$. The net flux through a Gaussian cube of side length $\sqrt{3} \text{ cm}$, if the wire passes through any two corners of the cube, that are maximally displaced from each other, would be $x\text{Nm}^2\text{C}^{-1}$, where x is: [Neglect any edge effects and use $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ units}$]

[JEE MAINS April 08, 2025 (II)]

- (1) 0.72π (2) 1.44π
(3) 6.48π (4) 2.16π

98. The electric field in a region is given by $\vec{E} = (2\hat{i} + 4\hat{j} + 6\hat{k}) \times 10^3 \text{ N/C}$. The flux of the field through a rectangular surface parallel to x - z plane is $6.0 \text{ Nm}^2\text{C}^{-1}$. The area of the surface is _____ cm^2 .

[JEE MAINS April 7, 2025 (II)]

99. Match List-I with List-II.

List-I		List-II	
(A)	Electric field inside (distance $r > 0$ from center) of a uniformly charged spherical shell with surface charge density σ , and radius R .	(I)	$\frac{\sigma}{\epsilon_0}$
(B)	Electric field at distance $r > 0$ from a uniformly charged infinite plane sheet with surface charge density σ .	(II)	$\frac{\sigma}{2\epsilon_0}$
(C)	Electric field outside (distance $r > 0$ from center) of a uniformly charged spherical shell with surface charge density σ , and radius R .	(III)	0
(D)	Electric field between 2 oppositely charged infinite plane parallel sheets with uniform surface charge density σ .	(IV)	$\frac{\sigma}{\epsilon_0 r^2}$

Choose the correct answer from the options given below:

[JEE MAINS Jan. 29, 2025 (I)]

- (1) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)
(2) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
(3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(4) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

100. A point charge causes an electric flux of $-2 \times 10^4 \text{ Nm}^2\text{C}^{-1}$ to pass through a spherical Gaussian surface of 8.0 cm radius, centred on the charge. The value of the point charge is:

(Given, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$).

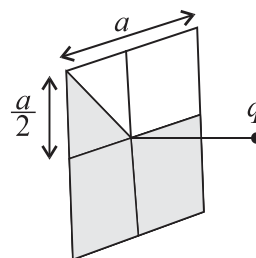
[JEE MAINS Jan. 29, 2025 (II)]

- (1) $-17.7 \times 10^{-8} \text{ C}$ (2) $-15.7 \times 10^{-8} \text{ C}$
(3) $17.7 \times 10^{-8} \text{ C}$ (4) $15.7 \times 10^{-8} \text{ C}$

101. A square loop of sides $a = 1 \text{ m}$ is held normally in front of a point charge $q = 1 \text{ C}$. The flux of the electric field through the shaded region is $\frac{5}{P} \times \frac{1}{\epsilon_0} \frac{\text{Nm}^2}{\text{C}}$,

where the value of P is _____.

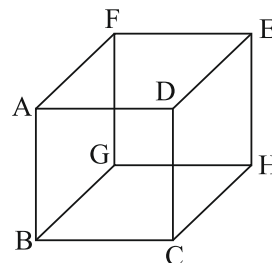
[JEE MAINS Jan. 24, 2025 (I)]



102. A line charge of length $\frac{a'}{2}$ is kept at the center of an edge BC of a cube ABCDEFGH having edge length ' a ' as shown in the figure. If the density of line is $\lambda \text{ C}$ per unit length, then the total electric flux through all the faces of the cube will be _____.

(Take ϵ_0 as the free space permittivity)

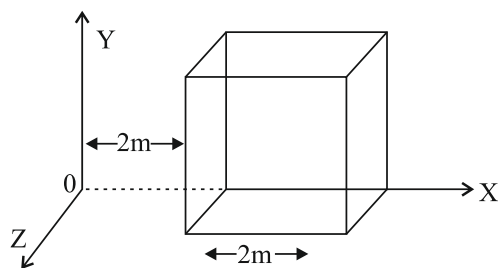
[JEE MAINS Jan. 22, 2025 (I)]



- (1) $\frac{\lambda a}{8\epsilon_0}$ (2) $\frac{\lambda a}{16\epsilon_0}$
(3) $\frac{\lambda a}{2\epsilon_0}$ (4) $\frac{\lambda a}{4\epsilon_0}$

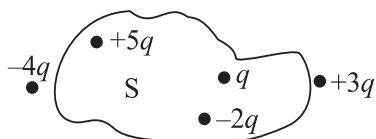
103. An electric field $\vec{E} = (2x\hat{i}) \text{ NC}^{-1}$ exists in space. A cube of side 2m is placed in the space as per figure given below. The electric flux through the cube is _____ Nm^2/C .

[JEE MAINS April 9, 2024 (II)]



104. Five charges $+q$, $+5q$, $-2q$, $+3q$ and $-4q$ are situated as shown in the figure. The electric flux due to this configuration through the surface S is:

[JEE MAINS April 09, 2024 (II)]



- (1) $\frac{5q}{\epsilon_0}$ (2) $\frac{4q}{\epsilon_0}$
 (3) $\frac{3q}{\epsilon_0}$ (4) $\frac{q}{\epsilon_0}$
105. An electric field, $\vec{E} = \frac{2\hat{i} + 6\hat{j} + 8\hat{k}}{\sqrt{6}}$ passes through the surface of 4m^2 area having unit vector $\hat{n} = \left(\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}\right)$. The electric flux for that surface is _____ V m . [JEE MAINS April 08, 2024 (I)]

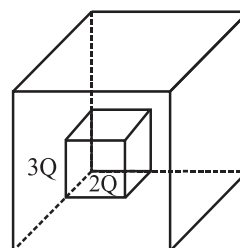
106. A charge q is placed at the center of one of the surface of a cube. The flux linked with the cube is:

[JEE MAINS April 04, 2024 (II)]

- (1) $\frac{q}{4\epsilon_0}$ (2) $\frac{q}{2\epsilon_0}$
 (3) $\frac{q}{8\epsilon_0}$ (4) Zero

107. C_1 and C_2 are two hollow concentric cubes enclosing charges $2Q$ and $3Q$ respectively as shown in figure. The ratio of electric flux passing through C_1 and C_2 is:

[JEE MAINS Feb. 01, 2024 (II)]



- (1) 3 : 2 (2) 5 : 2
 (3) 2 : 5 (4) 2 : 3
108. Two charges of $5Q$ and $-2Q$ are situated at the points $(3a, 0)$ and $(-5a, 0)$ respectively. The electric flux through a sphere of radius ' $4a$ ' having center at origin is:

[JEE MAINS Jan. 29, 2024 (I)]

- (1) $\frac{3Q}{\epsilon_0}$ (2) $\frac{5Q}{\epsilon_0}$
 (3) $\frac{7Q}{\epsilon_0}$ (4) $\frac{2Q}{\epsilon_0}$

109. An electric field is given by $(6\hat{i} + 5\hat{j} + 3\hat{k}) \text{ N/C}$. The electric flux through a surface area $30\hat{i} \text{ m}^2$ lying in YZ -plane (in SI unit) is:

[JEE MAINS Jan. 29, 2024 (II)]

- (1) 180 (2) 90
 (3) 150 (4) 60

110. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: If an electric dipole of dipole moment $30 \times 10^{-5} \text{ cm}$ is enclosed by a closed surface, the net flux coming out of the surface will be zero.

Reason R: Electric dipole consists of two equal and opposite charges [JEE MAINS April 12, 2023 (I)]

In the light of above, statements, choose the correct answer from the options given below:

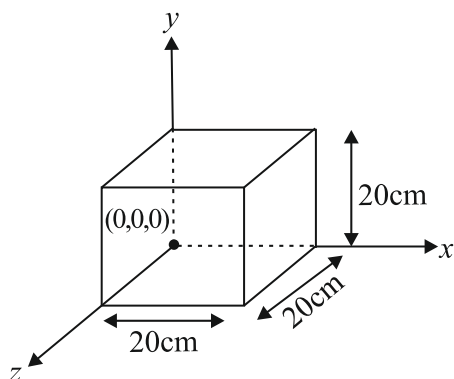
- (1) Both A and R are true and R is the correct explanation of A
 (2) A is true but R is false
 (3) Both A and R are true but R is NOT the correct explanation of A
 (4) A is false but R is true

111. A cubical volume is bounded by the surfaces $x = 0$, $x = a$, $y = 0$, $y = a$, $z = 0$, $z = a$. The electric field in the region is given by $\vec{E} = E_0 x \hat{i}$. Where $E_0 = 4 \times 10^4 \text{ NC}^{-1}\text{m}^{-1}$. If $a = 2 \text{ cm}$, the charge contained in the cubical volume is $Q \times 10^{-14} \text{ C}$. The value of Q is _____. (Take $\epsilon_0 = 9 \times 10^{-12} \text{ C}^2/\text{Nm}^2$).

[JEE MAINS Feb. 01, 2023 (II)]

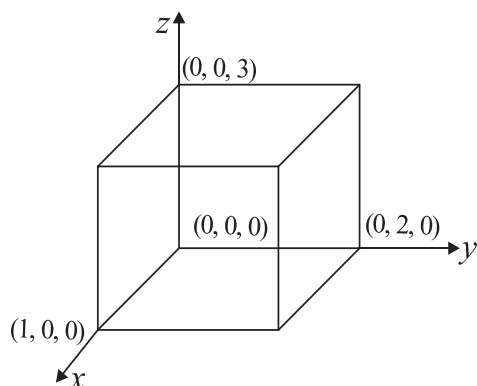
112. Expression for an electric field is given by $\vec{E} = 4000x^2 \hat{i} \frac{\text{V}}{\text{m}}$. The electric flux through the cube of side 20 cm when placed in electric field (as shown in the figure) is _____ V cm.

[JEE MAINS Jan. 31, 2023 (I)]



113. As shown in figure, a cuboid lies in a region with electric field $\vec{E} = 2x^2 \hat{i} - 4y \hat{j} + 6z \hat{k}$. The magnitude of charge within the cuboid is $n\epsilon_0 \text{ CN/C}$. The value of n is _____ (if dimension of cuboid is $1 \times 2 \times 3 \text{ m}^3$)

[JEE MAINS Jan. 30, 2023 (II)]



114. In a cuboid of dimension $2L \times 2L \times L$, a charge q is placed at the centre of the surface 'S' having area of $4L^2$. The flux through the opposite surface to 'S' is given by: [JEE MAINS Jan. 29, 2023 (I)]

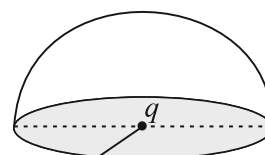
- (1) $\frac{q}{12\epsilon_0}$ (2) $\frac{q}{3\epsilon_0}$
(3) $\frac{q}{2\epsilon_0}$ (4) $\frac{q}{6\epsilon_0}$

115. The volume charge density of a sphere of radius 6 m is $2 \mu\text{C cm}^{-3}$. The number of lines of force per unit surface area coming out from the surface of the sphere is _____ $\times 10^{10} \text{ NC}^{-1}$. (Given: Permittivity of vacuum $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$).

[JEE MAINS July 25, 2022 (I)]

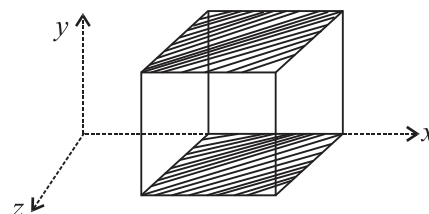
116. If a charge q is placed at the centre of a closed hemispherical non-conducting surface, the total flux passing through the flat surface would be:

[JEE MAINS June 27, 2022 (II)]



- (1) $\frac{q}{\epsilon_0}$ (2) $\frac{q}{2\epsilon_0}$
(3) $\frac{q}{4\epsilon_0}$ (4) $\frac{q}{2\pi\epsilon_0}$

117. A cube is placed inside an electric field, $\vec{E} = 150y^2 \hat{j}$. The side of the cube is 0.5 m and is placed in the field as shown in the given figure. The charge inside the cube is: [JEE MAINS Sep. 01, 2021 (I)]



- (1) $3.8 \times 10^{-11} \text{ C}$ (2) $8.3 \times 10^{-11} \text{ C}$
(3) $3.8 \times 10^{-12} \text{ C}$ (4) $8.3 \times 10^{-12} \text{ C}$

118. Choose the incorrect statement:

- (A) The electric lines of force entering into a Gaussian surface provide negative flux.
- (B) A charge 'q' is placed at the centre of a cube. The flux through all the faces will be the same.
- (C) In a uniform electric field net flux through a closed Gaussian surface containing no net charge, is zero.
- (D) When electric field is parallel to a Gaussian surface, it provides a finite non-zero flux.

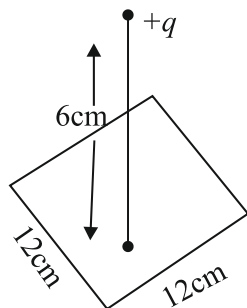
Choose the most appropriate answer from the options given below. [JEE MAINS Aug. 31, 2021 (II)]

- (1) (C) and (D) only
- (2) (B) and (D) only
- (3) (D) only
- (4) (A) and (C) only

119. The electric field in a region is given $\vec{E} = \frac{2}{5} E_0 \hat{i} + \frac{3}{5} E_0 \hat{j}$ with $E_0 = 4.0 \times 10^3 \frac{\text{N}}{\text{C}}$. The flux of this field through a rectangular surface area 0.4 m^2 parallel to the Y-Z plane is _____ $\text{Nm}^2 \text{C}^{-1}$.

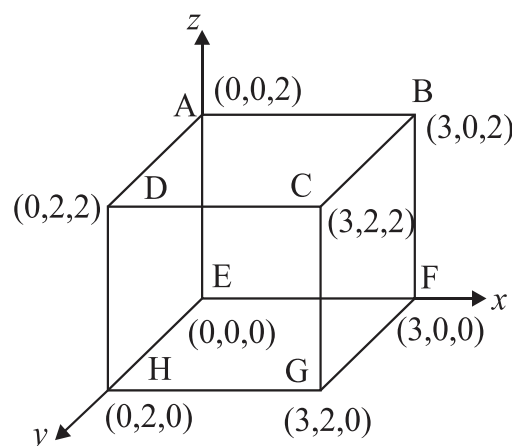
[JEE MAINS March 17, 2021 (I)]

120. A point charge of $+12 \mu\text{C}$ is at a distance 6 cm vertically above the centre of side of a square of side 12 cm as shown in figure. The magnitude of the electric flux through the square will be _____ $\times 10^3 \text{ Nm}^2/\text{C}$. [JEE MAINS Feb. 24, 2021 (II)]



121. The electric field in a region is given by $\vec{E} = \left(\frac{3}{5} E_0 \hat{i} + \frac{4}{5} E_0 \hat{j} \right) \frac{\text{N}}{\text{C}}$. The ratio of flux of reported field through the rectangular surface of area 0.2 m^2 (parallel to y-z plane) to that of the surface of area 0.3 m^2 (parallel to x-z plane) is $a : b$, where $a =$ _____. [JEE MAINS Feb. 25, 2021 (I)]

122. An electric field $\vec{E} = 4x\hat{i} - (y^2 + 1)\hat{j} \text{ N/C}$ passes through the box shown in figure. The flux of the electric field through surfaces ABCD and BCGF are marked as ϕ_1 , and ϕ_{11} respectively. The difference between $(\phi_1 - \phi_{11})$ is (in Nm^2/C) _____. [JEE MAINS Jan. 09, 2020 (II)]

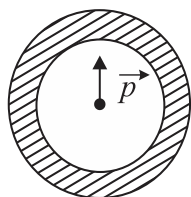


123. In finding the electric field using Gauss law the formula $|\vec{E}| = \frac{q_{enc}}{\epsilon_0 |A|}$ is applicable. In the formula ϵ_0 is permittivity of free space, A is the area of Gaussian surface and q_{enc} is charge enclosed by the Gaussian surface. This equation can be used in which of the following situation?

[JEE MAINS Jan. 08, 2020 (I)]

- (1) Only when the Gaussian surface is an equipotential surface.
- (2) Only when the Gaussian surface is an equipotential surface and $|\vec{E}|$ is constant on the surface.
- (3) Only when $|\vec{E}| = \text{constant}$ on the surface.
- (4) For any choice of Gaussian surface.

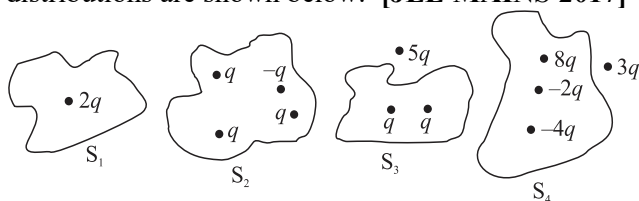
124. Shown in the figure is a shell made of a conductor. It has inner radius a and outer radius b , and carries charge Q . At its centre is a dipole $\vec{p}$ as shown. In this case: [JEE MAINS April 12, 2019 (I)]



- (1) surface charge density on the inner surface is uniform and equal to $\frac{Q/2}{4\pi a^2}$
 - (2) electric field outside the shell is the same as that of a point charge at the centre of the shell.
 - (3) surface charge density on the outer surface depends on $|\vec{p}|$
 - (4) surface charge density on the inner surface of the shell is zero everywhere.
125. Let a total charge $2Q$ be distributed in a sphere of radius R , with the charge density given by $\rho(r) = kr$, where r is the distance from the centre. Two charges A and B , of $-Q$ each, are placed on diametrically opposite points, at equal distance, a , from the centre. If A and B do not experience any force, then.

[JEE MAINS April 12, 2019 (II)]

- (1) $a = 8^{-1/4} R$
 - (2) $a = \frac{3R}{2^{1/4}}$
 - (3) $a = 2^{-1/4} R$
 - (4) $a = R/\sqrt{3}$
126. Four closed surfaces and corresponding charge distributions are shown below. [JEE MAINS 2017]



Let the respective electric fluxes through the surfaces be ϕ_1, ϕ_2, ϕ_3 , and ϕ_4 . Then:

- (1) $\phi_1 < \phi_2 = \phi_3 > \phi_4$
- (2) $\phi_1 > \phi_2 > \phi_3 > \phi_4$
- (3) $\phi_1 = \phi_2 = \phi_3 = \phi_4$
- (4) $\phi_1 > \phi_3 ; \phi_2 > \phi_4$

127. An electric dipole is placed at an angle of 30° to a non-uniform electric field. The dipole will experience [JEE MAINS 2006]

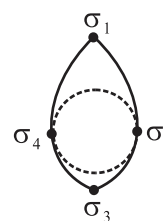
- (1) a translational force only in the direction of the field
- (2) a translational force only in a direction normal to the direction of the field
- (3) a torque as well as a translational force
- (4) a torque only

128. If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 , and ϕ_2 , the electric charge inside the surface will be: [JEE MAINS 2003]

- (1) $(\phi_2 - \phi_1)\epsilon_0$
- (2) $(\phi_1 - \phi_2)\epsilon_0$
- (3) $(\phi_2 - \phi_1)\epsilon_0$
- (4) $(\phi_1 - \phi_2)\epsilon_0$

ELECTRIC FIELD OF A CONTINUOUS CHARGE DISTRIBUTION

129. Electric charge is transferred to an irregular metallic disk as shown in figure. If $\sigma_1, \sigma_2, \sigma_3$ and σ_4 are charge densities at given points then, choose the correct answer from the options given below:



- (A) $\sigma_1 > \sigma_3; \sigma_2 = \sigma_4$
- (B) $\sigma_1 > \sigma_2; \sigma_3 > \sigma_4$
- (C) $\sigma_1 > \sigma_3 > \sigma_2 = \sigma_4$
- (D) $\sigma_1 < \sigma_3 < \sigma_2 = \sigma_4$
- (E) $\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4$

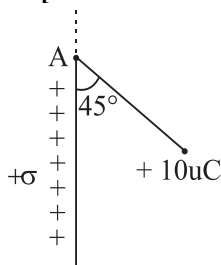
[JEE MAINS April 8, 2025 (II)]

- (1) A, B and C Only
- (2) A and C Only
- (3) D and E Only
- (4) B and C Only

130. A small bob of mass 100 mg and charge $+10 \mu\text{C}$ is connected to an insulating string of length 1 m. It is brought near to an infinitely long non-conducting sheet of charge density ' σ ' as shown in figure. If string subtends an angle of 45° with the sheet at equilibrium the charge density of sheet will be:

(Given $\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{F}}{\text{m}}$ add acceleration due to gravity, $g = 10 \text{ m/s}^2$).

[JEE MAINS April 2, 2025 (I)]



- (1) 0.885 nC/m^2 (2) 17.7 nC/m^2
(3) 885 nC/m^2 (4) 1.77 nC/m^2
131. Suppose a uniformly charged wall provides a uniform electric field of $2 \times 10^4 \text{ N/C}$ normally. A charged particle of mass 2g being suspended through a silk thread of length 20 cm and remain stayed at a distance of 10 cm from the wall. Then the charge on the particle will be $\frac{1}{\sqrt{x}} \mu\text{C}$ where $x =$ _____.
[use $g = 10 \text{ m/s}^2$].

[JEE MAINS Feb. 1, 2024 (II)]

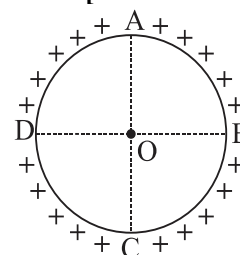
132. Charge is distributed within a sphere of radius R with a volume charge density $\rho(r) = \frac{A}{r^2} e^{-2r/a}$ where A and a are constants. If Q is the total charge of this charge distribution, the radius R is:

[JEE MAINS Jan. 9, 2019 (II)]

- (1) $a \log \left(1 - \frac{Q}{2\pi a A} \right)$ (2) $\frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$
(3) $a \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$ (4) $\frac{a}{2} \log \left(1 - \frac{Q}{2\pi a A} \right)$

133. A metallic ring is uniformly charged as shown in figure. AC and BD are two mutually perpendicular diameters. Electric field due to arc AB to 'O' is 'E' is magnitude. What would be the magnitude of electric field at 'O' due to arc ABC?

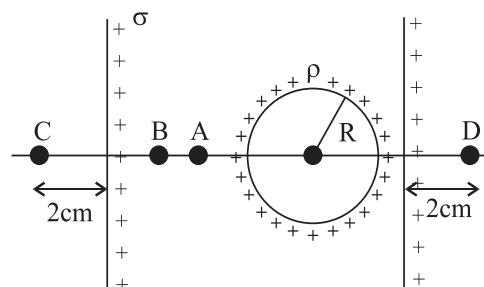
[JEE MAINS April 4, 2025 (II)]



- (1) $2E$ (2) $\sqrt{2}E$
(3) $\frac{E}{2}$ (4) zero

134. Two infinite identical charged sheets and a charged spherical body of charge density ' ρ ' are arranged as shown in figure. Then the correct relation between the electrical fields at A, B, C and D points is:

[JEE MAINS April 4, 2025]



- (1) $\vec{E}_A = \vec{E}_B$; $\vec{E}_C = \vec{E}_D$
(2) $\vec{E}_A > \vec{E}_B$; $\vec{E}_C = \vec{E}_D$
(3) $\vec{E}_C \neq \vec{E}_D$; $\vec{E}_A > \vec{E}_B$
(4) $|\vec{E}_A| = |\vec{E}_B|$; $\vec{E}_C > \vec{E}_D$

135. Two small spherical balls of mass 10g each with charges $-2\mu\text{C}$ and $2\mu\text{C}$, are attached to two ends of very light rigid rod of length 20 cm. The arrangement is now placed near an infinite non-conducting charge sheet with uniform charge density of $100 \mu\text{C/m}^2$ such that length of rod makes an angle of 30° with electric field generated by charge sheet. Net torque acting on the rod is:

[JEE MAINS April 4, 2025 (I)]

(Take $\epsilon_0 : 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$)

- (1) 112 Nm (2) 1.12 Nm
(3) 2.24 Nm (4) 11.2 Nm

136. Consider a circular loop that is uniformly charged and has a radius $a\sqrt{2}$. Find the position along the positive z -axis of the cartesian coordinate system where the electric field is maximum if the ring was assumed to be placed in xy -plane at the origin:

[JEE MAINS April 2, 2025 (II)]

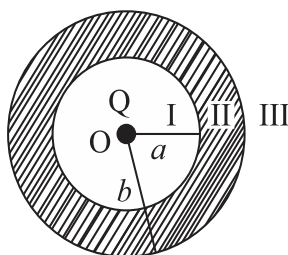
- (1) $\frac{a}{\sqrt{2}}$ (2) $\frac{a}{2}$
(3) a (4) 0

137. A thin infinite sheet charge and an infinite line charge of respective charge densities $+\sigma$ and $+\lambda$ are placed parallel at 5 m distance from each other. Points 'P' and 'Q' are at $\frac{3}{\pi}$ m and $\frac{4}{\pi}$ m perpendicular distance from line charge towards sheet charge, respectively. ' E_P ' and ' E_Q ' are the magnitudes of resultant electric field intensities at point 'P' and 'Q', respectively. If $\frac{E_P}{E_Q} = \frac{4}{a}$ for $2|\sigma| = |\lambda|$. Then the value of a is _____.

[JEE MAINS April 13, 2023 (I)]

138. As shown in the figure, a point charge Q is placed at the centre of conducting spherical shell of inner radius a and outer radius b . The electric field due to charge Q in three different regions I, II and III is given by: (I : $r < a$, II : $a < r < b$, III : $r > b$).

[JEE MAINS Jan. 30, 2023 (II)]

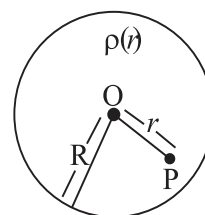


- (1) $E_I = 0, E_{II} = 0, E_{III} \neq 0$
(2) $E_I \neq 0, E_{II} = 0, E_{III} \neq 0$
(3) $E_I \neq 0, E_{II} = 0, E_{III} = 0$
(4) $E_I = 0, E_{II} = 0, E_{III} = 0$

139. A spherically symmetric charge distribution is considered with charge density varying as:

[JEE MAINS July 29, 2022 (I)]

$$\rho(r) = \begin{cases} \rho_0 \left(\frac{3}{4} - \frac{r}{R} \right) & \text{for } r \leq R \\ \text{zero} & \text{for } r > R \end{cases}$$



Where, $r (r < R)$ is the distance from the centre O (as shown in figure). The electric field at point P will be:

- (1) $\frac{\rho_0 r}{4\epsilon_0} \left(\frac{3}{4} - \frac{r}{R} \right)$ (2) $\frac{\rho_0 r}{3\epsilon_0} \left(\frac{3}{4} - \frac{r}{R} \right)$
(3) $\frac{\rho_0 r}{4\epsilon_0} \left(1 - \frac{r}{R} \right)$ (4) $\frac{\rho_0 r}{5\epsilon_0} \left(1 - \frac{r}{R} \right)$

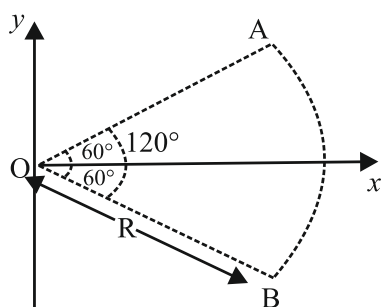
140. A uniformly charged disc of radius R having surface charge density σ is placed in the xy plane with its center at the origin. Find the electric field intensity along the z -axis at a distance Z from origin:

[JEE MAINS Aug. 27, 2021 (I)]

- (1) $E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$
(2) $E = \frac{\sigma}{2\epsilon_0} \left(1 + \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$
(3) $E = \frac{2\epsilon_0}{\sigma} \left(\frac{Z}{(Z^2 + R^2)^{1/2}} + Z \right)$
(4) $E = \frac{\sigma}{2\epsilon_0} \left(\frac{Z}{(Z^2 + R^2)} + \frac{1}{Z^2} \right)$

141. Figure shows a rod AB, which is bent in a 120° circular arc of radius R . A charge $(-Q)$ is uniformly distributed over rod AB. What is the electric field $\vec{E}$ at the centre of curvature O ?

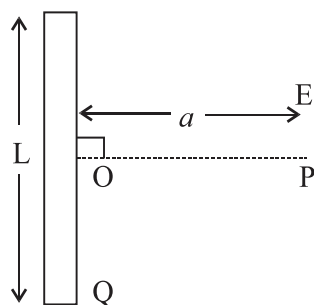
[JEE MAINS Aug. 27, 2021 (II)]



- (1) $\frac{3\sqrt{3}Q}{8\pi\epsilon_0 R^2}(\hat{i})$ (2) $\frac{3\sqrt{3}Q}{8\pi^2\epsilon_0 R^2}(\hat{i})$
 (3) $\frac{3\sqrt{3}Q}{16\pi^2\epsilon_0 R^2}(\hat{i})$ (4) $\frac{3\sqrt{3}Q}{8\pi^2\epsilon_0 R^2}(-\hat{i})$

142. Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q . The distance of the point P from the centre of the rod is a $\frac{\sqrt{3}}{2}L$.

[JEE MAINS Feb. 26, 2021 (I)]



- (1) $\frac{Q}{3\pi\epsilon_0 L^2}$ (2) $\frac{Q}{4\pi\epsilon_0 L^2}$
 (3) $\frac{\sqrt{3}Q}{4\pi\epsilon_0 L^2}$ (4) $\frac{Q}{2\sqrt{3}\pi\epsilon_0 L^2}$

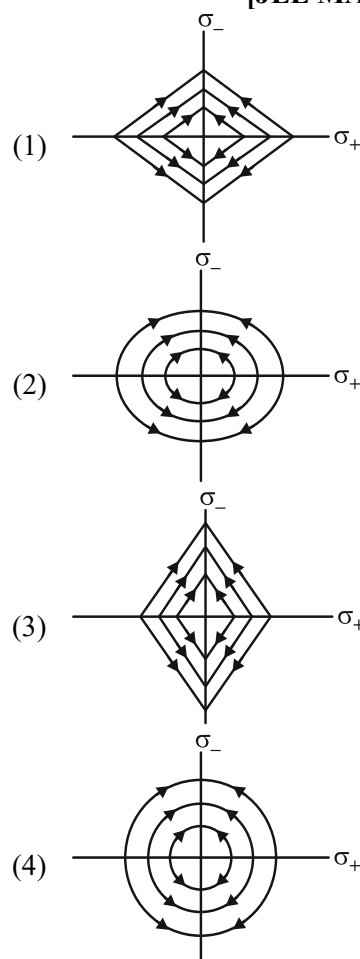
143. Consider the force F on a charge ' q ' due to a uniformly charged spherical shell of radius R carrying charge Q distributed uniformly over it. Which one of the following statements is true for F , if ' q ' is placed at distance from the centre of the shell?

[JEE MAINS Sep. 06, 2020 (II)]

- (1) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for $r < R$
 (2) $\frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2} > F > 0$ for $r < R$
 (3) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$ for $r > R$
 (4) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for all r

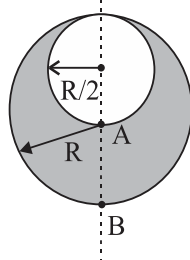
144. Two charged thin infinite plane sheets of uniform surface charge density σ_+ , and σ_- , where $|\sigma_+| > |\sigma_-|$, intersect at right angle. Which of the following best represents the electric field lines for this system?

[JEE MAINS Sep. 04, 2020 (I)]



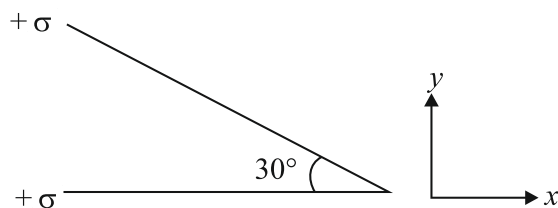
145. Consider a sphere of radius R which carries a uniform charge density ρ . If a sphere of radius $\frac{R}{2}$ is carved out of it, as shown, the ratio $\frac{|\vec{E}_A|}{|\vec{E}_B|}$ of magnitude of electric field $\vec{E}_A$ and $\vec{E}_B$, respectively, at points A and B due to the remaining portion is:

[JEE MAINS Jan. 9, 2020 (I)]



146. Two infinite planes each with uniform surface charge density $+\sigma$ are kept in such a way that the angle between them is 30° . The electric field in the region shown between them is given by:

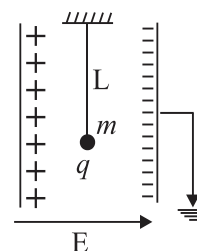
[JEE MAINS Jan. 07, 2020 (I)]



- (1) $\frac{\sigma}{2\epsilon_0} \left[(1+\sqrt{3})\hat{y} - \frac{\hat{x}}{2} \right]$
 (2) $\frac{\sigma}{\epsilon_0} \left[\left(1 + \frac{\sqrt{3}}{2} \right) \hat{y} + \frac{\hat{x}}{2} \right]$
 (3) $\frac{\sigma}{2\epsilon_0} \left[(1+\sqrt{3})\hat{y} + \frac{\hat{x}}{2} \right]$
 (4) $\frac{\sigma}{2\epsilon_0} \left[\left(1 - \frac{\sqrt{3}}{2} \right) \hat{y} - \frac{\hat{x}}{2} \right]$

147. A simple pendulum of length L is placed between the plates of a parallel plate capacitor having electric field E , as shown in figure. Its bob has mass m and charge q . The time period of the pendulum is given by:

[JEE MAINS April 10, 2019 (II)]



- (1) $2\pi \sqrt{\frac{L}{\left(g + \frac{qE}{m}\right)}}$ (2) $2\pi \sqrt{\frac{L}{\left(g^2 - \frac{q^2 E^2}{m^2}\right)}}$
 (3) $2\pi \sqrt{\frac{L}{\left(g - \frac{qE}{m}\right)}}$ (4) $2\pi \sqrt{\frac{L}{\left(g^2 + \left(\frac{qE}{m}\right)^2\right)}}$

148. The bob of a simple pendulum has mass $2g$ and a charge of $5.0 \mu\text{C}$. It is at rest in a uniform horizontal electric field of intensity 2000 V/m . At equilibrium, the angle that the pendulum makes with the vertical is:

[JEE MAINS April 08, 2019 (I)]

(take $g = 10 \text{ m/s}^2$)

- (1) $\tan^{-1}(2.0)$ (2) $\tan^{-1}(0.2)$
 (3) $\tan^{-1}(5.0)$ (4) $\tan^{-1}(0.5)$

149. For a uniformly charged ring of radius R , the electric field on its axis has the largest magnitude at a distance h from its centre. Then value of h is:

[JEE MAINS Jan. 09, 2019 (I)]

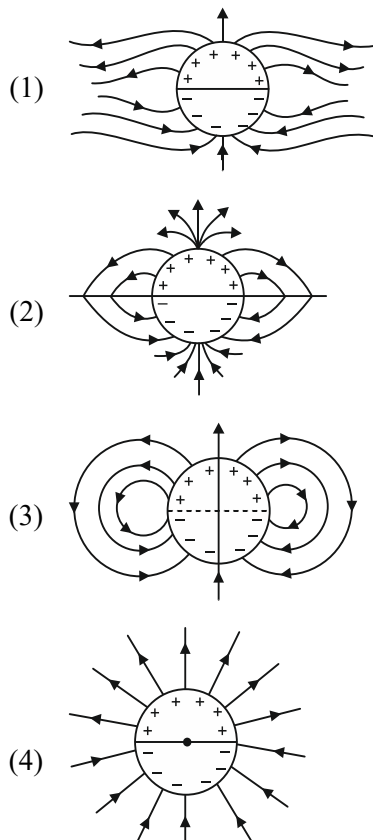
- (1) $\frac{R}{\sqrt{5}}$ (2) $\frac{R}{\sqrt{2}}$
 (3) R (4) $R\sqrt{2}$

150. A solid ball of radius R has a charge density ρ given by $\rho = \rho_0 \left(1 - \frac{r}{R}\right)$ for $0 \leq r \leq R$. The electric field outside the ball is: [JEE MAINS 2018]

- (1) $\frac{\rho_0 R^3}{\epsilon_0 r^2}$ (2) $\frac{4\rho_0 R^3}{3\epsilon_0 r^2}$
 (3) $\frac{3\rho_0 R^3}{4\epsilon_0 r^2}$ (4) $\frac{\rho_0 R^3}{12\epsilon_0 r^2}$

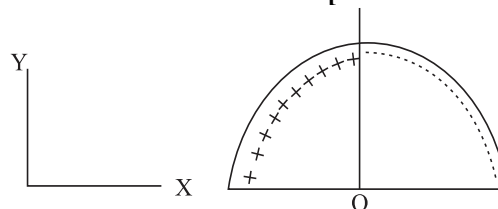
151. A long cylindrical shell carries positive surface charge σ in the upper half and negative surface charge $-\sigma$ in the lower half. The electric field lines around the cylinder will look like figure given in: (figures are schematic and not drawn to scale).

[JEE MAINS 2015]



152. A wire of length L ($= 20$ cm), is bent into a semicircular arc. If the two equal halves of the arc were each to be uniformly charged with charges $\pm Q$, [$Q = 10^3 \epsilon_0$ Coulomb where ϵ_0 is the permittivity (in SI units) of free space] the net electric field at the centre O of the semicircular arc would be:

[JEE MAINS 2015]



- (1) $(50 \times 10^3 \text{ N/C}) \hat{j}$ (2) $(50 \times 10^3 \text{ N/C}) \hat{i}$
 (3) $(25 \times 10^3 \text{ N/C}) \hat{j}$ (4) $(25 \times 10^3 \text{ N/C}) \hat{i}$

153. The magnitude of the average electric field normally present in the atmosphere just above the surface of the Earth is about 150 N/C , directed inward towards the center of the Earth. This gives the total net surface charge carried by the Earth to be:

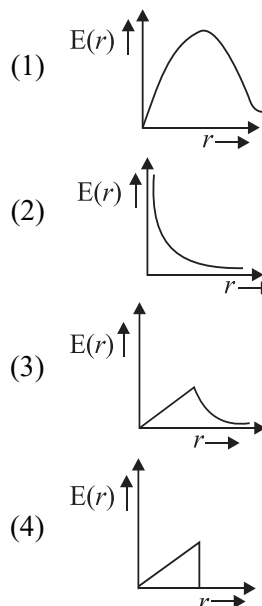
[Given $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N-m}^2$, $R_E = 6.37 \times 10^6 \text{ m}$]

[JEE MAINS 2013]

- (1) $+670 \text{ kC}$ (2) -670 kC
 (3) $+680 \text{ kC}$ (4) $+680 \text{ kC}$

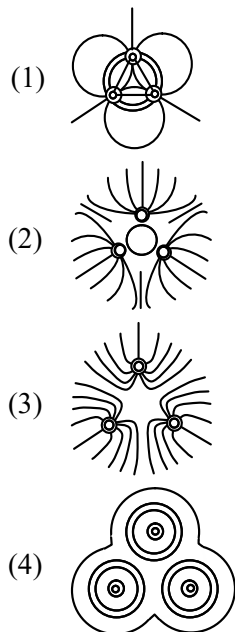
154. In a uniformly charged sphere of total charge Q and radius R , the electric field E is plotted as function of distance from the centre, The graph which would correspond to the above will be:

[JEE MAINS 2012]



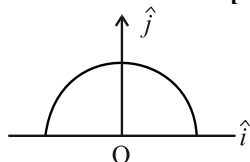
155. Three positive charges of equal value q are placed at vertices of an equilateral triangle. The resulting lines of force should be sketched as in,

[JEE MAINS 2012]



156. A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is.

[JEE MAINS 2010]



- (1) $\frac{q}{4\pi^2\epsilon_0 r^2} \hat{j}$ (2) $-\frac{q}{4\pi^2\epsilon_0 r^2} \hat{j}$
(3) $-\frac{q}{2\pi^2\epsilon_0 r^2} \hat{j}$ (4) $\frac{q}{2\pi^2\epsilon_0 r^2} \hat{j}$

157. Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0 \left(\frac{5}{4} - \frac{r}{R} \right)$ upto $r = R$, and $\rho(r) = 0$ for $r > R$, where r is the distance from the origin. The electric field at a distance r ($r < R$) from the origin is given by

[JEE MAINS 2010]

- (1) $\frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$ (2) $\frac{4\pi\rho_0 r}{3\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$
(3) $\frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$ (4) $\frac{\rho_0 r}{3\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$

158. Let $\rho(r) = \frac{Q}{\pi R^4}$ be the charge density distribution

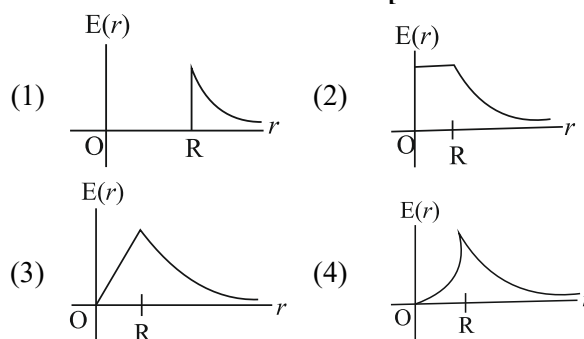
for a solid sphere of radius R and total charge Q . For a point 'P' inside the sphere at distance r_1 , from the centre of the sphere, the magnitude of electric field is:

[JEE MAINS 2009]

- (1) $\frac{Q}{4\pi\epsilon_0 r_1^2}$ (2) $\frac{Q r_1^2}{4\pi\epsilon_0 R^4}$
(3) $\frac{Q r_1^2}{3\pi\epsilon_0 R^4}$ (4) 0

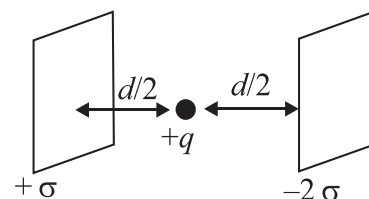
159. A thin spherical shell of radius R has charge spread uniformly over its surface. Which of the following graphs most closely represents the electric field $E(r)$ produced by the shell in the range $0 \leq r < \infty$ where r is the distance from the centre of the shell?

[JEE MAINS 2008]



160. Consider two infinitely large plane parallel conducting plates as shown below. The plates are uniformly charged with a surface charge density $+\sigma$ and -2σ . The force experienced by a point charge $+q$ placed at the mid-point between two plates will be:

[JEE MAINS April 2, 2025 (I)]



- (1) $\frac{\sigma q}{4\epsilon_0}$ (2) $\frac{3\sigma q}{2\epsilon_0}$
(3) $\frac{3\sigma q}{4\epsilon_0}$ (4) $\frac{\sigma q}{2\epsilon_0}$

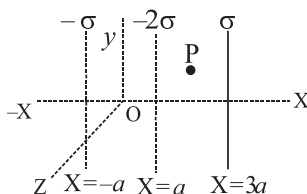
161. σ is the uniform surface charge density of a thin spherical shell of radius R . The electric field at any point on the surface of the spherical shell is:

[JEE MAINS April 06, 2024 (I)]

- (1) $\frac{\sigma}{\epsilon_0 R}$
- (2) $\frac{\sigma}{2\epsilon_0}$
- (3) $\frac{\sigma}{\epsilon_0}$
- (4) $\frac{\sigma}{4\epsilon_0}$

162. Three infinitely long charged thin sheets are placed as shown in figure. The magnitude of electric field at the point P is $\frac{x\sigma}{\epsilon_0}$. The value of x is _____ (all quantities are measured in SI units).

[JEE MAINS April 06, 2024 (I)]

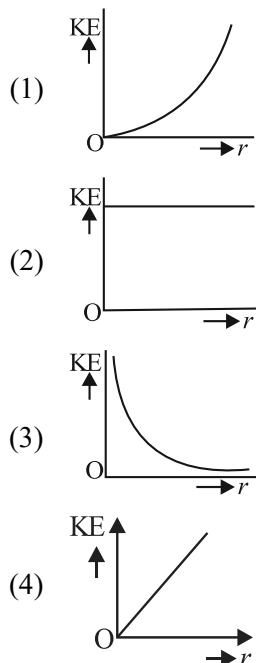


163. An infinite plane sheet of charge having uniform surface charge density $+\sigma_s$ C/m² is placed on x - y plane. Another infinitely long line charge having uniform linear charge density $+\lambda_c$ C/m is placed at $z = 4$ m plane and parallel to y -axis. If the magnitude values $|\sigma_s| = 2 |\lambda_c|$ then at point $(0, 0, 2)$, the ratio of magnitudes of electric field values due to sheet charge to that of line charge is $\pi\sqrt{n} : 1$. The value of n is _____.

[JEE MAINS April 04, 2024 (I)]

164. An infinitely long positively charged straight thread has a linear charge density λ Cm⁻¹. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is:

[JEE MAINS April 04, 2024 (I)]



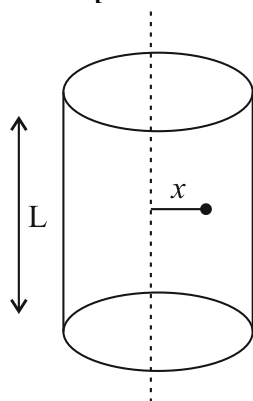
165. A particle of charge ' $-q$ ' and mass ' m ' moves in a circle of radius ' r ' around an infinitely long line charge of linear charge density ' $+\lambda$ '. Then time period will be given as: (Consider k as Coulomb's constant)

[JEE MAINS Jan. 30, 2024 (II)]

- (1) $T^2 = \frac{4\pi^2 m}{2k\lambda q} r^3$
- (2) $T = 2\pi r \sqrt{\frac{m}{2k\lambda q}}$
- (3) $T = \frac{1}{2\pi r} \sqrt{\frac{m}{2k\lambda q}}$
- (4) $T = \frac{1}{2\pi} \sqrt{\frac{2k\lambda q}{m}}$

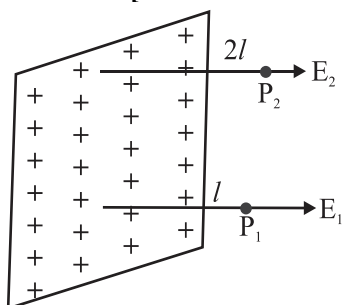
166. A long cylindrical volume contains a uniformly distributed charge of density $\rho \text{ Cm}^{-3}$. The electric field inside the cylindrical volume at a distance $x = \frac{2\epsilon_0}{\rho} \text{ m}$ from its axis is _____ Vm^{-1} .

[JEE MAINS July 27, 2022 (I)]



167. In the figure, a very large plane sheet of positive charge is shown. P_1 and P_2 are two points at distance l and $2l$ from the charge distribution. If σ is the surface charge density, then the magnitude of electric fields E_1 and E_2 at P_1 and P_2 respectively are:

[JEE MAINS June 25, 2022 (I)]



- (1) $E_1 = \sigma/\epsilon_0, E_2 = \sigma/2\epsilon_0$
- (2) $E_1 = 2\sigma/\epsilon_0, E_2 = \sigma/\epsilon_0$
- (3) $E_1 = E_2 = \sigma/2\epsilon_0$
- (4) $E_1 = E_2 = \sigma/\epsilon_0$

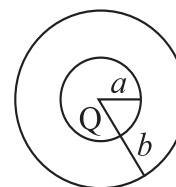
168. A long cylindrical volume contains a uniformly distributed charge of density ρ . The radius of cylindrical volume is R . A charge particle (q) revolves around the cylinder in a circular path. The kinetic energy of the particle is:

[JEE MAINS June 24, 2022 (II)]

- (1) $\frac{\rho q R^2}{4\epsilon_0}$
- (2) $\frac{\rho q R^2}{2\epsilon_0}$
- (3) $\frac{q\rho}{4\epsilon_0 R^2}$
- (4) $\frac{4\epsilon_0 R^2}{q\rho}$

169. The region between two concentric spheres of radii ' a ' and ' b ', respectively (see figure), have volume charge density $\rho = \frac{A}{r}$, where A is a constant and r is the distance from the centre. At the centre of the spheres is a point charge Q . The value of A such that the electric field in the region between the spheres will be constant, is:

[JEE MAINS 2016]



- (1) $\frac{2Q}{\pi(a^2 - b^2)}$
- (2) $\frac{2Q}{\pi a^2}$
- (3) $\frac{Q}{2\pi a^2}$
- (4) $\frac{Q}{2\pi(b^2 - a^2)}$



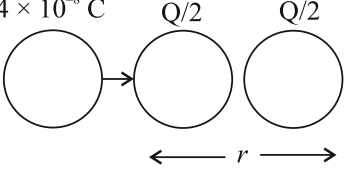
Answer Key

1. (1)	30. (4)	59. (16)	88. (1)	117. (2)	146. (4)
2. (5)	31. (2)	60. (2)	89. (3)	118. (3)	147. (4)
3. (2)	32. (4)	61. (3)	90. (8)	119. (640)	148. (4)
4. (3)	33. (2)	62. (2)	91. (3)	120. (226)	149. (2)
5. (2)	34. (4)	63. (6)	92. (1)	121. (1)	150. (4)
6. (2)	35. (2)	64. (3)	93. (2)	122. (-48)	151. (3)
7. (3)	36. (2)	65. (2)	94. (3)	123. (2)	152. (4)
8. (2)	37. (3)	66. (2)	95. (1)	124. (2)	153. (3)
9. (5440)	38. (3)	67. (4)	96. (2)	125. (1)	154. (3)
10. (300)	39. (2)	68. (1)	97. (4)	126. (3)	155. (3)
11. (2)	40. (5)	69. (3)	98. (15)	127. (3)	156. (3)
12. (24)	41. (1)	70. (4)	99. (4)	128. (1)	157. (1)
13. (1)	42. (1)	71. (3)	100. (1)	129. (1)	158. (2)
14. (2)	43. (1)	72. (3)	101. (48)	130. (4)	159. (1)
15. (1)	44. (2)	73. (3)	102. (1)	131. (3)	160. (2)
16. (2)	45. (3)	74. (45)	103. (16)	132. (2)	161. (3)
17. (4)	46. (1)	75. (2)	104. (2)	133. (2)	162. (2)
18. (4)	47. (4)	76. (2)	105. (12)	134. (3)	163. (16)
19. (2)	48. (1)	77. (4)	106. (2)	135. (2)	164. (2)
20. (1)	49. (2)	78. (2)	107. (3)	136. (3)	165. (2)
21. (2)	50. (17)	79. (1)	108. (2)	137. (6)	166. (1)
22. (2)	51. (1)	80. (3)	109. (1)	138. (2)	167. (3)
23. (2)	52. (2)	81. (2)	110. (1)	139. (3)	168. (1)
24. (12)	53. (2)	82. (4)	111. (288)	140. (1)	169. (3)
25. (36)	54. (4)	83. (3)	112. (640)	141. (2)	
26. (1)	55. (3)	84. (2)	113. (12)	142. (4)	
27. (4)	56. (3)	85. (3)	114. (4)	143. (3)	
28. (1)	57. (4)	86. (1)	115. (45)	144. (3)	
29. (4)	58. (4)	87. (2)	116. (2)	145. (2)	

Solution

1. (1)

$Q = 4 \times 10^{-8} \text{ C}$



$$F = \frac{k \left(\frac{Q}{2} \right) \left(\frac{Q}{2} \right)}{r^2}$$

$$9 \times 10^{-3} = \frac{9 \times 10^9 \times (4 \times 10^{-8}) \times (4 \times 10^{-8})}{4 \times r^2}$$

$$\therefore r = 2 \times 10^{-2} \text{ m} = 2 \text{ cm}$$

2. (5)

The expression for electrostatic force

$$F_e = \frac{k q_1 q_2}{r_2}$$

The expression for gravitational force $F_g = \frac{G m_1 m_2}{r_2}$

$$\frac{F_e}{F_g} = \frac{9 \times 10^9 \times 6.67 \times 10^{-19} \times 9.6 \times 10^{-10}}{6.67 \times 10^{-11} \times 19.2 \times 10^{-27} \times 9 \times 10^{-27}}$$

$$= 5 \times 10^{44}$$

3. (2)

Using Coulomb's law,

$$F = K \frac{Q^2}{R^2}$$

$$F_{PS} \propto Q^2$$

$$F_{PS} = 16 \text{ N}$$

If P & R are brought in contact then charges will equally divide.

$$Q_P = Q_R = \frac{Q}{2}$$

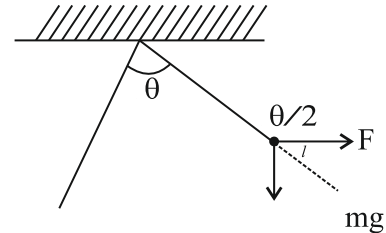
Now, If S and R are brought in contact then

$$Q_S = \frac{3Q}{4}, Q_R = \frac{3Q}{4}$$

Force between P & S is,

$$F_{PS} \propto \frac{Q}{2} \times \frac{3Q}{4} = \frac{3}{8} \times 16 = 6 \text{ N}$$

4. (3)



If spheres are in air,

$$\tan \frac{\theta}{2} = \frac{F}{mg} = \frac{q^2}{4\pi\epsilon_0 r^2 mg}$$

If spheres are in water,

$$\tan \frac{\theta}{2} = \frac{F'}{mg'} = \frac{q^2}{4\pi k \epsilon_0 r^2 mg'}$$

Since angle remains same.

$$\Rightarrow \frac{q^2}{4\pi\epsilon_0 r^2 mg} = \frac{q^2}{4\pi k \epsilon_0 r^2 mg'}$$

$$\Rightarrow g = kg' = g \left(1 - \frac{1}{1.5} \right) k$$

$$\therefore k = 3$$

5. (2)

Electrostatic force acting between two point charges in vacuum

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

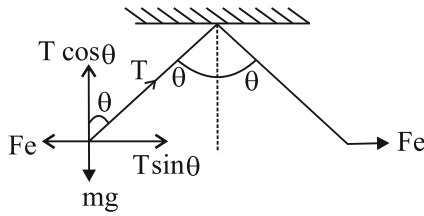
Electrostatic force acting between two point charges in a medium of dielectric constant K is

$$F' = \frac{1}{4\pi(K\epsilon_0)} \frac{q_1 q_2}{(r')^2} = \frac{25}{4\pi(5\epsilon_0)} \frac{q_1 q_2}{(r)^2} \Rightarrow F' = 5F$$

6. (2)

Density of material of the sphere, $\rho_B = 1.4 \text{ g/cm}^3$

Density of liquid, $\rho_L = 0.7 \text{ g/cm}^3$



From the free body diagram

$$T \cos \theta = mg$$

$$T \sin \theta = F_e$$

$$\tan \theta = \frac{F_e}{mg}$$

If spheres are in air,

$$\tan \theta = \frac{F_e}{\rho_B V_g} \quad \dots\dots(i)$$

If spheres are in liquid,

$$\tan \theta = \frac{F_e}{k(\rho_B - \rho_L)V_g} \quad \dots\dots(ii)$$

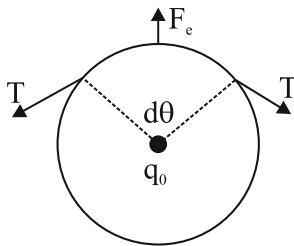
From Eq. (i) & (ii),

$$\rho_B V_g = (\rho_B - \rho_L) k V_g$$

$$\rho_B = (\rho_B - \rho_L) k$$

$$\text{Dielectric constant, } k = \frac{1.4}{1.4 - 0.7} = 2$$

7. (3)



$$\text{Here, } 2T \sin \frac{d\theta}{2} = \frac{kq_0}{R^2} \cdot \lambda R d\theta$$

$$\left[\because \text{Linear charge density, } \lambda = \frac{Q}{2\pi R} \right]$$

$$\Rightarrow T = \frac{K q_0 Q}{(R^2) \times 2\pi}$$

$$= \frac{(9 \times 10^9)(2\pi \times 30 \times 10^{-12})}{(0.30)^2 \times 2\pi} = \frac{9 \times 10^{-3} \times 30}{9 \times 10^{-2}} = 3\text{N}$$

8. (2)

Let charge $q = 10 \mu\text{C}$ is divided into two parts x and $q - x$.

From coulomb's law, $F = k \frac{q_1 q_2}{r^2}$

$$F = \frac{K_x (q - x)}{r^2}$$

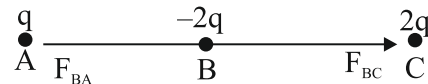
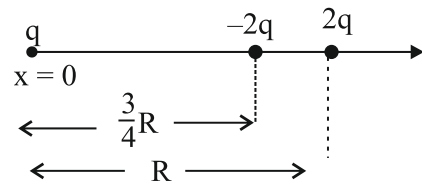
For F to be maximum

$$\frac{dF}{dx} = \frac{K}{r^2} (q - 2x) = 0 \quad \therefore x = \frac{q}{2}$$

Hence charge of the two parts are $5 \mu\text{C}$ and $5 \mu\text{C}$

9. (5440)

From coulomb's law



$$F_{BA} = \frac{Kq(2q)}{\left(\frac{3}{4}R\right)^2} = \frac{32Kq^2}{9R^2} \text{ and}$$

$$F_{BC} = \frac{K(2q)(2q)}{\left(\frac{R}{4}\right)^2} = \frac{64Kq^2}{R^2}$$

$$F_B = F_{BC} - F_{BA} = \frac{544Kq^2}{9R^2} B$$

$$= \frac{544 \times 9 \times 10^9 \times (2 \times 10^{-6})^2}{9 \times (2 \times 10^{-2})^2} = 5440\text{N}$$

10. (300)

Given, Mass of charge, $m = 20 \text{ g} = 0.02 \text{ kg}$

Charge, $q_0 = +2 \text{ Mc}$

For equilibrium

$$mg \sin \theta = F_e$$

$$\Rightarrow mg \sin \theta = \frac{1}{4\pi\epsilon_0} \times \frac{q_0^2}{(h \operatorname{cosec} 30^\circ)^2}$$

$$\therefore h^2 = \frac{1}{4\pi\epsilon_0} \times \frac{q_0^2}{mg \operatorname{cosec} 30^\circ}$$

$$= 9 \times 10^9 = \frac{(2 \times 10^{-6})^2}{0.02 \times 10 \times 2}$$

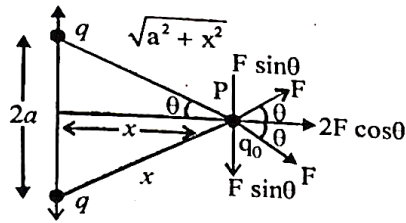
$$\Rightarrow h = 3 \times 10^4 \times \frac{2 \times 10^{-6}}{0.2} = 0.3 \text{ m} = 300 \text{ mm}$$

11. (2)

Force due to both charge at point P

$$F' = 2F \cos \theta$$

$$\text{Here, } F = \frac{Kqq_0}{a^2 + x^2}; F' = \frac{Kqq_0}{a^2 + x^2} \frac{x}{(a^2 + x^2)^{1/2}}$$



$$F' = \frac{2Kqq_0x}{(a^2 + x^2)^{3/2}}$$

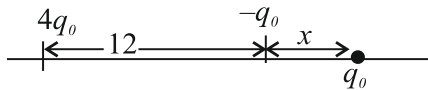
$$\text{For } F' \text{ to be maximum } \frac{dF'}{dx} = 0$$

$$2Kqq_0 \left[\frac{\frac{3x}{2}(x^2 + a^2)^{1/2} \times 2x - (x^2 + a^2)^{3/2}}{(x^2 + a^2)^3} \right] = 0$$

$$\therefore x = \frac{a}{\sqrt{2}}$$

12. (24)

Let x be the distance from charge $-q_0$ where electric force is zero.



Using the Coulomb's law

$$\frac{1}{4\pi\epsilon_0} \cdot \frac{q_0^2}{x^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{4q_0^2}{(x+12)^2}$$

$$\Rightarrow x + 12 = 2x$$

$$x = 12$$

$$\text{Distance from origin} = x + 12 = 24 \text{ cm.}$$

13. (1)

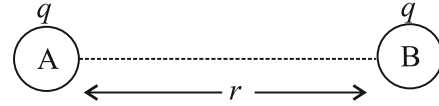
As,

$$F_{\text{air}} = F_{\text{med}}$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{d_{\text{air}}^2} = \frac{1}{K} \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{d^2} \Rightarrow d_{\text{air}} = d\sqrt{K}$$

14. (2)

Consider two metallic spheres A and B both have charge q .



The force between two sphere will be

$$F = \frac{kq^2}{r^2}$$

When sphere C is placed in contact with sphere A, charge on sphere A and sphere C will be $= \frac{q}{2}$

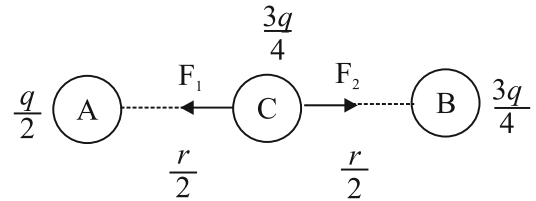
Now sphere C is placed in contact with sphere B, charge on sphere B and sphere C will be

$$= \frac{q + \frac{q}{2}}{2} = \frac{3q}{4}$$

Now,

$$F_1 = \frac{\frac{kq}{2} \cdot \frac{3q}{4}}{\frac{r^2}{4}}$$

$$F_2 = \frac{\frac{k3q}{4} \cdot \frac{3q}{4}}{\frac{r^2}{4}}$$

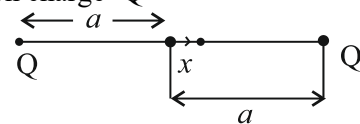


The force experienced by sphere C,

$$F' = F_1 - F_2 = \frac{\left(k \frac{q}{2} - k \frac{3q}{4} \right)}{\frac{r^2}{4}} \cdot \frac{3q}{4} = \frac{3kq^2}{4r^2} = \frac{3F}{4}$$

15. (1)

Force on charge 'Q'



$$F = \frac{KQq_0}{(a-x)^2} - \frac{KQq_0}{(a+x)^2}$$

$$= KQq_0 \left(\frac{4ax}{(a^2 - x^2)^2} \right) \text{ as } a < x$$

$$\text{So, } F = \frac{KQq_0 \times 4ax}{a^4} = \frac{4Kxq_0Q}{a^3}$$

$$\text{As, } F = m\omega^2 x$$

$$\text{So, } m\omega^2 = \frac{4KQq_0}{a^3}$$

$$\omega = \sqrt{\frac{4KQq_0}{ma^3}} \Rightarrow T = \sqrt{\frac{4\pi^3 \epsilon_0 ma^3}{q_0 Q}}$$

16. (2)

Let distance between the two divided charges be r .
From Coulomb's Law, force between two charge,

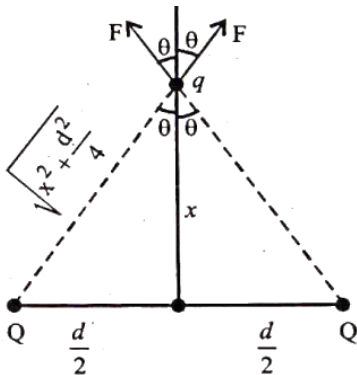
$$F = \frac{Kq(4-q)}{r^2}$$

For F to be maximum,

$$\frac{dF}{dq} = \frac{K}{r^2} [4 - 2q] = 0 \Rightarrow q = 2\mu\text{C}$$

So, other charge will $(4 - 2)\mu\text{C}$ i.e., $2\mu\text{C}$

17. (4)



We have, from the given figure

$$F_{net} = 2F \cos \theta$$

$$F_{net} = 2 \cdot \frac{KQq}{x^2 + \frac{d^2}{4}} \cdot \frac{x}{\left(x^2 + \frac{d^2}{4}\right)^{1/2}}$$

$$F_{net} = \frac{2KQqx}{\left(x^2 + \frac{d^2}{4}\right)^{3/2}}$$

For maximum F_{net}

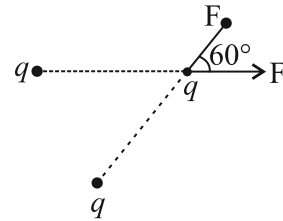
$$\frac{dF_{net}}{dx} = 0$$

$$\Rightarrow x \times -\frac{3}{2} \left(x^2 + \frac{d^2}{4}\right)^{-5/2} \cdot 2x + \left(x^2 + \frac{d^2}{4}\right)^{-3/2} = 0$$

$$\Rightarrow \left(x^2 + \frac{d^2}{4}\right)^{-5/2} \left[-3x^2 + x^2 + \frac{d^2}{4}\right] = 0$$

$$\Rightarrow 2x^2 = \frac{d^2}{4} \Rightarrow x^2 = \frac{d^2}{8} \Rightarrow x = \frac{d}{2\sqrt{2}}$$

18. (4)



Let force between two charge be F . Then

$$F_{net} = \sqrt{F^2 + F^2 + 2F \cdot F \cos 60^\circ} = \sqrt{3} F$$

$$\text{So, required ratio} = \frac{F_{net}}{F} = \frac{\sqrt{3} F}{F} = \sqrt{3}$$

19. (2)

From law of conservation of charge $q_i = q_f$

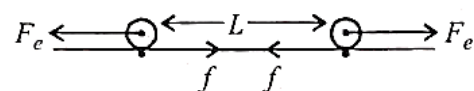
$$\Rightarrow 64q = Q \Rightarrow \frac{Q}{q} = 64 \text{ and, } \frac{4}{3} \pi R^3 = 64 \times \frac{4}{3} \pi r^3$$

$$\Rightarrow R = 4r \Rightarrow \frac{r}{R} = \frac{1}{4}$$

$$\text{So, } \frac{\sigma_{bigger}}{\sigma_{smaller}} = \frac{\frac{Q}{4\pi R^2}}{\frac{q}{4\pi r^2}} = \frac{Q}{q} \times \left(\frac{r}{R}\right)^2$$

$$= 64 \times \frac{1}{16} = \frac{4}{1}$$

20. (1)



For charge to stay in equilibrium $F_e = f$

$$\frac{kq^2}{L^2} = \mu mg$$

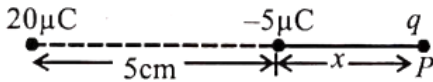
$$L = \sqrt{\frac{kq^2}{\mu mg}} = \sqrt{\frac{k}{\mu mg}} \cdot q = \sqrt{\frac{9 \times 10^9}{0.25 \times 10^{-2} \times 10}} \times 2 \times 10^{-7}$$

$$= 12 \text{ cm}$$

21. (2)

Let, charge q be placed at P.

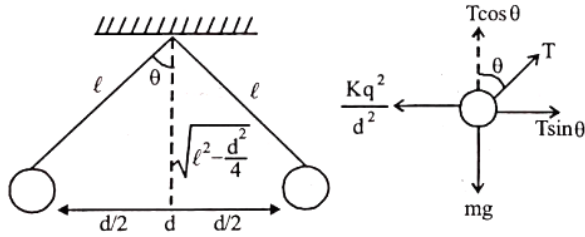
At point P forces due to $20 \mu\text{C}$ & $-5 \mu\text{C}$ should be in opposite direction



For net force $\vec{F} = 0$ & from coulomb's law force

$$k \frac{20q}{(5+x)^2} = \frac{k5q}{x^2} \Rightarrow x = 5 \text{ cm}$$

22. (2)



$$T \cos \theta = mg \quad \dots\dots(i)$$

$$\text{Force due to charges} = \frac{kq^2}{d^2}$$

$$T \sin \theta = \frac{kq^2}{d^2} \quad \dots\dots(ii)$$

Form (i) and (ii) we get

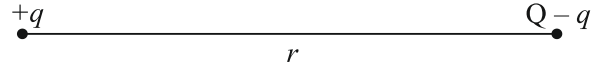
$$\tan \theta = \frac{kq^2}{mg d^2}$$

$$\text{as } \tan \theta \approx \sin \theta \approx \frac{d}{2l}$$

$$\frac{kq^2}{mg d^2} = \frac{d}{2l} \Rightarrow d^3 = \frac{2kq^2 l}{mg}$$

$$\Rightarrow d = \left(\frac{2kq^2 l}{mg} \right)^{1/3} = \left(\frac{q^2 l}{2\pi\epsilon_0 mg} \right)^{1/3}$$

23. (2)



Let two parts of charge Q be divided into q and $Q - q$

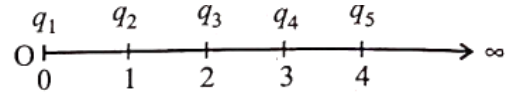
$$F = \frac{K q_1 q_2}{r^2} = \frac{K(q)(Q-q)}{r^2}$$

For maximum repulsion

$$\frac{dF}{dq} = 0 \text{ or, } \frac{d}{dq} \left[\frac{Kq(Q-q)}{r^2} \right] = 0$$

$$\Rightarrow Q - 2q = 0 \Rightarrow Q = 2q$$

24. (12)



$$F_{\text{total}} = \left[\frac{kq_1 q_2}{r_1^2} + \frac{kq_1 q_3}{r_2^2} + \frac{kq_1 q_4}{r_3^2} + \dots \right]$$

$$\Rightarrow F = (k)(10^{-6}) \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{4^2} + \dots \right]$$

$$= \frac{(k)10^{-6}}{1 - \frac{1}{4}} = \frac{(9 \times 10^9) \times 4 \times 10^{-6} \text{ N}}{3}$$

$$= 12 \times 10^3 \text{ N}$$

$$\therefore x = 12$$

25. (36)

When two spheres charges $q_1' = 2.1 \text{ nC}$ and $q_2' = -0.1 \text{ nC}$ are brought into contact and then separated by a distance $r = 0.5 \text{ m}$ then,

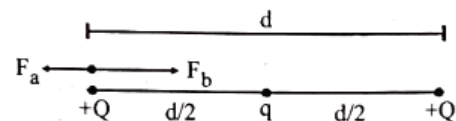
$$q_1' = q_2' = \frac{Q_1 + Q_2}{2} = 1 \text{ nC}$$

Electrostatic force between the two charged sphere,

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1' q_2'}{r^2} = 9 \times 10^9 \times \frac{10^{-9} \times 10^{-9}}{(0.5)^2}$$

$$= 36 \times 10^{-9} \text{ N}$$

26. (1)



Force due to charge +Q, $F_a = \frac{KQQ}{d^2}$

Force due to charge q , $F_b = \frac{KQq}{\left(\frac{d}{2}\right)^2}$

For equilibrium, $\vec{F}_a + \vec{F}_b = 0$

$$\Rightarrow \frac{kQQ}{d^2} + \frac{kQq}{(d/2)^2} = 0 \quad \therefore q = -\frac{Q}{4}$$

27. (4)

Spheres A and B carry equal charges say 'q'

$\therefore$ Force between them, $F = \frac{kqq}{r^2}$

When A and C are touched, charge on both

$$q_A = q_C = \frac{q}{2}$$

Then when B and C are touched, charge on B.

$$q_B = \frac{\frac{q}{2} + q}{2} = \frac{3q}{4}$$

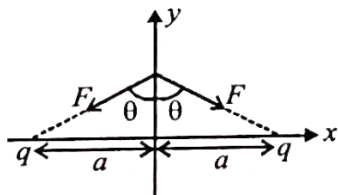
Now, the force between charge q_A and q_B

$$F' = \frac{kq_A q_B}{r^2} = \frac{k \times \frac{q}{2} \times \frac{3q}{4}}{r^2} = \frac{3}{8} \frac{kq^2}{r^2} = \frac{3}{8} F$$

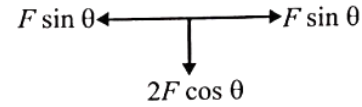
28. (1)

$$F_{\text{net}} = 2F \cos \theta$$

$$F_{\text{net}} = \frac{2kq\left(\frac{q}{2}\right)}{\left(\sqrt{y^2 + a^2}\right)^2} \cdot \frac{y}{\sqrt{y^2 + a^2}}$$



$$F_{\text{net}} = \frac{2kq\left(\frac{q}{2}\right)y}{\left(y^2 + a^2\right)^{3/2}} \quad (\because y < a)$$



$$\Rightarrow \frac{kq^2 y}{a^3}$$

$$\text{So, } F \propto y$$

29. (4)

From figure

$$T \cos \theta = mg$$

$$T \sin \theta = F_e$$

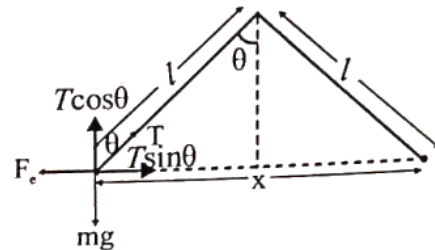
Dividing equation (ii) by (i), we get

$$\Rightarrow \frac{\sin \theta}{\cos \theta} = \frac{F_e}{mg}$$

$$\Rightarrow F_e = mg \tan \theta$$

$$\Rightarrow \frac{kq^2}{x^2} = mg \tan \theta$$

$$\Rightarrow q^2 = \frac{x^2 mg \tan \theta}{k}$$



Since θ is small

$$\therefore \tan \theta \approx \sin \theta = \frac{x}{2l}$$

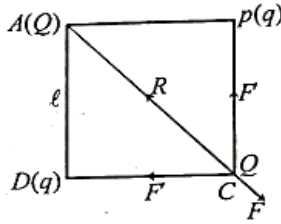
$$\therefore q^2 = \frac{x^3 mg}{2kl} \Rightarrow q^2 \propto x^{3/2}$$

$$\Rightarrow \frac{dq}{dt} \propto \frac{3}{2} \sqrt{x} \frac{dx}{dt} = \frac{3}{2} \sqrt{x} v$$

$$\text{Since } \frac{dq}{dt} = \text{const.} \Rightarrow v \propto x^{-1/2}$$

30. (4)

Let F be the force between Q and Q . The force between q and Q should be attractive for net force on Q to be zero. Let F be the force between Q and q . The resultant of F' and F' is R . For equilibrium Net force on Q and C is zero.



$$\therefore \vec{R} + \vec{F} = 0 \Rightarrow \sqrt{2} F' = -F$$

$$\Rightarrow \sqrt{2} \times k \frac{Qq}{l^2} = -k \frac{Q^2}{(\sqrt{2}l)^2}$$

$$\Rightarrow \frac{Q}{q} = -2\sqrt{2}$$

31. (2)

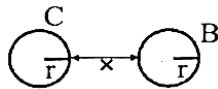
It is obvious that by charge conservation law, electronic charge does not depend on acceleration due to gravity as it is a universal constant.

So, electronic charge on earth = electronic charge on moon

$\therefore$ Required ratio = 1

32. (4)

$$\text{Initial force, } F = K \frac{Q_B Q_C}{x^2}$$



x is distance between the spheres. When third spherical conductor comes in contact with B charge on B is halved i.e., $\frac{Q}{2}$ and charge on third sphere

becomes $\frac{Q}{2}$. Now it is touched to C, charge then equally distributes themselves to make potential same, hence charge on C becomes $\left(Q + \frac{Q}{2}\right) \frac{1}{2} = \frac{3Q}{4}$.

$$\therefore F_{\text{new}} = k \frac{Q'_C Q'_B}{x^2} = k \frac{\left(\frac{3Q}{4}\right) \left(\frac{Q}{2}\right)}{x^2}$$

$$= k \frac{3Q^2}{8x^2} = \frac{3}{8} F$$

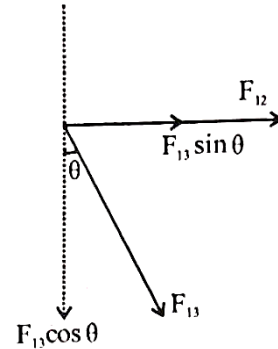
33. (2)

Force applied by charge q_2 and q_1 ,

$$F_{12} = k \frac{q_1 q_2}{b^2}$$

Force applied by charge q_3 and q_1 ,

$$F_{13} = k \frac{q_1 q_3}{a^2}$$



The X-component of net force (F_x) on q_1 is $F_{12} + F_{13} \sin \theta$

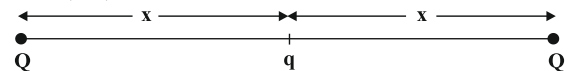
$$\therefore F_x = k \frac{q_1 q_2}{b^2} + k \frac{q_1 q_3}{a^2} \sin \theta$$

$$\therefore F_x \propto \frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta$$

34. (4)

At equilibrium net force is zero,

$$\therefore k \frac{Q \times Q}{(2x)^2} + k \frac{Qq}{x^2} = 0$$



$$\Rightarrow \frac{kQ}{x^2} \left(\frac{Q}{4} + q \right) = 0 \Rightarrow q = -\frac{Q}{4}$$

35. (4)

The electrostatic field will balance the weight of oil drop,

$$m_{\text{oil}} \times g_{\text{oil}} = qE$$

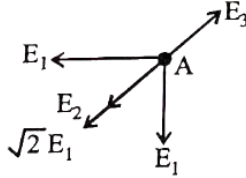
$$= \frac{4}{3} \pi r^3 \times \rho \times g = neE \quad (n = \text{no. of excess electrons})$$

$$\Rightarrow n = \frac{\frac{4}{3} \pi r^3 \rho g}{eE} = \frac{\frac{4}{3} \times \pi \times (2 \times 10^{-3})^3 \times (3 \times 10^3) \times 9.81}{1.6 \times 10^{-19} \times 3.55 \times 10^5}$$

$$= 173.65 \times 10^8 \approx 1.73 \times 10^{10}$$

36. (2)

For the system to be equilibrium, net field at A should be zero.

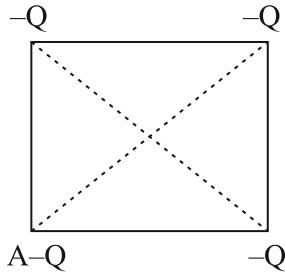


$$\sqrt{2}E_1 + E_2 = E_3$$

$$\therefore \frac{kQ \times \sqrt{2}}{a^2} + \frac{kQ}{(\sqrt{2}a)^2} = \frac{kq}{\left(\frac{a}{\sqrt{2}}\right)^2}$$

$$\Rightarrow \frac{Q\sqrt{2}}{1} + \frac{Q}{2} = 2q$$

$$\Rightarrow q = \frac{Q}{4}(2\sqrt{2} + 1)$$



37. (2)

Given, Electric field, $E = 3 \times 10^4$

Mass of the drop, $m = 9.9 \times 10^{-15}$ kg

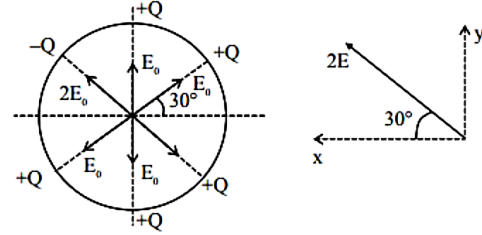
At equilibrium, coulomb force on drop balances weight of drop.

$$qE = mg$$

$$\Rightarrow q = \frac{mg}{E} \Rightarrow q = \frac{9.9 \times 10^{-15} \times 10}{3 \times 10^4} = 3.3 \times 10^{-18} \text{ C}$$

38. (3)

$$\text{Let } \frac{KQ}{r^2} = E_0$$



$$\vec{E}_{\text{net}} = 2E_0 \cos 30^\circ (-\hat{i}) + 2E_0 \sin 30^\circ (\hat{j})$$

$$= \frac{2kQ}{r^2} \left[\frac{\sqrt{3}}{2}(-\hat{i}) + \frac{1}{2}\hat{j} \right]$$

$$= \frac{-1Q}{4\pi\epsilon_0 r^2} (\sqrt{3}\hat{i} - \hat{j})$$

39. (2)

Let a point P between the charge q and $9q$ electric field is zero i.e., $E_P = 0$

$$\therefore \frac{kq}{x^2} = \frac{k9q}{(d-x)^2} \Rightarrow \frac{d-x}{x} = 3$$

$$\therefore x = \frac{d}{4}$$

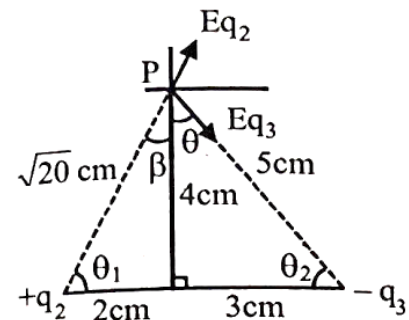
The point is between two charges q and $9q$ where the electric field vanishes is $\left(\frac{d}{4}, 0, 0\right)$

40. (5)

Since $E^p = 0$

$$\therefore E_{q2} = E_{q3}$$

$$\frac{Kq_2}{20} \cos \beta = \frac{Kq_3}{25} \cos \theta$$

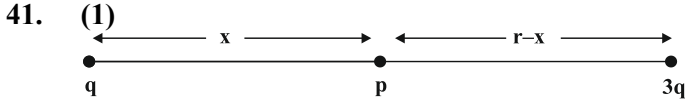


$$\frac{q_2}{20} \frac{4}{\sqrt{20}} = \frac{q_3}{25} \frac{4}{\sqrt{25}}$$

$$\frac{q_2}{q_3} = \frac{20}{25} \frac{\sqrt{20}}{\sqrt{25}} = \frac{8}{5\sqrt{x}}$$

On comparing & rearranging.

$$\sqrt{x} = \frac{8 \times 25 \sqrt{25}}{5 \times 20 \sqrt{20}} \Rightarrow x = 5$$

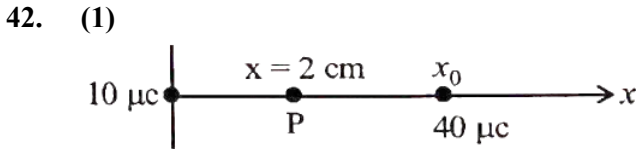


Net Electric field at P = 0

$$\therefore E_q = E_{3q} \Rightarrow \frac{kq}{x^2} = \frac{k \cdot 3q}{(r-x)^2}$$

$$\Rightarrow (r-x)^2 = 3x^2 \Rightarrow r-x = \sqrt{3}x$$

$$\Rightarrow x = \frac{r}{\sqrt{3}+1}$$

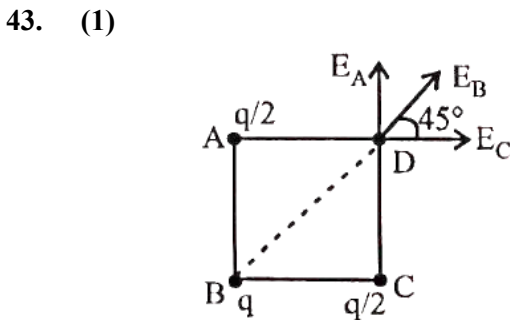


As,

$$E_P = 0 \Rightarrow K \times \frac{10 \times 10^{-6}}{2^2} \hat{i} - K \frac{(40 \times 10^{-6})}{(x_0 - 2)^2} \hat{i} = 0$$

$$\Rightarrow \frac{10 \times 10^{-6}}{40 \times 10^{-6}} = \frac{2^2}{(x_0 - 2)^2}$$

$$\Rightarrow \frac{1}{2} = \frac{2}{x_0 - 2} \Rightarrow x_0 - 2 = 4 \Rightarrow x_0 = 6 \text{ cm}$$

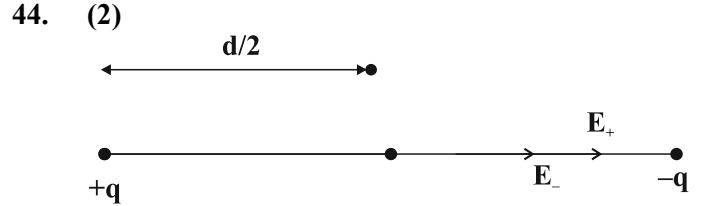


Here, sum of $\vec{E}_C$ and $\vec{E}_A$ will lie along $\vec{E}_B$.

$$\text{So, } |\vec{E}_{\text{net}}| = |\vec{E}_B| + |\vec{E}_C + \vec{E}_A|$$

$$= \frac{Kq}{(\sqrt{2}a)^2} + \left[\sqrt{2} \frac{kq}{2a^2} \right]$$

$$= \frac{Kq}{a^2} \left[\frac{1}{2} + \frac{1}{\sqrt{2}} \right]$$



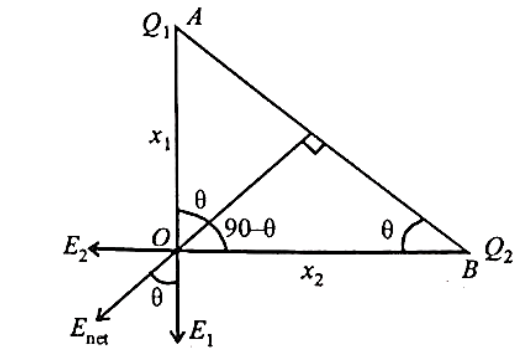
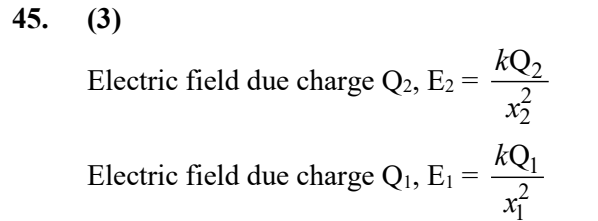
So, $E_{\text{net}} = E_+ + E_-$

$$= 2 \frac{kq}{\left(\frac{d}{2}\right)^2} = \frac{8kq}{d^2}$$

$$\Rightarrow 6.4 \times 10^4 = 8 \times 9 \times 10^9 \times \frac{8 \times 10^{-6}}{d^2}$$

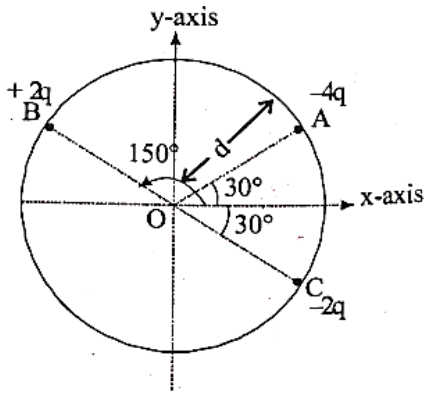
$$\Rightarrow d^2 = \frac{72 \times 8 \times 10^{+3}}{6.4 \times 10^4}$$

$$\Rightarrow d^2 = 9 \Rightarrow d = 3 \text{ m}$$



$$\Rightarrow \frac{Q_2 x_1^2}{Q_1 x_2^2} = \frac{x_1}{x_2} \Rightarrow \frac{Q_2}{Q_1} = \frac{x_2}{x_1} \text{ or, } \frac{Q_1}{Q_2} = \frac{x_1}{x_2}$$

46. (1)



Electric field due to charge $+2q$ at centre O

$$\vec{E}_1 = \frac{1}{4\pi\epsilon_0} \times \frac{2q}{d^2} \left[\frac{\sqrt{3}\hat{i} - \hat{j}}{2} \right]$$

Electric field due to charge $-2q$ at centre O

$$\vec{E}_2 = \frac{1}{4\pi\epsilon_0} \times \frac{2q}{d^2} \left[\frac{\sqrt{3}\hat{i} - \hat{j}}{2} \right]$$

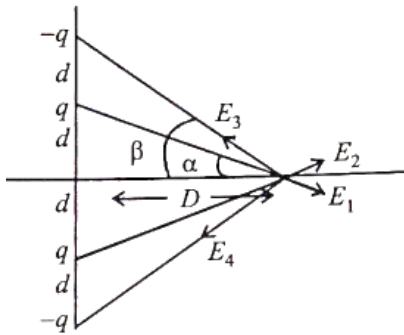
Electric field due to charge $-4q$ at centre O

$$\vec{E}_3 = \frac{1}{4\pi\epsilon_0} \times \frac{4q}{d^2} \left[\frac{\sqrt{3}\hat{i} + \hat{j}}{2} \right]$$

$\therefore$ Net electric field at point O

$$\vec{E}_0 = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 = \frac{\sqrt{3}q}{\pi\epsilon_0 d^2} \hat{i}$$

47. (4)



$$\vec{E} = (\vec{E}_1 + \vec{E}_2) + (\vec{E}_3 + \vec{E}_4) \text{ or } E = 2E \cos \alpha - 2E \cos \beta$$

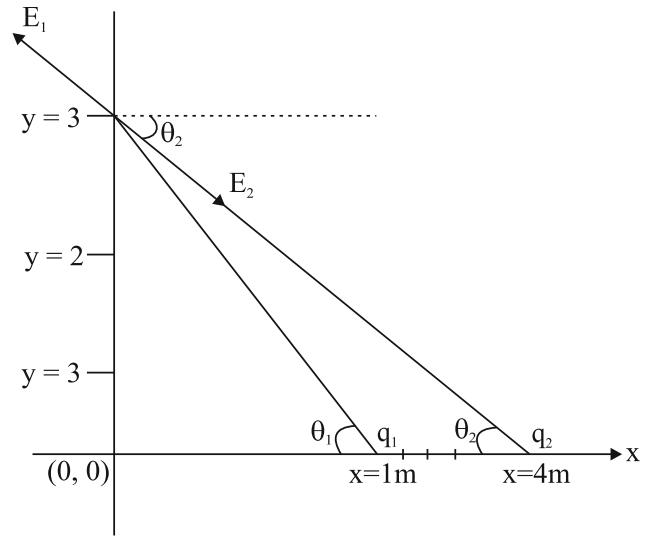
$$= \frac{2kq}{(D^2 + d^2)} \times \frac{D}{\sqrt{D^2 + d^2}} - \frac{2kq}{(D^2 + (2d)^2)} \times \frac{D}{\sqrt{D^2 + (2d)^2}}$$

$$= \frac{2kqD}{(D^2 + d^2)^{3/2}} - \frac{2kqD}{[D^2 + (2d)^2]^{3/2}}$$

For $d \ll D$

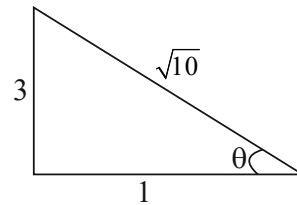
$$E \propto \frac{D}{D^3} \propto \frac{1}{D^2}$$

48. (1)



Let $\vec{E}_1$ and $\vec{E}_2$ are the of electric field due to charge, q_1 and q_2 respectively magnitude of

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1^2} = \frac{1}{4\pi\epsilon_0} \frac{\sqrt{10} \times 10^{-6}}{(1^2 + 3^2)}$$



$$= (9 \times 10^9) \times \sqrt{10} \times 10^{-7} = 9\sqrt{10} \times 10^2$$

$$\therefore \vec{E}_1 = 9\sqrt{10} \times 10^2 [\cos \theta_1 (-\hat{i}) + \sin \theta_1 \hat{j}]$$

$$\Rightarrow E_1 = 9 \times \sqrt{10} \times 10^2 \left[\frac{1}{\sqrt{10}} (-\hat{i}) + \frac{3}{\sqrt{10}} \hat{j} \right] \text{ A}$$

$$\Rightarrow E_1 = 9 \times 10^2 [-\hat{i} + 3\hat{j}] = [-9\hat{i} + 27\hat{j}] 10^2$$

Similarly, $E_2 = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2^2}$

$$E_2 = \frac{9 \times 10^9 \times (25) \times 10^{-6}}{(4^2 + 3^2)}$$

$$\Rightarrow E_2 = 9 \times 10^3 \text{ V/m}$$

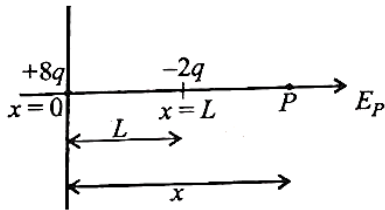
$$\therefore \vec{E}_2 = 9 \times 10^3 \left(\cos \theta_2 \hat{i} - \sin \theta_2 \hat{j} \right) \quad \because \tan \theta_2 = \frac{3}{4}$$

$$\therefore \vec{E}_2 = 9 \times 10^3 \left(\frac{4}{5} \hat{i} - \frac{3}{5} \hat{j} \right) = \left(72 \hat{i} - 54 \hat{j} \right) \times 10^2$$

$$\therefore \vec{E} = \vec{E}_1 + \vec{E}_2 = \left(63 \hat{i} - 27 \hat{j} \right) \times 10^2 \text{ V/m}$$

49. (2)

$$\text{At, P, } E_p = \frac{-K 2q}{(x-L)^2} + \frac{K 8q}{x^2} = 0$$



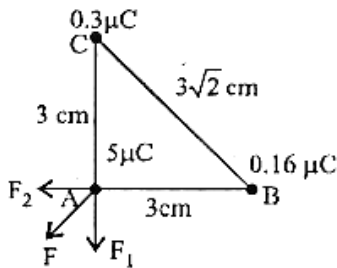
$$\Rightarrow \frac{1}{(x-L)^2} = \frac{4}{x^2}$$

$$\Rightarrow \frac{1}{x-L} = \frac{2}{x}$$

$$\Rightarrow x = 2x - 2L \text{ or } x = 2L$$

50. (17)

The force between $5 \mu\text{C}$ and $0.3 \mu\text{C}$ given by



$$F_1 = \frac{k \times 5 \times 0.3 \times 10^{-12}}{9 \times 10^{-4}} \quad (\because K = 9 \times 10^9)$$

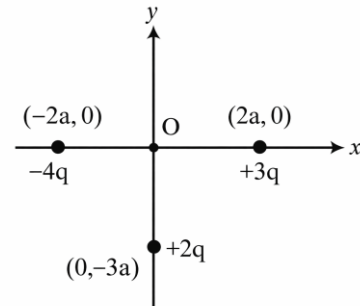
$$= \frac{9 \times 10^9 \times 5 \times 0.3 \times 10^{-12}}{9 \times 10^{-4}} = 15 \text{ N}$$

The force between $5 \mu\text{C}$ and $0.16 \mu\text{C}$ given by

$$F_2 = \frac{9 \times 10^9 \times 5 \times 0.16 \times 10^{-12}}{9 \times 10^{-4}} = 8 \text{ N}$$

$$\text{Net force experienced by charge at A} = \sqrt{F_1^2 + F_2^2} \\ = \sqrt{15^2 + 8^2} = \sqrt{289} = 17 \text{ N}$$

51. (1)

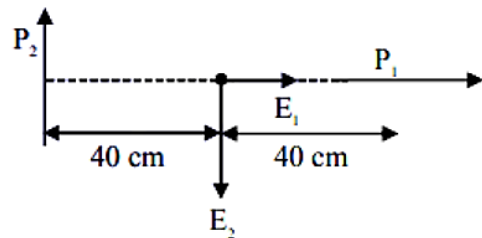


$$\vec{p} = q_1 \vec{r}_1 + q_2 \vec{r}_2 + q_3 \vec{r}_3$$

$$\vec{p} = (2q)(-3a)\hat{j} + (3q)(2a)\hat{i} + (-4q)(-2a)\hat{i}$$

$$\vec{p} = 2qa(7\hat{i} - 3\hat{j})$$

52. (2)



$$\vec{E}_2 = -\frac{KP_2}{r^3}; \vec{E}_1 = -\frac{2KP_1}{r^3}$$

$$P_1 = 2 \times 10^{-6} \times 10^{-2} = 2 \times 10^{-8}$$

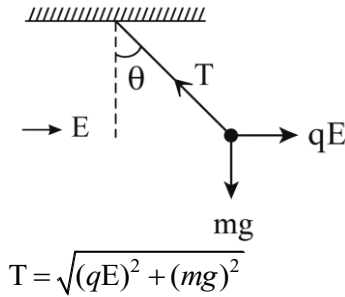
$$P_2 = 4 \times 10^{-6} \times 10^{-2} = 4 \times 10^{-8}$$

$$\vec{E}_{\text{net}} = \frac{2 \times 9 \times 10^9 \times 2 \times 10^{-8}}{(0.4)^3} \hat{i} - \frac{9 \times 10^9 \times 4 \times 10^{-8}}{(0.4)^3} \hat{j}$$

$$\vec{E}_{\text{net}} = \frac{9 \times 10^9 \times 4 \times 10^{-8}}{(0.4)^3} [\hat{i} - \hat{j}]$$

$$|\vec{E}_{\text{net}}| = \frac{9 \times 10^4}{16} (\sqrt{2})$$

53. (2)



54. (1)

Since polar dielectrics are oriented in random directions

$$\therefore \vec{P}_{\text{net}} = 0$$

In absence of an external electric field, polar dielectric remain polar and are randomly oriented.

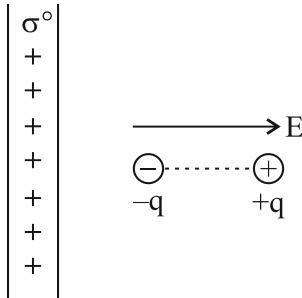
55. (3)

Since $\vec{r} \cdot \vec{p} = 0$

$\vec{E}$ must be antiparallel to $\vec{p}$

$$\therefore \hat{E} \text{ is parallel to } \left(\hat{i} + 3\hat{j} - 2\hat{k} \right)$$

56. (3)



$$\tau = PE \sin \theta$$

$$\tau = 0 \quad (\because \theta = 0^\circ)$$

The expression of electric field of infinite plane sheet is

$$E = \frac{\sigma_0}{2\epsilon_0}$$

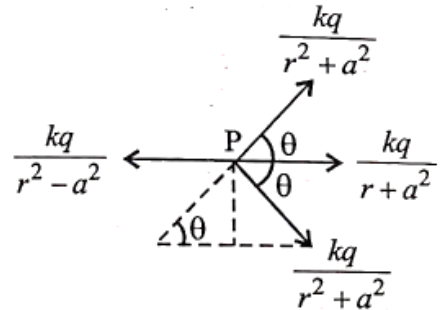
$$U = -\vec{P} \cdot \vec{E} = -P \left(\frac{\sigma_0}{2\epsilon_0} \right) (\text{minimum})$$

57. (4)

The time period of an electric dipole is

$$T = 2\pi \sqrt{\frac{I}{PE}} = 2\pi \sqrt{\frac{m \left(\frac{l}{2} \right)^2 \times 2}{(ql) \cdot E_0}} = 2\pi \sqrt{\frac{ml}{2qE_0}}$$

58. (4)



$$\frac{kq}{(r-a)^2} = \frac{kq}{(r+a)^2} + \frac{2kq}{(r^2+a^2)} \cos \theta$$

$$\frac{1}{(r-a)^2} = \frac{1}{(r+a)^2} + \frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} = \frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\frac{4ra}{(r^2-a^2)^2} = \frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{2r}{(r^2-a^2)^2} = \frac{1}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{4r^2}{(r^2-a^2)^4} = \frac{1}{(r^2+a^2)^3}$$

$$\Rightarrow 4r^2 (r^2+a^2)^3 = (r^2-a^2)^4$$

$$4r^8 \left(1 + \frac{a^2}{r^2} \right)^3 = r^8 \left(1 - \frac{a^2}{r^2} \right)^4$$

$$4 \left(1 + \frac{a^2}{r^2} \right)^3 = \left(1 - \frac{a^2}{r^2} \right)^4$$

$$\therefore \frac{a}{r} \approx 3$$

59. (16)

At axial line,

$$E_p = \frac{2KP}{r^3} = E$$

At equatorial line,

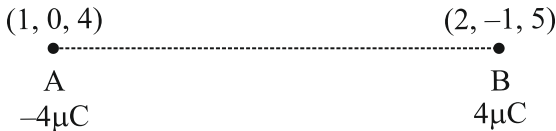
$$E_R = \frac{KP}{(2r)^3} = \frac{1}{16} \frac{KP}{r^3} = \frac{E}{16} = \frac{E}{x} \text{ (given)}$$

$$\therefore x = 16$$

60. (2)

Torque acting on the dipole,

$$\vec{\tau} = \vec{p} \times \vec{E}$$



$$\vec{p} = q\vec{\ell} = 4 \times (\hat{i} - \hat{j} + \hat{k})$$

$$= (4\hat{i} - 4\hat{j} + 4\hat{k}) \mu\text{C-m}$$

$$\therefore \vec{\tau} = \vec{p} \times \vec{E} = (4\hat{i} - 4\hat{j} + 4\hat{k}) \times (20\hat{i}) \times 10^{-6} \text{ Nm}$$

$$= (8\hat{k} + 8\hat{j}) \times 10^{-5} = 8\sqrt{2} \times 10^{-5} \text{ Nm}$$

$$= 8\sqrt{\alpha} \times 10^{-5} \text{ Nm (given)} \quad \therefore \alpha = 2$$

61. (3)

Electric field due to an electric dipole at distance r on

the equatorial plane, $E = \frac{kP}{r^3}$

62. (2)

Dipole moment, $|\vec{P}| = qd$

$$= 0.01 \times 0.4 \times 10^{-3} = 4 \times 10^{-6}$$

And torque, $|\vec{\tau}| = PE \sin \theta$

$$= 4 \times 10^{-6} \times 10 \times 10^{-5} \times \sin 30$$

$$\therefore |\vec{\tau}| = 2 \times 10^{-10}$$

63. (6)

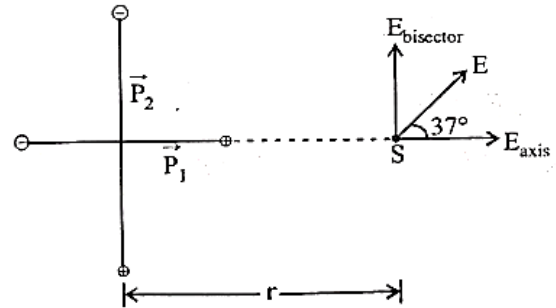
Maximum torque,

$$|\tau|_{\max} = PE \text{ or } \tau_1 = P_1 E_1 \text{ and } \tau_2 = P_2 E_2$$

$$\therefore \frac{\tau_1}{\tau_2} = \frac{P_1 E_1}{P_2 E_2} = \frac{1.2 \times 10^{-30} \times 5 \times 10^4}{2.4 \times 10^{-30} \times 15 \times 10^4} = \frac{1}{6} = \frac{1}{x}$$

$$\therefore x = 6$$

64. (3)



Electric field due to $\vec{P}_1$ at axis point S

$$E_{\text{axis}} = \frac{2kP_1}{r^3}$$

Electric field due to $\vec{P}_2$ at perpendicular bisector at point S.

$$E_{\text{bisector}} = \frac{kP_2}{r^3} \quad \dots\dots(ii)$$

$$\tan 37^\circ = \frac{E_{\text{bisector}}}{E_{\text{axis}}} \Rightarrow \frac{3}{4} = \frac{\frac{kP_2}{r^3}}{\frac{2kP_1}{r^3}} \Rightarrow \frac{P_2}{P_1} = \frac{3}{2}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{2}{3}$$

65. (2)

Let v be the speed of dipole.

Using energy conservation

$$K_i + U_i = K_f + U_f$$

$$\Rightarrow 0 - \frac{2k \cdot p_1}{r^3} p_2 \cos(180^\circ) = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + 0$$

$$\left(\because \text{Potential energy of interaction between dipole} \right)$$

$$-p_2 \cdot E_1 = \frac{-2p_1 p_2 \cos \theta}{4\pi\epsilon_0 r^3}$$

$$\Rightarrow mv^2 = \frac{2kp_1 p_2}{r^3} \Rightarrow v = \sqrt{\frac{2kp_1 p_2}{mr^3}}$$

When $p_1 = p_2 = p$ and $r = a$

$$v = \frac{p}{q} \sqrt{\frac{1}{2\pi\epsilon_0 ma}}$$

66. (2)

Potential energy of a dipole is given by

$$U = -\vec{P} \cdot \vec{E} = -PE \cos \theta$$

[Where θ = angle between dipole and electric field]

$$= -(10^{-29})(10^3) \cos 45^\circ$$

$$= -0.707 \times 10^{-26} \text{ J} = -7 \times 10^{-27} \text{ J}$$

67. (4)

Electric field of equatorial plane of dipole = $-\frac{K\vec{P}}{r^3}$

$$\therefore \text{At point P, } F = +\frac{KP}{y^3} Q$$

At point P',

$$F' = +\frac{KPQ}{(y/3)^3} = 27 F.$$

68. (1)

$T = PE \sin \theta$ Torque experienced by the dipole in an

electric field, $\vec{T} = \vec{P} \times \vec{E}$

$$\vec{p} = p \cos \theta \hat{i} + p \sin \theta \hat{j}; \vec{E}_1 = E \hat{i}$$

$$\vec{T}_1 = \vec{p} \times \vec{E}_1 \left(p \cos \theta \hat{i} + p \sin \theta \hat{j} \right) \times E (\hat{i})$$

$$\tau \hat{k} = pE \sin \theta \left(-\hat{k} \right) \dots\dots\dots(i)$$

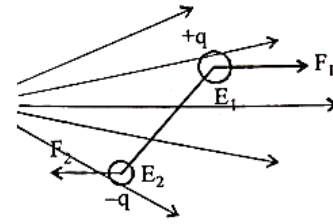
$$\vec{E}_2 = \sqrt{3}E_1 \hat{j}; \vec{T}_2 = p \cos \theta \hat{i} + p \sin \theta \hat{j} \times \sqrt{3}E_1 \hat{j}$$

$$-\tau \hat{k} = \sqrt{3}pE_1 \cos \theta \hat{k} \dots\dots\dots(ii)$$

From eqns. (i) and (ii)

$$pE \sin \theta = \sqrt{3}pE \cos \theta; \tan \theta = \sqrt{3} \quad \therefore \theta = 60^\circ$$

69. (3)



As the dipole is placed in non-uniform field, so the force acting on the dipole will not cancel each other. This will result in a translational force as well as torque.

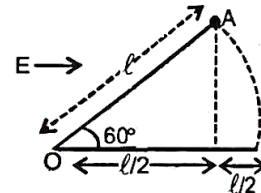
70. (4)

Using energy conservation

$$(2) \left(\frac{1}{2} mu^2 \right) - \frac{Gm^2}{4r} + \frac{KQ^2}{4r} = -\frac{Gm^2}{2r} + \frac{KQ^2}{2r}$$

$$u = \sqrt{\frac{1}{4mr} (KQ^2 - Gm^2)}$$

71. (3)



Use work energy theorem

$$W = K.E_f - K.E_i$$

$$qE \frac{l}{2} = \frac{1}{2} mv^2 - 0$$

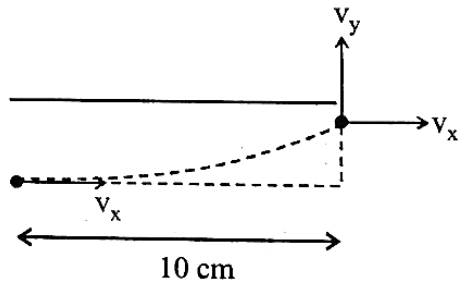
$$\Rightarrow v = \sqrt{\frac{qEl}{m}}$$

72. (3)

Given, length of metal plates, $l = 10 \text{ cm}$

Horizontal component of velocity, $V_x = 10^6 \text{ m/s}$

Electric field, $E = 9.1 \text{ V/cm}$



$\Rightarrow$ Time inside electric field,

$$t = \frac{l}{V_x} = \frac{10 \times 10^{-2}}{10^6} = 10^{-7}$$

Acceleration along y-axis, $a = \frac{eE}{m}$

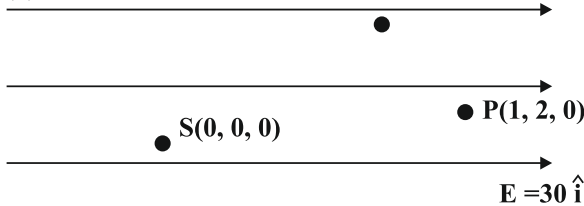
Using $v_y = u_y + a_y t$

$$\Rightarrow v_y = 0 + \frac{eE}{m} \times 10^{-7}$$

$$\Rightarrow v_y = \frac{1.6 \times 10^{-19}}{9.1 \times 10^{-31}} \times 9.1 \times 10^2 \times 10^{-7}$$

$$= 16 \times 10^6 \text{ m/s}$$

73. (3)



$$W = \vec{F} \cdot \vec{S} \Rightarrow W = q\vec{E} \cdot \vec{S}$$

$$W = 2 \times 10^{-2} \left[30 \hat{i} \cdot \left(-\hat{i} \right) \right] = 2 \times 10^{-2} (-30) = -600 \text{ mJ}$$

74. (45)

$$\text{Kinetic theory} = \frac{1}{2} m v_x^2$$

$$0.5e = \frac{1}{2} m v_x^2 \Rightarrow v_x = \sqrt{\frac{e}{m}} \quad \dots\dots(i)$$

$$\text{Along, } x, L = v_x t = \sqrt{\frac{e}{m}} t$$

$$\text{Along, } y, v_y = \frac{eE}{m} t \quad \dots\dots(ii)$$

Divide equation (ii) by (i)

$$\frac{v_y}{L} = E \sqrt{\frac{e}{m}} = E v_x$$

$$\Rightarrow \tan \theta = \frac{v_y}{v_x} = E \times L = 10 \times 0.1 = 1$$

$$\theta = 45^\circ$$

75. (2)

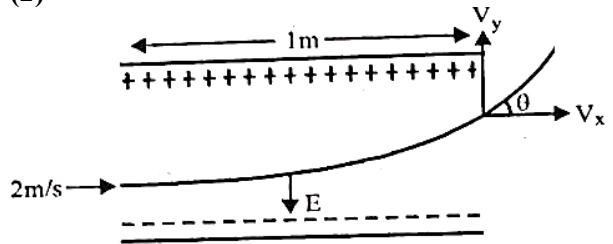
$$a = \frac{F}{m} = \frac{qE}{m} = 2 \times 10^{11} \times 1.8 \times 10^3 = 3.6 \times 10^{14} \text{ m/s}^2$$

$$\text{Time to cross the plates} = \frac{d}{v} = \frac{0.10}{3 \times 10^7} = \frac{1}{3} \times 10^{-8}$$

$$Y = \frac{1}{2} a t^2 = \frac{1}{2} \times 3.6 \times 10^{14} \times \frac{1}{9} \times 10^{-16}$$

$$= 2 \times 10^{-3} \text{ m} = 2 \text{ mm}$$

76. (2)



We have,

$$a_y = \frac{F_y}{m} = \frac{e(E)}{m} = \frac{e \left(\frac{8m}{e} \right)}{m} = 8 \text{ m/s}^2$$

and, $s_x = u_x t$

$$\Rightarrow 1 = 2 \times t \Rightarrow t = \frac{1}{2} \text{ sec}$$

and, $v_y = u_y + a_y t$

$$\Rightarrow v_y = 0 + 8 \times \frac{1}{2} \Rightarrow v_y = 4 \text{ m/s}$$

$$\tan \theta = \frac{v_y}{v_x} = \frac{4}{2} = 2 \Rightarrow \theta = \tan^{-1}(2)$$

77. (4)

Here,

$$a = \frac{qE}{m} = \frac{40 \times 10^{-6} \times 10^5}{100 \times 10^{-6}} = 40000 \text{ m/s}^2$$

As charge particle is moving opposite to direction of $\vec{E}$. So $\vec{E}$ will decelerate the charge and charge will come to rest.

$$\text{Now, } v^2 = u^2 + 2as$$

$$0 = 200^2 + 2 \times (-40 \times 10^3) \times s$$

$$80 \times 10^3 \text{ s} = 4 \times 10^4$$

$$\therefore s = \frac{4 \times 10^4}{80 \times 10^3} = \frac{40}{80} = 0.5 \text{ m}$$

78. (2)

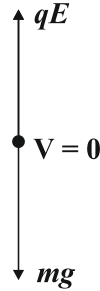
As water droplet is at rest

$$\text{So, } \vec{F}_{\text{net}} = 0$$

$$\Rightarrow mg = qE \Rightarrow q = \frac{mg}{E}$$

$$\Rightarrow q = \frac{0.1 \times 10^{-3} \times 9.8}{4.9 \times 10^5}$$

$$\Rightarrow q = 2 \times 10^{-9} \text{ C}$$



79. (1)

Electric field, $E = 100 \text{ N/m}$,

Charge to mass ratio, $\frac{q}{m} = 8 \mu\text{C/g}$

Force, $F = ma = qE$

$$\Rightarrow ma = qE \Rightarrow a = \frac{qE}{m} = \frac{8 \times 10^{-6}}{10^{-3}} \times 100$$

$$= 0.8 \text{ m/s}^2$$

As electric field is switched on, ball first strikes to wall and returns back one oscillation.

$$\text{Thus } s = ut + \frac{1}{2}at_1^2 \Rightarrow 0.1 = \frac{1}{2} \times 0.8t_1^2$$

($\because$ initial velocity, $u = 0$)

$$\Rightarrow t_1 = \frac{1}{2} \text{ s}$$

$$\text{Thus time period } T = 2 \times \frac{1}{2} = 1 \text{ sec.}$$

80. (3)

Given, Electric Field, $E = E_0(1 - ax^2)$

$\therefore$ Force, $F = qE = qE_0(1 - ax^2)$

$$\text{Also, } F = ma = m \frac{dv}{dx} \quad \left(\because a = v \frac{dv}{dx} \right)$$

$$\therefore m \frac{dv}{dx} = qE_0(1 - ax^2) \Rightarrow v dv = \frac{qE_0(1 - ax^2)}{m} dx$$

Integrating both sides we get,

$$\Rightarrow \int_0^v v dv = \int_0^x \frac{qE_0(1 - ax^2)}{m} dx$$

$$\Rightarrow \frac{v^2}{2} = \frac{qE_0}{m} \left(x - \frac{ax^3}{3} \right) = 0$$

$$\Rightarrow x = \sqrt{\frac{3}{a}}$$

81. (2)

$$F_x = 0, a_x = 0, (v)_x = v_0 = \text{constant}$$

$$\text{Time taken to reach at 'P' } = \frac{d}{v_0} = t_0 \text{ (let)(i)}$$

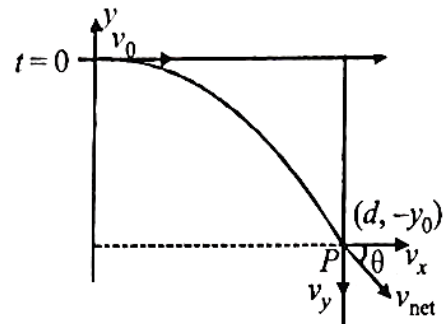
$$\text{(Along -y), } v_y = \frac{qE}{m} t_0 \text{(ii)}$$

$$\tan \theta = \frac{v_y}{v_x} = \frac{qEt_0}{m.v_0}, \left(t_0 = \frac{d}{v_0} \right)$$

$$\tan \theta = \frac{qEd}{m.v_0^2}, \text{ slope} = \frac{-qEd}{mv_0^2}$$

$$\text{No electric field } \Rightarrow F_{\text{net}} = 0, \vec{v} = \text{const.}$$

$$y = (\text{slope})x + c, \left\{ \begin{array}{l} \text{slope} = -\frac{qEd}{mv_0^2} \\ (d, -y_0) \end{array} \right\}$$



$$-y_0 = \frac{-qEd}{mv_0^2}, d + c$$

$$\Rightarrow c = -y_0 + \frac{qEd^2}{mv_0^2}$$

$$y = \frac{-qEd}{mv_0^2} x - y_0 + \frac{qEd^2}{mv_0^2}$$

$$\text{Also, } y_0 = \frac{1}{2} \cdot \frac{qE}{m} \left(\frac{d}{v_0} \right)^2 = \frac{1}{2} \frac{qEd^2}{mv_0^2}$$

$$y = \frac{-qEdx}{mv_0^2} - \frac{1}{2} \frac{qEd^2}{mv_0^2} + \frac{qEd^2}{mv_0^2}$$

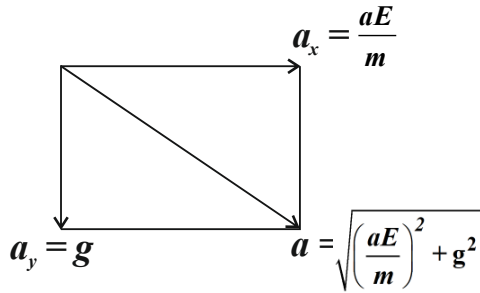
$$y = \frac{-qEd}{mv_0^2} + \frac{1}{2} \frac{qEd^2}{mv_0^2} \Rightarrow y = \frac{qEd}{mv_0^2} \left(\frac{d}{2} - x \right)$$

82. (4)

Net force acting on the particle,

$$\vec{F} = qE\hat{i} + mg\hat{j}$$

Net acceleration of particle is constant, initial velocity is zero therefore path is straight line.



83. (3)

(A) By work energy theorem

$$W_{\text{mag}} + W_{\text{ele}} = \frac{1}{2} m (2v)^2 - \frac{1}{2} m (v)^2$$

$$0 + qE_0 2a = \frac{3}{2} mv^2 \Rightarrow E_0 = \frac{3}{4} \frac{mv^2}{qa}$$

(B) Rate of work done at P = power of electric force

$$= qE_0 v = \frac{3}{4} \frac{mv^3}{a}$$

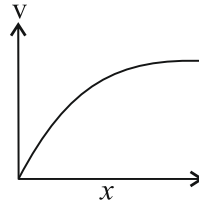
(C) At Q, $\frac{dw}{dt} = 0$ for both the fields

(D) The difference of magnitude of angular momentum of the particle at P and Q.

$$\Delta \vec{L} = \left(-m2v2a\hat{k} \right) - \left(-mva\hat{k} \right)$$

$$|\Delta \vec{L}| = 3mva$$

84. (2)



Using

$$v^2 - u^2 = 2As$$

Here, $u = 0$, $s = x$

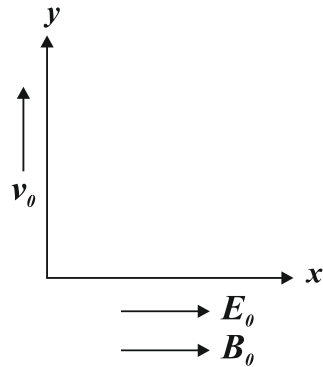
Also, $F_{\text{electric}} = ma$

$$\Rightarrow qE = ma \Rightarrow a = \frac{qE}{m} \Rightarrow a = \frac{qE}{m}$$

Substituting the values in (i) we get

$$v^2 = \frac{2qE}{m} x$$

85. (3)



In the x direction

$$F_x = qE_0$$

$$\Rightarrow ma_x = qE_0$$

$$\Rightarrow a_x = \frac{E_0 q}{m}$$

For speed to be double,

$$v_0^2 + v_x^2 = (2v_0)^2$$

$$\Rightarrow v_x = \sqrt{3}v_0 = a_x t$$

$$\Rightarrow \sqrt{3}v_0 = \frac{qE_0 t}{m} \Rightarrow t = \frac{\sqrt{3}v_0 m}{E_0 q}$$

86. (1)

Equilibrium position will shift to point where resultant force = 0

$$kx_{eq} = qE \Rightarrow x_{eq} = \frac{qE}{k}$$

$$\text{Total energy} = \frac{1}{2}m\omega^2 A^2 + \frac{1}{2}kx_{eq}^2$$

$$\text{Total energy} = \frac{1}{2}m\omega^2 A^2 + \frac{1}{2} \frac{q^2 E^2}{k}$$

87. (2)

$$F = qE = mg \quad (q = 6e = 6 \times 1.6 \times 10^{-19})$$

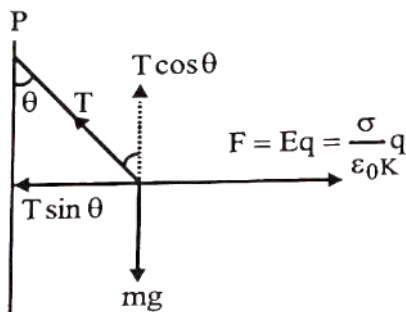
$$\text{Density } (d) = \frac{\text{mass}}{\text{volume}} = \frac{m}{\frac{4}{3}\pi r^3} \text{ or } r^3 = \frac{m}{\frac{4}{3}\pi d}$$

Putting the value of d and $m \left(= \frac{qE}{g} \right)$ and solving we get $r = 7.8 \times 10^{-7} \text{ m}$

88. (1)

In conservative Field work done is independent of path and work done by force over a closed loops is zero.

89. (3)



$$T \sin \theta = qE \quad \dots\dots(i)$$

$$T \cos \theta = mg \quad \dots\dots(ii)$$

Dividing (i) by (ii),

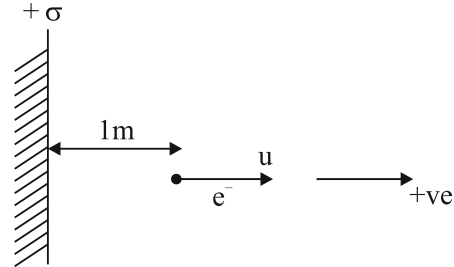
$$\tan \theta = \frac{qE}{mg}$$

$$= \frac{q}{mg} \left(\frac{\sigma}{\epsilon_0 K} \right) = \frac{\sigma q}{\epsilon_0 K \cdot mg}$$

$$\therefore \sigma \propto \tan \theta$$

90. (8)

For maximum value of σ , initially, electron must move away and strikes the plate due to retardation.



Acceleration of electron,

$$a = \frac{F}{m} = -\frac{eE}{m} = -\frac{e\sigma}{2\epsilon_0 m}$$

$$\text{Now, } s = ut + \frac{1}{2}at^2$$

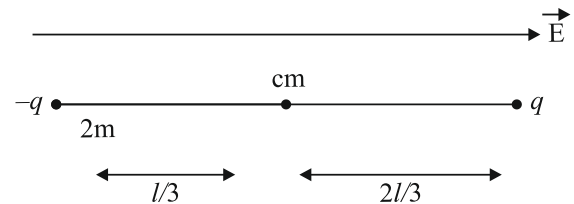
$$t = 1\text{s}, u = 1 \text{ m/s}, s = -1\text{m}$$

$$-1 = 1 \times 1 - \frac{1}{2} \left(\frac{e\sigma}{2\epsilon_0 m} \right) \times 1^2$$

$$\therefore \sigma_{\max} = \left(\frac{8m\epsilon_0}{e} \right) \frac{C}{m^2} = \alpha \left(\frac{m\epsilon_0}{e} \right) \frac{C}{m^2} \text{ (given)}$$

$$\therefore \alpha = 8$$

91. (1)



If released, it will oscillate about centre of mass.

$$\tau_{\text{com}} = qEl \sin \theta$$

For small ' θ '

$$\tau_{\text{com}} = qEl\theta$$

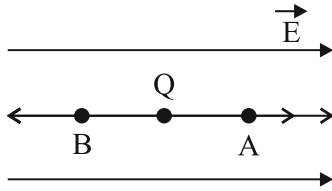
Using $T = I\alpha$

$$\Rightarrow qEl.\theta = \left[2m \frac{l^2}{9} + m \frac{4l^2}{9} \right] \alpha$$

$$\Rightarrow qEl.\theta = \frac{2ml^2}{3} \alpha \Rightarrow \alpha = \frac{3qE\theta}{2ml}$$

$$\Rightarrow \Rightarrow \omega = \sqrt{\frac{3qE}{2ml}}$$

92. (1)



At point 'A' electric field will increase because field due to charge 'Q' will add up in external electric field. At point 'B' electric field will decrease because field due to charge 'Q' will oppose the external electric field. In non-uniform electric field, when dipole is placed at a point, then net force experienced by it is not zero.

93. (2)

$$\tau = -PE \sin \theta$$

$$I\alpha = -PE(\theta) \Rightarrow \alpha = \frac{PE}{I}(-\theta)$$

On comparing with

$$\alpha = -\omega^2 \theta$$

$$\omega = \sqrt{\frac{PE}{I}} = \sqrt{\frac{qdE}{2m\left(\frac{d}{2}\right)^2}} = \sqrt{\frac{2qE}{md}}$$

94. (3)

Static electric field lines do NOT form closed loops (they start on positive and end on negative charges), making statement A false.

95. (1)

$$v = ar^3 + b$$

$$E = -\frac{dv}{dr} = -3ar^2$$

$$\phi_{\text{closed}} = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$q_{\text{enc}} = \epsilon_0 \cdot E \cdot A$$

$$= \epsilon_0 (-3a \cdot (1)^2) 4\pi(1)^2$$

$$= -12\pi a \epsilon_0$$

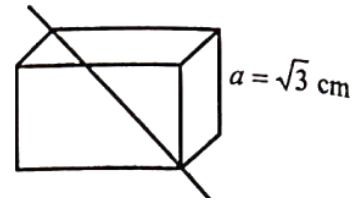
$$\therefore x = -12$$

96. (2)

$$\phi = \frac{Q_{\text{in}}}{\epsilon_0}$$

$$\phi = \frac{\frac{q}{4} + \frac{q}{2}}{\epsilon_0} = \frac{3q}{4\epsilon_0}$$

97. (4)



Linear charge density

$$\lambda = \frac{q}{l}$$

Net flux,

$$\phi = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{\lambda \cdot \sqrt{3}a}{\epsilon_0}$$

$$= 2 \times 10^{-9} \times \sqrt{3} \times \sqrt{3} \times 10^{-2} \times 36\pi \times 10^9 \text{ Nm}^2\text{C}^{-1}$$

$$= 2.16\pi \text{ Nm}^2\text{C}^{-1}$$

98. (15)

$$\phi = \vec{E} \cdot \vec{A} = (2\hat{i} + 4\hat{j} + 6\hat{k}) \times 10^3 \cdot A (0\hat{i} + \hat{j} + 0\hat{k})$$

Given: $\phi = 6 \text{ Nm}^2/\text{C}$

$$\therefore 6 = 4 \times 10^3 A$$

$$\therefore A = 1.5 \times 10^{-3} \text{ m}^2 = 15 \text{ cm}^2$$

99. (4)

For spherical shell,

$$E_{\text{inside}} = 0, E_{\text{outside}} = \frac{\sigma}{\epsilon_0} \left(\frac{R}{r} \right)^2$$

For infinite plane sheet,

$$E = \frac{\sigma}{2\epsilon_0}$$

For two oppositely charged infinite parallel sheets,

$$E = \frac{\sigma}{\epsilon_0}$$

100. (1)

$$\text{Given, } \phi = -2 \times 10^4 \frac{\text{Nm}^2}{\text{C}}$$

Radius, $r = 8.0 \text{ cm}$

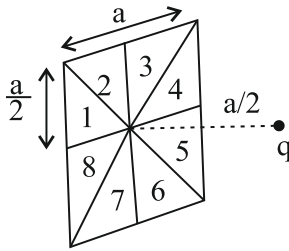
Flux is given by

$$\phi = \frac{q}{\epsilon_0} \Rightarrow q = \epsilon_0 \phi$$

$$= (8.85 \times 10^{-12}) \times (-2 \times 10^4)$$

$$q = -17.7 \times 10^{-8} \text{ C}$$

101. (48)



$$\text{Total flux through square} = \phi_T = \frac{q}{\epsilon_0} \left(\frac{1}{6} \right)$$

$$\therefore \text{Flux through shaded portion is } \phi_{\text{shaded}} = \frac{5}{8} \phi_T$$

$$= \frac{5}{8} \times \frac{q}{\epsilon_0} \frac{1}{6} = \frac{5}{48} \frac{q}{\epsilon_0} = \frac{5}{8} \times \frac{1}{\epsilon_0} \text{ (given)} \quad \therefore P = 48$$

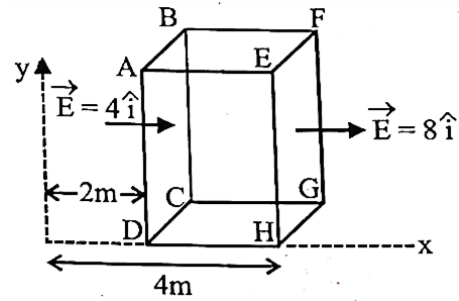
102. (1)

Total charge inside the given cube

$$q_{\text{in}} = \frac{\lambda \frac{a}{2}}{4} = \frac{\lambda a}{8}$$

$$\therefore \text{Total flux } \phi = \frac{q_{\text{in}}}{\epsilon_0} = \frac{\lambda a}{8\epsilon_0}$$

103. (16)



Electric field, $\vec{E} = 2 \times \hat{i}$

Electric flux,

$$\phi = \vec{E} \cdot \vec{A}$$

Only those surfaces having area vector parallel to x -axis will contribute to the flux.

Total flux entering

$$\phi_1 = -2 \times 2 \times 4 = -16 \text{ Nm}^2/\text{c}$$

Total flux leaving,

$$\phi_2 = 8 \times 4 = 32 \text{ Nm}^2/\text{c}$$

$$\phi_{\text{Total}} = \phi_1 + \phi_2 = -16 + 32 = 16 \text{ Nm}^2/\text{c}$$

104. (2)

Using Gauss's law, $\phi = \frac{q}{\epsilon_0}$

Here, q = charge inside the closed surface

$$\therefore \phi = \frac{q + (-2q) + 5q}{\epsilon_0}$$

$$\Rightarrow \phi = \frac{4q}{\epsilon_0}$$

105. (12)

Given,

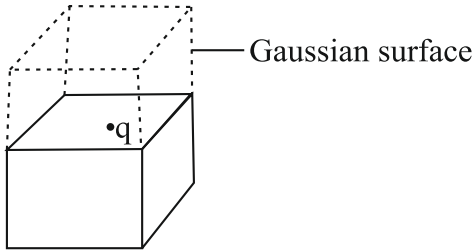
Area of surface, $A = 4\text{m}^2$

Electric flux, $\phi = \vec{E} \cdot \vec{A} = \vec{E} \cdot \hat{n}(A)$

$$= \left(\frac{2\hat{i} + 6\hat{j} + 8\hat{k}}{\sqrt{6}} \right) \cdot 4 \left(\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}} \right)$$

$$= 4 \times \frac{(4 + 6 + 8)}{6} = 12 \text{ Vm}$$

106. (2)



Total flux emerges from the system $\phi_T = \frac{q}{\epsilon_0}$

Therefore the flux linked with the cube $\phi_{\text{cube}} = \frac{q}{2\epsilon_0}$

107. (3)

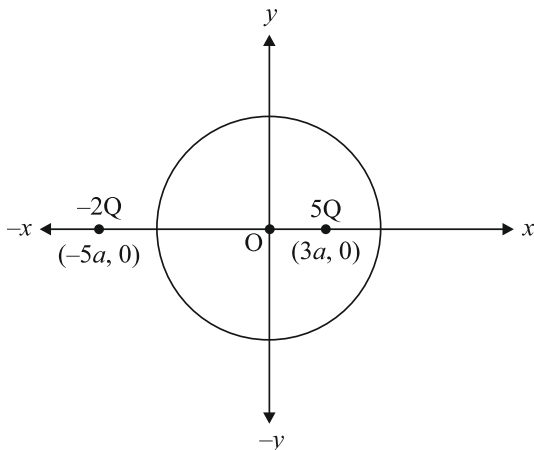
Flux is given by $\phi = \frac{q}{\epsilon_0}$

$$\phi_{\text{Inner cube}} = \frac{q_1}{\epsilon_0} = \frac{2Q}{\epsilon_0}$$

$$\phi_{\text{Outer cube}} = \frac{q_2}{\epsilon_0} = \frac{5Q}{\epsilon_0}$$

$$\therefore \frac{\phi_{\text{Inner cube}}}{\phi_{\text{Outer cube}}} = \frac{2Q}{\epsilon_0} \times \frac{\epsilon_0}{5Q} = \frac{2}{5}$$

108. (2)



Using Gauss law,

$$\text{Electric flux, } \phi = \frac{q_{\text{enclosed}}}{\epsilon_0} = \frac{5Q}{\epsilon_0}$$

109. (1)

$$\vec{E} = (6\hat{i} + 5\hat{j} + 3\hat{k}) \text{ N/C}$$

$$\vec{A} = 30\hat{i} \text{ m}^2$$

Electric flux, $\phi = \vec{E} \cdot \vec{A}$

$$= (6\hat{i} + 5\hat{j} + 3\hat{k}) \cdot (30\hat{i}) = 6 \times 30 = 180 \text{ V-m}$$

110. (1)

Given,

Dipole moment,

$$\vec{P} = 30 \times 10^{-5} \text{ cm}$$

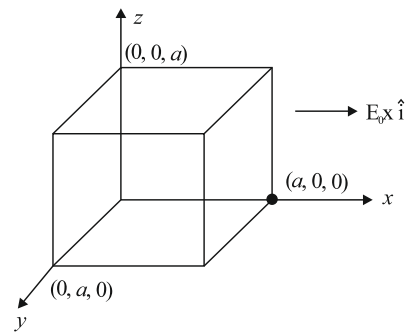
An electric dipole consists of two equal and opposite charges.

$\therefore$ Net charge enclosed by a closed surface is zero.

Using Gauss Law

$$\phi = \frac{Q_{\text{in}}}{\epsilon_0} \Rightarrow \phi = 0$$

111. (288)



For plane, $x = a$

$$\phi(\text{flux}) = \iiint E_0 a \hat{i} \cdot d\vec{y} d\vec{z} \hat{i} = \int_0^a \int_0^a E_0 a dy dz$$

$$\text{By Gauss law } \phi = \oint \vec{E} \cdot \vec{dA} = \frac{q_{\text{in}}}{\epsilon_0}$$

For plane $x = 0, y = 0, y = a, z = 0$ and $z = a$, we will have $\int \vec{E} \cdot d\vec{A} = 0$ and there will be flux through $x = a$ only

$$\Rightarrow \int_0^a \int_0^a E_0 a dy dz = \frac{q_{in}}{\epsilon_0}$$

$$\Rightarrow E_0 a^3 = \frac{q_{in}}{\epsilon_0} \Rightarrow q_{in} = \epsilon_0 E_0 a^3$$

$$= 9 \times 10^{-12} \times 4 \times 10^4 \times (2 \times 10^{-2})^3 = 288 \times 10^{-14} \text{ C}$$

112. (640)

Flux through surface $x = 0, y = 0, y = 20 \text{ cm}, z = 0$ and $z = 20 \text{ cm}$ will be zero as for these surface $\vec{E} \perp \vec{A}$
Flux through surface ($x = 0.2 \text{ m}$) = $4000 \times 0.2^2 \times 0.2^2$
= $6.4 \text{ Vm} = 640 \text{ Vcm}$

113. (12)

Given, electric field

$$\vec{E} = 2x^2 \hat{i} - 4y \hat{j} + 6z \hat{k}$$

Electric flux for face, $x = 1, \phi = 2 \times (1)^2 \times 2 \times 3 = 12$

Electric flux for face, $y = 2, \phi = -4 \times 2 \times 3 \times 1 = -24$

Electric flux for face, $z = 3, \phi = 6 \times 1 \times 2 = 12$

Electric flux for face, $x = 0, y = 0$, is zero

Electric flux for face, $z = 0, \phi = -12$

$$\phi_{net} = -24 + 12 = -12$$

From the Gauss's theorem

$$-12 = \frac{q}{\epsilon_0}$$

$$|q| = 12\epsilon_0$$

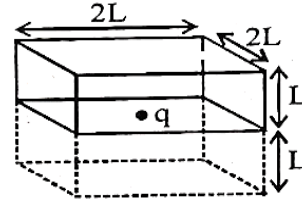
114. (4)

After placing similar cubic at bottom, we get cube of side $2L$ having charge q at its centre,

From the Gauss's law electric flux through whole cube, $\phi = \frac{q}{\epsilon_0}$.

Flux passing through shaded face

$$\phi = \frac{q / \epsilon_0}{6} = \frac{q}{6\epsilon_0}$$



115. (45)

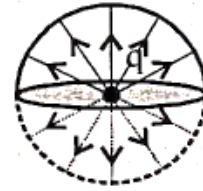
Here,

$$\phi_e = \int \vec{E} \cdot d\vec{A} \Rightarrow \frac{\phi_e}{A} = E$$

$$\Rightarrow \frac{\phi_e}{A} = \frac{\rho R}{3\epsilon_0} = \frac{2 \times 10^0 \times 6}{3 \times 8.85 \times 10^{-12}} = 45 \times 10^{10} \text{ NC}^{-1}$$

116. (2)

Total flux through complete spherical surface is $\frac{q}{\epsilon_0}$.



So flux, through curved surface will be $\frac{q}{2\epsilon_0}$.

The flux through flat surface will be zero.

117. (2)

At bottom surface, electric field is zero as $y = 0$

$\therefore$ Electric flux, $\phi_1 = 0$; At top surface, $y = 0.5$

$\therefore$ Electric flux, $\phi_2 = EA = (150y^2)(0.5)^2$

$$= 150 \times (0.5)^2 \times (0.5)^2 = \frac{150}{16}$$

And flux through all other surface is zero because

$\vec{E} \perp \vec{A}$ for each of them

$$\therefore \phi_{total} = \phi_1 + \phi_2 = \frac{150}{16}$$



Using Gauss's law $\phi_{\text{total}} = \frac{Q_{\text{in}}}{\epsilon_0} \Rightarrow \frac{150}{16} = \frac{Q_{\text{in}}}{\epsilon_0}$

$$\Rightarrow Q_{\text{in}} = \frac{150}{16} \times 8.85 \times 10^{-12} = 8.3 \times 10^{-11} \text{ C}$$

118. (3)

When electric field is parallel to surface, it makes 90° angle with area vector so flux = 0

$$\text{As } \phi = \vec{E} \cdot \vec{A} = E A \cos 90^\circ = 0$$

119. (640)

$$\begin{aligned} \text{Flux, } \phi &= \vec{E} \cdot \vec{A} = \left[\frac{2}{5} E_0 \hat{i} + \frac{3}{5} E_0 \hat{j} \right] \cdot \vec{A} \\ &= \frac{E_0}{5} (2\hat{i} + 3\hat{j}) \cdot (0.4\hat{i}) = \frac{4000}{5} (2 \times 0.4) \\ &= 640 \text{ Nm}^2 \text{C}^{-1} \end{aligned}$$

120. (226)

$$\phi_{\text{total}} = \frac{q}{\epsilon_0},$$

From symmetry electric flux,

$$\begin{aligned} \phi_{S_q} &= \frac{1}{6} \left(\frac{q}{\epsilon_0} \right) = \frac{12 \times 10^{-6}}{6 \times 8.85 \times 10^{-12}} = 225.98 \times 10^3 \\ &= 226 \times 10^3 \frac{\text{Nm}^2}{\text{C}} \end{aligned}$$

121. (1)

Given:

$$\text{Electric field, } \vec{E} = \frac{3}{5} E_0 \hat{i} + \frac{4}{5} E_0 \hat{j} \frac{\text{N}}{\text{C}}$$

$$\text{Area (Parallel to } y\text{-}z \text{ plane), } A_a = 0.2 \hat{i} \text{ m}^2$$

$$\text{Area (Parallel to } x\text{-}z \text{ plane), } A_b = 0.3 \hat{j} \text{ m}^2$$

$$\text{Flux, } \phi_a = \vec{E} \cdot \vec{A}_a = \left(\frac{3}{5} E_0 \hat{i} + \frac{4}{5} E_0 \hat{j} \right) \cdot 0.2 \hat{i} = \frac{6}{50} E_0$$

$$\phi_b = \vec{E} \cdot \vec{A}_b = \left(\frac{3}{5} E_0 \hat{i} + \frac{4}{5} E_0 \hat{j} \right) \cdot 0.3 \hat{j} = \frac{12}{50} E_0$$

$$\therefore \frac{\phi_a}{\phi_b} = \frac{6/50 E_0}{12/50 E_0} = \frac{1}{2}.$$

So $a = 1$

122. (-48)

Flux of electric field $\vec{E}$ through any area $\vec{A}$ is defined as

$$\phi = \int E \cdot A \cos \theta$$

Here, θ = angle between electric field and area vector of a surface

For surface ABCD Angle, $\theta = 90^\circ$

$$\therefore \phi_1 = \int E \cdot A \cos 90^\circ = 0$$

$$\text{For surface BCGF } \phi_n = \int \vec{E} \cdot d\vec{A}$$

$$\therefore \phi_{11} = \left[4x \hat{i} - (y^2 + 1) \hat{j} \right] \cdot 4 \hat{i} = 16x$$

$$\phi_{11} = 48 \frac{\text{NM}^2}{\text{C}}; \phi_1 - \phi_{11} = -48 \frac{\text{NM}^2}{\text{C}}$$

123. (2)

By Gauss law,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

If $E = \text{cons.}$ And $\vec{E} \parallel \vec{A}$, then $|E||A| = \frac{q_{\text{in}}}{\epsilon_0}$

$\Rightarrow |E| = \frac{q_{\text{in}}}{\epsilon_0 A}$ And $\vec{E}$ is always $\perp$ to equipotential

surface so, gaussian surface is equipotential.

124. (2)

By Gauss Law,

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{in}}}{\epsilon_0} = \frac{Q + q - q}{\epsilon_0} = \frac{Q}{\epsilon_0}$$

$$\text{or } E \times 4\pi r^2 = \frac{Q}{\epsilon_0} \Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}, r \geq R$$

So, electric field outside is same as point charge inside the shell.

125. (1)

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$\text{or } E \times 4\pi r^2 = \frac{1}{\epsilon_0} \int \delta(4\pi r^2) dr$$

$$\Rightarrow E \times 4\pi r^2 = \frac{1}{\epsilon_0} \int_0^r (kr) (4\pi r^2) dr$$

$$\text{or } E \times 4\pi r^2 = \frac{4\pi k}{\epsilon_0} \left(\frac{r^4}{4} \right)$$

$$\therefore E = \frac{k}{4\epsilon_0} r^2 \quad \dots\dots(i)$$

$$\text{Also } 2Q = \int_0^R (kr) (4\pi r^2) dr = 4\pi k \left[\frac{r^4}{4} \right]_0^R$$

$$Q = \frac{\pi k R^4}{2} \quad \dots\dots(ii)$$

From above equations,

$$E = \frac{Qr^2}{2\pi\epsilon_0 R^4}. \text{ For } r = a, E = \frac{Qa^2}{2\pi\epsilon_0 R^4} \quad \dots\dots(iii)$$

According to given condition

$$QE = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{(2a)^2}$$

$$= Q \cdot \frac{Qa^2}{2\pi\epsilon_0 R^4} = \frac{Q^2}{4\pi\epsilon_0 \cdot 4a^2} \Rightarrow R = a8^{1/4} \Rightarrow a = R8^{-1/4}$$

126. (3)

The net flux linked with closed surfaces S_1 , S_2 , S_3 & S_4 are

$$\text{For surface } S_1, \phi_1 = \frac{1}{\epsilon_0} (2q)$$

$$\text{For surface } S_2, \phi_2 = \frac{1}{\epsilon_0} (q + q + q - q) = \frac{1}{\epsilon_0} 2q$$

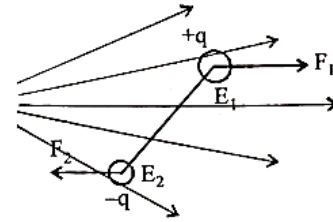
$$\text{For surface } S_3, \phi_3 = \frac{1}{\epsilon_0} (q + q) = \frac{1}{\epsilon_0} (2q)$$

$$\text{For surface } S_4, \phi_4 = \frac{1}{\epsilon_0} (8q - 2q - 4q) = \frac{1}{\epsilon_0} (2q)$$

Hence, $\phi_1 = \phi_2 = \phi_3 = \phi_4$ i.e. net electric flux is same for all surfaces.

Keep in mind, the electric field due to a charge outside (S_3 and S_4), the Gaussian surface contributes zero net flux through the surface, because as the lines due to that charge enter the surface is same as leave it.

127. (3)



As the dipole is placed in non-uniform field, so the force acting on the dipole will not cancel each other. This will result in a translational force as well as torque.

128. (1)

The electric flux ϕ_1 entering an enclosed surface is taken as negative and the electric flux ϕ_2 leaving the surface is taken as positive, by convention. Therefore the net flux leaving the enclosed surface, $\phi = \phi_2 - \phi_1$

$$\phi = \frac{q}{\epsilon_0} \Rightarrow q = \epsilon_0 \phi = \epsilon_0 (\phi_2 - \phi_1)$$

129. (1)

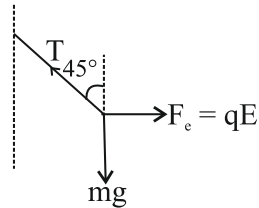
Charge accumulated maximum at sharp end.

$$\text{So, } \sigma_1 > \sigma_3 > \sigma_2 = \sigma_4.$$

130. (4)

At equilibrium

$$q_E = mg$$

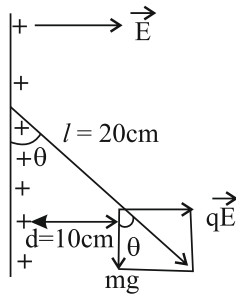


$$q \left[\frac{\sigma}{2\epsilon_0} \right] = mg$$

$$\sigma = \frac{2\epsilon_0 mg}{q} = \frac{2 \times 8.85 \times 10^{-12} \times 100 \times 10^{-3} \times 10}{10 \times 10^{-6}}$$

$$\therefore \sigma = 1.77 \text{ nC/m}^2$$

131. (3)



$$\text{Here, } \sin \theta = \frac{d}{l} = \frac{10}{20} = \frac{1}{2} \therefore \theta = 30^\circ$$

$$\tan \theta = \frac{F_e}{F_g} = \frac{qE}{mg}$$

Substituting values,

$$\tan 30^\circ = \frac{q \times 2 \times 10^4}{2 \times 10^{-3} \times 10} \Rightarrow q = \frac{1}{\sqrt{3}} \mu\text{C} = \frac{1}{\sqrt{x}} \mu\text{C}$$

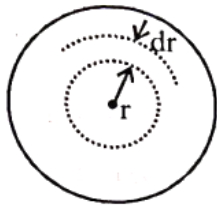
(given)

$$\therefore x = 3$$

132. (2)

$$Q = \int \rho dv = \int_0^R \frac{A}{r^2} e^{-2r/a} (4\pi r^2 dr)$$

$$\Rightarrow Q = 4\pi A \int_0^R e^{-2r/a} dr = 4\pi A \left[\frac{e^{-2r/a}}{-\frac{2}{a}} \right]_0^R$$



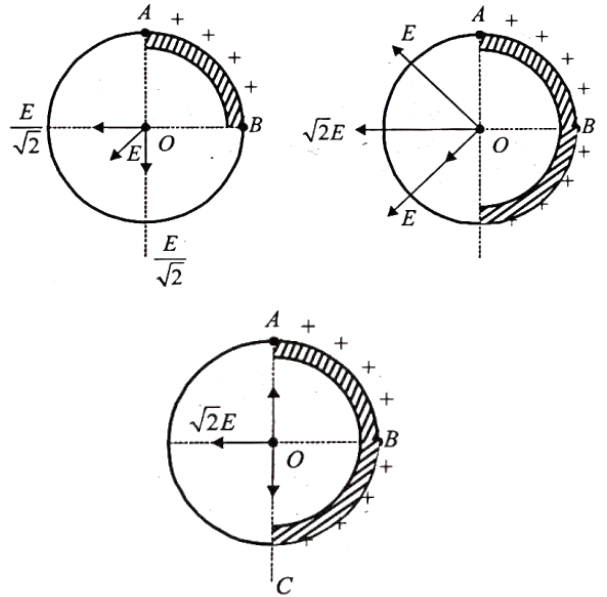
$$\Rightarrow Q = 4\pi A \left(-\frac{a}{2} \right) (e^{-2R/a} - 1)$$

$$\Rightarrow Q = 2\pi a A (1 - e^{-2R/a})$$

$$\Rightarrow R = \frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$$

133. (2)

The E.F. at the center of a full uniformly charged ring is zero but when only a $\frac{1}{4}$ th arc remains (from $\theta = 0$ to $\frac{\pi}{2}$), the symmetry is broken.



134. (3)

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{due to sheet})$$

$$\Rightarrow E = \frac{\rho R^3}{3\epsilon_0 r^2} \quad (\text{due to sphere})$$

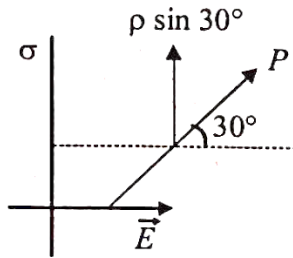
E (due to sheet) will be same at point C and D but E (due to sphere) will be maximum at point D. Similarly, for point A and B.

135. (2)

$$E = \frac{\sigma}{2\epsilon_0}$$

$$\vec{\tau} = \vec{P} \times \vec{E}$$

$$\tau = PE \sin \theta$$



$$= \left[\left(2 \times 10^{-6} \right) \left(\frac{2}{10} \right) \right] \left[\frac{100 \times 10^{-6}}{2 \times 8.85 \times 10^{-12}} \right] \left(\frac{1}{2} \right)$$

$$= \frac{10}{8.85} = 1.12 \text{ Nm.}$$

136. (3)

$$\text{Electric field } E = \frac{KQr}{(x^2 + R^2)^{3/2}}$$

$$\frac{dE}{dx} = 0 \therefore x = \frac{R}{\sqrt{2}} = \frac{\sqrt{2}a}{\sqrt{2}} = a$$

137. (6)

Electric field intensity

$$E_P = \frac{\lambda}{2\pi\epsilon_0 r_P} - \frac{\sigma}{2\epsilon_0} \left\{ r_P = \frac{3}{\pi} \right\} = \frac{1}{2\epsilon_0} \left[\frac{\lambda}{3} - \sigma \right]$$

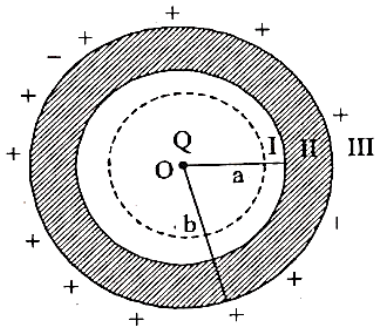
$$E_Q = \frac{\lambda}{2\pi\epsilon_0 r_Q} - \frac{\sigma}{2\epsilon_0} \left\{ r_Q = \frac{4}{\pi} \right\} = \frac{1}{2\epsilon_0} \left[\frac{\lambda}{4} - \sigma \right]$$

$$\frac{E_P}{E_Q} = \frac{4}{3} \left(\frac{\lambda - 3\sigma}{\lambda - 4\sigma} \right) = \frac{4}{3} \left[\frac{2\sigma - 3\sigma}{2\sigma - 4\sigma} \right] = \frac{4}{3} \left[\frac{-\sigma}{-2\sigma} \right] = \frac{4}{6}$$

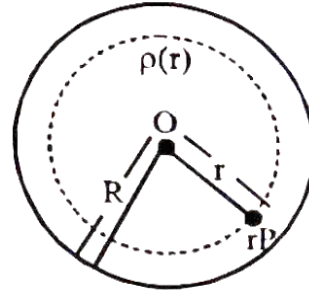
$$\therefore a = 6$$

138. (2)

In region I Electric field will be due to charge Q
 $E_{II} = 0$. Since electric field inside conductor is zero.
 $E_{III} \neq 0$, Electric field will be due to charge Q only
 or just due to charge.



139. (3)



$$\text{By Gauss law} = \oint \vec{E} \cdot \vec{da} = \frac{q_{in}}{\epsilon_0}$$

$$\text{Here, } E \cdot 4\pi r^2 = \frac{\int_0^r \rho_0 \left(\frac{3}{4} - \frac{r}{R} \right) 4\pi r^2 dr}{\epsilon_0}$$

$$\Rightarrow E \cdot 4\pi r^2 = \frac{\rho_0 4\pi}{\epsilon_0} \left(\frac{3}{4} \frac{r^3}{3} - \frac{r^4}{4R} \right)$$

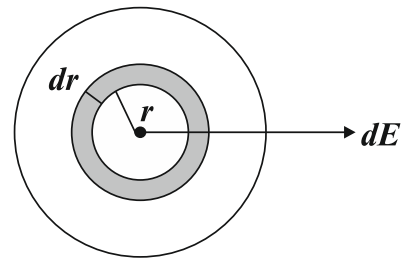
$$\Rightarrow E r^2 = \frac{\rho_0 r^3}{4\epsilon_0} \left(1 - \frac{r}{R} \right)$$

$$\Rightarrow E = \frac{\rho_0 r}{4\epsilon_0} \left(1 - \frac{r}{R} \right)$$

140. (1)

Consider a small ring of radius r and thickness dr on disc.

$$dq = \sigma 2\pi r dr$$



$$dE = \frac{K dq z}{(r^2 + z^2)^{3/2}} = \frac{K \sigma (2\pi r) dr z}{(r^2 + z^2)^{3/2}}$$

$$E = \int dE = \int dE$$

$$= \int_0^R \frac{K \sigma 2\pi r dr z}{(z^2 + r^2)^{3/2}} = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{R^2 + z^2}} \right]$$

141. (2)

Charge per unit length of the rod,

$$\lambda = \left(\frac{-Q}{R\theta} \right) = \left(\frac{-Q}{R \frac{2\pi}{3}} \right) = \frac{-3Q}{2\pi R}$$

$$E = \int dE \cos \theta \Rightarrow E = \int_{-\pi/3}^{\pi/3} \frac{K \times (Q)}{\frac{2\pi}{3} R} \times \frac{R d\theta}{R^2} \cos \theta$$

$$\Rightarrow E = \frac{3}{2\pi} \frac{KQ}{R^2} [\sin \theta]_{-\pi/3}^{\pi/3} = \frac{3}{2\pi} \frac{KQ}{R^2} \times \frac{2\sqrt{3}}{2}$$

$$\Rightarrow E = \frac{3\sqrt{3}Q}{8\pi^2 \epsilon_0 R^2} \left(\hat{i} \right)$$

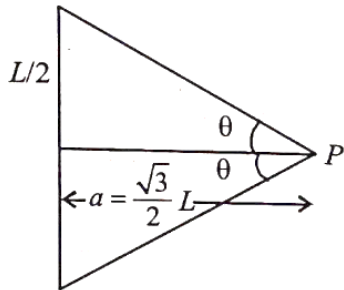
142. (4)

Electric field at point P

$$E = \frac{k\lambda}{a} (\sin \theta_1 + \sin \theta_2)$$

$$= E = \frac{1}{4\pi\epsilon_0} \times \frac{Q}{L} \times \frac{1}{\left(\frac{\sqrt{3}L}{2} \right)} \times 2 \sin \theta \quad \left[\because \lambda = \frac{Q}{L} \right]$$

From Figure,



$$\tan \theta = \frac{\frac{L}{2}}{\frac{\sqrt{3}L}{2}} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = 30^\circ$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \times \frac{2Q}{\sqrt{3}L^2} \times \left(\frac{2 \times 1}{2} \right) \quad \left[\because \sin 30^\circ = \frac{1}{2} \right]$$

$$\Rightarrow E = \frac{Q}{2\sqrt{3}\pi\epsilon_0 L^2}$$

143. (3)

For spherical shell

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad (\text{if } r \geq R)$$

$$= 0 \quad (\text{if } r < R)$$

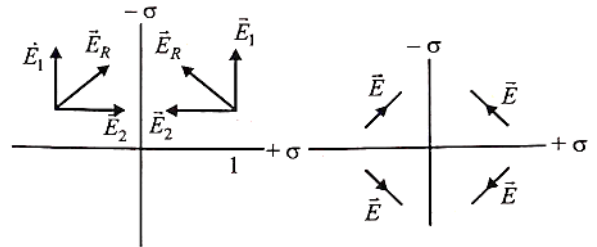
Force on charge in electric field, $F = qE$

$$\therefore F = 0 \quad (\text{For } r < R)$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} \quad (\text{For } r > R)$$

144. (3)

The electric field produced due to uniformly charged infinite plane is uniform. So option (b) and (d) are wrong. And +ve charge density σ_+ is bigger in magnitude so its field along Y direction will be bigger than field of -ve charge density σ_- in X direction. Hence option (c) is correct.



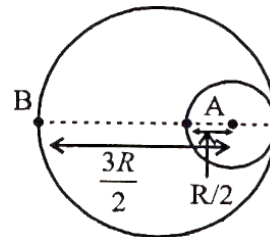
145. (2)

Electric field at A $\left(R' = \frac{R}{2} \right)$

$$E_A \cdot ds = \frac{q}{\epsilon_0}$$

$$\Rightarrow E_A = \frac{\rho \times \frac{4}{3} \pi \left(\frac{R}{2} \right)^3}{\epsilon_0 \cdot 4\pi \left(\frac{R}{2} \right)^2}$$

$$\Rightarrow E_A = \frac{\sigma(R/2)}{3\epsilon_0} = \left(\frac{\sigma R}{3\epsilon_0} \right)$$



Electric fields at 'B'

$$\Rightarrow E_B = \frac{k \times \rho \times \frac{4}{3} \pi R^3}{R^2} - \frac{k \times \rho \times \frac{4}{3} \pi \left(\frac{R}{2}\right)^3}{\left(\frac{3R}{2}\right)^2}$$

$$\Rightarrow E_B = \frac{\sigma q}{3\epsilon_0} - \left(\frac{1}{4\pi\epsilon_0}\right) \frac{(\sigma)}{\left(\frac{3R}{2}\right)^2} \frac{4\pi \left(\frac{R}{2}\right)^3}{3}$$

$$\Rightarrow E_B = \frac{\sigma R}{3\epsilon_0} - \frac{\sigma R}{54\epsilon_0} \Rightarrow E_B = \frac{17}{54} \left(\frac{\sigma R}{\epsilon_0}\right)$$

$$\left|\frac{E_A}{E_B}\right| = \frac{1 \times 54}{6 \times 17} = \left(\frac{9}{17}\right) = \frac{9}{17} \times \frac{2}{2} = \frac{18}{34}$$

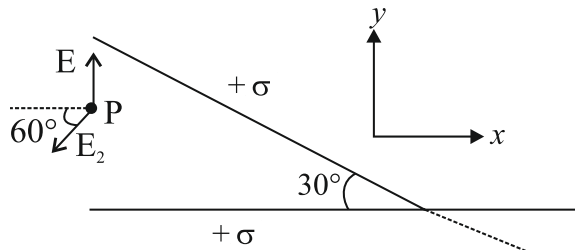
146. (4)

From figure,

$$\vec{E}_1 = \frac{\sigma}{2\epsilon_0} \hat{y} \text{ and}$$

$$\vec{E}_2 = \frac{\sigma}{2\epsilon_0} \left(-\cos 60^\circ \hat{x} - \sin 60^\circ \hat{y} \right)$$

$$= \frac{\sigma}{2\epsilon_0} \left(-\frac{1}{2} \hat{x} - \frac{\sqrt{3}}{2} \hat{y} \right)$$



Electric field in the region shown in figure (P)

$$\vec{E}_P = \vec{E}_1 + \vec{E}_2 = \frac{\sigma}{2\epsilon_0} \left[-\frac{1}{2} \hat{x} + \left(1 - \frac{\sqrt{3}}{2} \right) \hat{y} \right]$$

$$\text{or, } \vec{E}_P = \frac{\sigma}{2\epsilon_0} \left[\left(1 - \frac{\sqrt{3}}{2} \right) \hat{y} - \frac{\hat{x}}{2} \right]$$

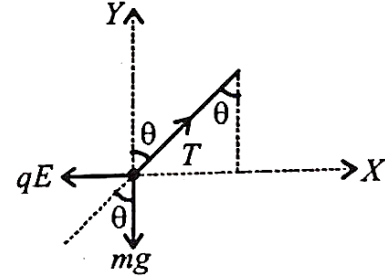
147. (4)

Time period of the pendulum (T) is given by

$$T = 2\pi \sqrt{\frac{L}{g_{\text{eff}}}} \Rightarrow g_{\text{eff}} = \frac{\sqrt{(mg)^2 + (qE)^2}}{m}$$

$$\Rightarrow g_{\text{eff}} = \sqrt{g^2 + \left(\frac{qE}{m}\right)^2} \Rightarrow T = 2\pi \sqrt{\frac{L}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2}}}$$

148. (4)



At equilibrium resultant force on bob must be zero, so

$$T \cos \theta = mg \quad \dots\dots(i)$$

$$T \sin \theta = qE \quad \dots\dots(ii)$$

Solving (i) and (ii) we get

$$\tan \theta = \frac{qE}{mg}$$

$$\tan \theta = \frac{5 \times 10^{-6} \times 2000}{2 \times 10^{-3} \times 10} = \frac{1}{2}$$

[Here, $q = 5 \times 10^{-6} \text{ C}$, $E = 2000 \text{ v/m}$, $m = 2 \times 10^{-3} \text{ kg}$]

$$\Rightarrow \theta = \tan^{-1} \left(\frac{1}{2} \right)$$

149. (2)

Electric field on the axis of a ring of radius R at a distance h from the centre,

$$E = \frac{kQh}{(h^2 + R^2)^{3/2}}$$

Condition: for maximum electric field $\frac{dE}{dh} = 0$

$$\Rightarrow \frac{d}{dh} \left[\frac{kQh}{(R^2 + h^2)^{3/2}} \right] = 0$$

By using the concept of maxima and minima we get,

$$h = \frac{R}{\sqrt{2}}$$

150. (4)

Charge density, $\rho = \rho_0 \left(1 - \frac{r}{R}\right)$

$$dq = \rho dv$$

$$q_{in} = \int dq = \int \rho dv$$

$$\int \rho_0 \left(1 - \frac{r}{R}\right) 4\pi r^2 dr \quad (\because dv = 4\pi r^2 dr)$$

$$= 4\pi\rho_0 \int_0^R \left(1 - \frac{r}{R}\right) r^2 dr = 4\pi\rho_0 \int_0^R r^2 dr - \frac{r^3}{R} dr$$

$$= 4\pi\rho_0 \left[\left[\frac{r^3}{3} \right]_0^R - \left[\frac{r^4}{4R} \right]_0^R \right] = 4\pi\rho_0 \left[\frac{R^3}{3} - \frac{R^4}{4R} \right]$$

$$= 4\pi\rho_0 \left[\frac{R^3}{3} - \frac{R^3}{4} \right] = 4\pi\rho_0 \left[\frac{R^3}{12} \right]$$

$$q_{in} = \frac{\pi\rho_0 R^3}{3}$$

Using Gauss law, $\oint \vec{E} \cdot \vec{dA} = \frac{q_{in}}{\epsilon_0}$

$$\Rightarrow E \cdot 4\pi r^2 = \left(\frac{\pi\rho_0 R^3}{3\epsilon_0} \right)$$

$$\therefore \text{Electric field outside the ball, } E = \frac{\rho_0 R^3}{12\epsilon_0 r^2}$$

151. (3)

Field lines originate perpendicular from positive charge and terminate perpendicular at negative charge. Further this system can be treated as an electric dipole.

152. (4)

Given: Length of wire $L = 20$ cm

$$\text{Charge } Q = 10^3 \epsilon_0$$

We know, electric field at the centre of the semicircular arc

$$E = \frac{2K\lambda}{r} \Rightarrow E = \frac{2K \left(\frac{2Q}{\pi r} \right)}{r} \left[\text{As } \lambda = \frac{2Q}{\pi r} \right]$$

$$= \frac{4KQ}{\pi r^2} = \frac{4KQ\pi^2}{\pi L^2} = \frac{4\pi KQ}{L^2} = 25 \times 10^3 \text{ N/C } \hat{i}$$

153. (3)

Given,

Electric field $E = 150$ N/C

Total surface charge carried by earth $q = ?$

By Gauss law

$$\frac{q_{in}}{\epsilon_0} = \int \vec{E} \cdot \vec{da} \Rightarrow \frac{q_{in}}{\epsilon_0} = |E| \times 4\pi r^2$$

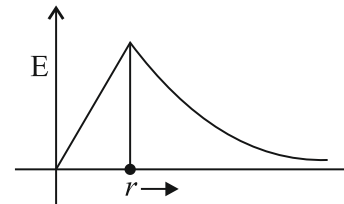
$$\Rightarrow q_{in} = 4\pi\epsilon_0 |E| r^2$$

$$= 4\pi \times 8.85 \times 10^{-12} \times 150 \times (6.37 \times 10^6)^2 \simeq 680 \text{ Kc.}$$

As electric field directed inward hence

$$q = -680 \text{ kC}$$

154. (3)



$$|E| \propto r, r < 0$$

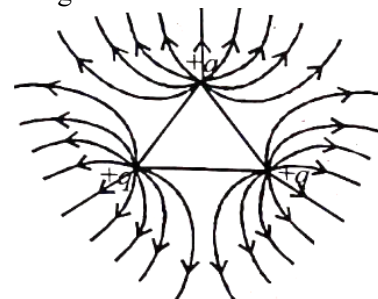
$$|E| \propto \frac{1}{r^2}, r > 0$$

So the graph will be

155. (3)

Electric lines of force due to a positive charge is spherically symmetric.

All the charges are positive and equal in magnitude. So repulsion takes place. Due to which no lines of force are present inside the equilateral triangle and the resulting lines of force obtained as shown:



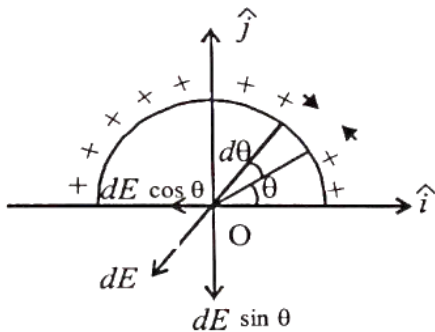
156. (3)

Let us consider a differential element dl subtending at angle $d\theta$ at the centre Q as shown in the figure. Linear charge density.

$$\lambda = \frac{q}{Qr}$$

Charge on the element,

$$dq = \left(\frac{q}{\pi r} \right) dl$$



$$= \frac{q}{\pi r} (rd\theta) = \left(\frac{q}{\pi} \right) d\theta \quad (\because dl = rd\theta)$$

Electric field at the center O due to dq is.

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{\pi r^2} d\theta$$

Resolving dE into two rectangular component, we find the component $dE \cos \theta$ will be counter balanced by another element on left portion. Hence resultant field at O is resultant of the component $dE \sin \theta$ only.

$$\therefore E = \int dE \sin \theta = \int_0^\pi \frac{q}{4\pi^2 r^2 \epsilon_0} \sin \theta d\theta$$

$$= \frac{q}{4\pi^2 r^2 \epsilon_0} [-\cos \theta]_0^\pi = \frac{q}{4\pi^2 r^2 \epsilon_0} (+1+1) = \frac{q}{2\pi^2 r^2 \epsilon_0}$$

The direction of E is towards negative y-axis.

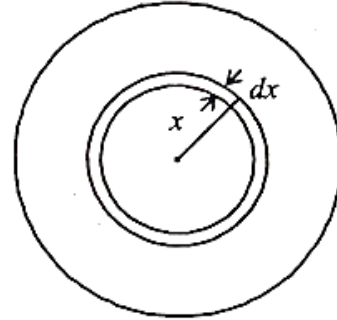
$$\therefore \vec{E} = -\frac{q}{2\pi^2 r^2 \epsilon_0} \hat{j}$$

157. (1)

Let us consider a spherical shell of radius x and thickness dx .

Due to spherically symmetric charge distribution, the charge on the spherical surface of radius x is.

$$dq = \rho dv = \rho \cdot 4\pi x^2 dx = \rho_0 \left(\frac{5}{4} - \frac{x}{R} \right) \cdot 4\pi x^2 dx$$



$\therefore$ Total charge in the spherical region from centre to r ($r < R$) is

$$\begin{aligned} q_{in} &= \int dq = 4\pi\rho_0 \int_0^r \left(\frac{5}{4} - \frac{x}{R} \right) x^2 dx \\ &= 4\pi\rho_0 \left[\frac{5}{4} \cdot \frac{r^3}{3} - \frac{1}{R} \cdot \frac{r^4}{4} \right] \\ &= \pi\rho_0 r^3 \left(\frac{5}{3} - \frac{r}{R} \right) \end{aligned}$$

By Gauss law

$$\frac{q_{in}}{\epsilon_0} = \oint \vec{E} \cdot d\vec{a}$$

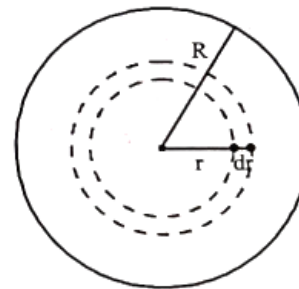
$$\Rightarrow \frac{\pi\rho_0 r^3 \left(\frac{5}{3} - \frac{r}{R} \right)}{\epsilon_0} = |E| \times 4\pi r^2$$

$$\Rightarrow E = \frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$$

158. (2)

Let us consider a spherical shell of thickness dr and radius r . The area of this spherical shell $= 4\pi r^2$.

The volume of this spherical shell $= 4\pi r^2 dr$. The charge enclosed within shell



$$dq = \left[\frac{Qr}{\pi R^4} \right] [4\pi r^2 dr] = \frac{4Q}{R^4} r^3 dr$$

The charge enclosed in a sphere of radius r_1 can be calculated by

$$Q = \int dq = \frac{4Q}{R^4} \int_0^{r_1} r^3 dr = \frac{4Q}{R^4} \left[\frac{r^4}{4} \right]_0^{r_1} = \frac{Q}{R^4} r_1^4$$

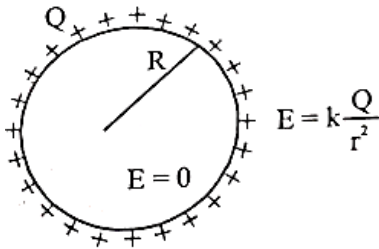
By Gauss law,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0} \Rightarrow |E| \times 4\pi r_1^2 = \frac{1}{\epsilon_0} \left(\frac{Q r_1^4}{R^4} \right)$$

$$\Rightarrow |E| = \frac{Q r_1^2}{4\pi \epsilon_0 R^4}$$

159. (1)

The electric field inside a thin spherical shell of radius R has charge Q spread uniformly over its surface is zero.

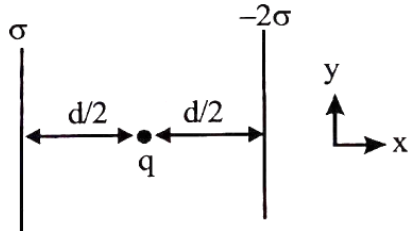


Outside the shell the electric field is $E = k \frac{Q}{r^2}$. These characteristics are represented by graph (a).

160. (2)

Here net electric field

$$E = \frac{\sigma}{2\epsilon_0} + \frac{2\sigma}{2\epsilon_0} = \frac{3\sigma}{2\epsilon_0}$$



$$\therefore F_{net} = qE = \frac{3\sigma}{2\epsilon_0} q$$

161. (3)

Using Gauss Law

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$EdA = \frac{\sigma \times dA}{\epsilon_0} \quad [\because Q = \sigma dA]$$

$$\Rightarrow E = \frac{\sigma}{\epsilon_0}$$

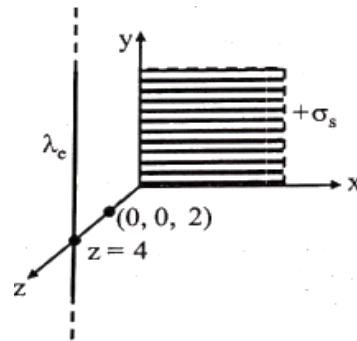
162. (2)

Electric field at point P is,

$$\vec{E} = \left(\frac{\sigma}{2\epsilon_0} + \frac{2\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} \right) (-\hat{i}) = -\frac{2\sigma}{\epsilon_0} \hat{i}$$

$$\therefore x = 2.$$

163. (16)



Here,

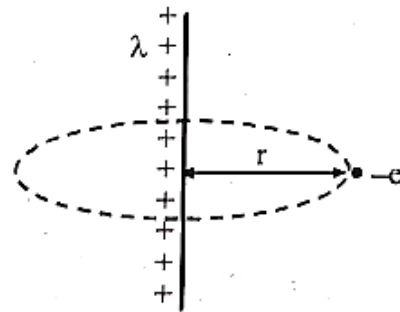
$$E_s = \frac{\sigma_s}{2\epsilon_0} \text{ and } E_\ell = \frac{\lambda_e}{2\pi r \epsilon_0}$$

$$\therefore \frac{E_s}{E_\ell} = \frac{\sigma}{2\epsilon_0} \times \frac{2\pi \epsilon_0 r}{\lambda}$$

$$= \frac{\pi \times \sigma r}{\lambda} = \frac{\pi \times 2\lambda \times 2}{\lambda} = \frac{4\pi}{1}$$

$$\therefore 4\pi : 1 = \pi \sqrt{n} : 1 \quad \therefore n = 16$$

164. (2)



Electric field E at a distance r due to infinite long wire

$$E = \frac{2k\lambda}{r} \text{ where } \lambda = \text{linear charge density}$$

Force on electron $F = eE$

$$F = e \left(\frac{2k\lambda}{r} \right) = \frac{2k\lambda e}{r}$$

This force will provide required centripetal force

$$F = \frac{mv^2}{r} = \frac{2k\lambda e}{r} \text{ or, } v = \sqrt{\frac{2k\lambda e}{m}}$$

$$\therefore KE = \frac{1}{2}mv^2 = \frac{1}{2}m \left(\frac{2k\lambda e}{m} \right) = k\lambda e \text{ (constant)}$$

So option (b) represents correct KE-r graph.

165. (2)

Electric field due to infinitely long wire, $E = \frac{\lambda}{2\pi\epsilon_0 r}$

$$F = Qe$$

$$= \frac{\lambda \cdot q}{2\pi\epsilon_0 r} = m\omega^2 r \quad \left(\because k = \frac{1}{4\pi\epsilon_0} \right)$$

$$\frac{2k\lambda q}{r} = m\omega^2 r \Rightarrow \omega^2 = \frac{2k\lambda q}{mr^2}$$

$$\left(\frac{2\pi}{T} \right)^2 = \frac{2k\lambda q}{mr^2} \Rightarrow T = 2\pi r \sqrt{\frac{m}{2k\lambda q}}$$

166. (1)

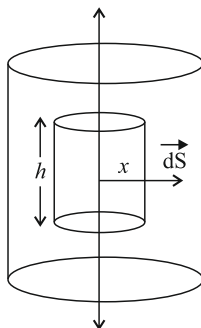
By Gauss's law

$$\int E ds \cos 0 = \frac{q}{\epsilon_0}$$

$$\Rightarrow E \cdot 2\pi x h = \frac{\rho \times \pi x^2 h}{\epsilon_0}$$

$$\Rightarrow E = \frac{\rho x}{2\epsilon_0}$$

$$\Rightarrow E = \frac{\rho}{2\epsilon_0} \times \frac{2\epsilon_0}{\rho} = 1 \text{ Vm}^{-1}$$



167. (3)

Electric field due to infinite sheet is given by

$$E = \frac{\sigma}{2\epsilon_0}, \text{ clearly } E \text{ is independent of distance}$$

$$\text{So, } E_1 = E_2 = \frac{\sigma}{2\epsilon_0}$$

168. (1)

As charge particle is moving in circular path. So,

$$qE = \frac{mv^2}{r}$$

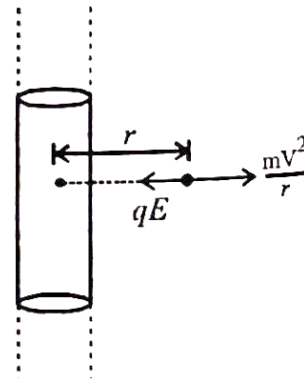
$$\Rightarrow mv^2 = qEr$$

$$\therefore K.E = \frac{mv^2}{2} = \frac{1}{2}qEr \quad \dots(i)$$

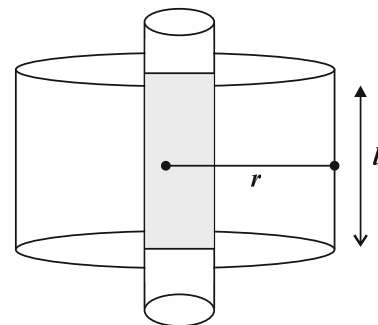
Now, by Gauss's law

$$E \times 2\pi r l = \frac{q_{in}}{\epsilon_0}$$

$$E \times 2\pi r l = \frac{\rho \times \pi R^2 l}{\epsilon_0}$$



$$E = \frac{\rho R^2}{2\epsilon_0 r}$$





From (i) and (ii), we get

$$K.E = \frac{1}{2}mv^2 = \frac{1}{2}q \times \frac{\rho R^2}{2\epsilon_0 r} \times r = \frac{\rho q R^2}{4\epsilon_0}$$

169. (3)

Applying Gauss's law

$$\oint_S \vec{E} \cdot \vec{ds} = \frac{Q + q}{\epsilon_0};$$

$$q = \int_a^r \frac{A}{r} 4\pi r^2 dr = 2\pi A [r^2 - a^2]$$

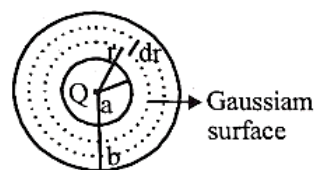
$$\therefore E \times 4\pi r^2 = \frac{Q + 2\pi A r^2 - 2\pi A a^2}{\epsilon_0}$$

$$E \times 4\pi r^2 = (Q - 2\pi A a^2) + 2\pi A r^2$$

For E to be independent of 'r'

$$Q - 2\pi A a^2 = 0$$

$$\Rightarrow A = \frac{Q}{2\pi a^2}$$



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>